

Random Matrix Theory and L -Functions:
From Gaussian Ensembles to Low-Lying Zeros,
Compact Symmetry, Universality, and the Ten-Fold Way

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Based on the author's 2013 thesis submitted to the
Facultad de Ciencias, Universidad Nacional Autónoma de México

2026

*Para el Dr. Eduardo Dueñez Guzmán,
maestro y guía:*

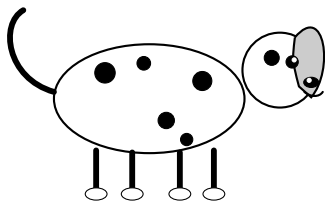
*por las conversaciones que empezaron con una pregunta
y nunca terminaron del todo;*

*por enseñarme que los ceros de las funciones L
y los valores propios de las matrices aleatorias
no son coincidencia — son un secreto compartido
entre el orden y el azar;*

y por confiar en mí para buscarlo.

Para mis alumnas y alumnos.

Para mi inspiración, Tarski:



*y para mi sol, Grecia
—el amor de mi vida.*

Abstract

This monograph grows out of the author's 2013 thesis and recasts that earlier work as a broader study of the ways in which local spectral statistics recur across analysis, number theory, geometry, and representation theory. Its central theme is that random matrices and L -functions are not linked by analogy alone. They are connected by a common pattern: one starts from an exact finite-dimensional law or trace formula, identifies a mechanism of local repulsion or symmetry, and then passes to a limit in which model-dependent structure gives way to a small number of universal kernels or compact symmetry types.

The analytic core develops this pattern for Gaussian ensembles through eigenvalue densities, orthogonal-polynomial and Pfaffian formalisms, Fredholm determinants, Dyson's theorem, and bulk universality. A second block treats compact groups and circular ensembles via Weyl integration and exact finite- N formulas. The arithmetic chapters then follow the same local questions into the theory of zeros of L -functions: Montgomery's pair correlation, the Rudnick–Sarnak bulk universality theorem, the Katz–Sarnak philosophy of low-lying zeros, Deligne-style equidistribution, the Sarnak–Shin–Templier invariants theorem, and a detailed case study on quadratic Dirichlet L -functions. A final chapter extends the symmetry dictionary through Dyson's threefold way to the full Altland–Zirnbauer ten-fold way, developing Dueñez's framework of random matrix ensembles associated to compact symmetric spaces and showing how restricted-root multiplicities determine the Jacobi weights and Bessel hard-edge kernels that govern the distribution of the first zero above the central point in arithmetic families.

The exposition is intended to be both explanatory and mathematically precise. Proofs are written so that the underlying mechanism remains visible throughout: finite- N algebra, comparison arguments, arithmetic averaging, geometric monodromy, and the Weyl integration formula on symmetric spaces are treated not as isolated techniques but as parts of a single spectral narrative whose arc runs from Wigner's original random Hamiltonian model to the modern Langlands-theoretic classification of symmetry types.

Abstract

Resumen. Esta monografía surge de la tesis del autor de 2013 y la reescribe como un estudio más amplio de la manera en que las estadísticas espectrales locales reaparecen a través del análisis, la teoría de números, la geometría y la teoría de representaciones. La idea rectora es que las matrices aleatorias y las funciones- L no están unidas sólo por una analogía, sino por un mismo patrón: se parte de una ley exacta en dimensión finita o de una fórmula de trazas, se identifica un mecanismo de repulsión local o de simetría, y luego se pasa a un límite en el que la estructura particular del modelo es sustituida por un pequeño conjunto de núcleos universales o tipos de simetría compacta. El capítulo final extiende el diccionario de simetría desde el camino triple de Dyson hasta el camino décuple de Altland–Zirnbauer, desarrollando el marco de Dueñez para ensembles de matrices aleatorias asociados a espacios simétricos compactos.

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1. Introduction

“This is the kind of mathematics that I enjoy, with many conjectures and few proofs. Deep mysteries remain, and the best is still to come”. (F. Dyson)

The theory of random matrices originated in the late 1950s with the physicist Eugene Wigner, who introduced them to model the spectra of heavy atoms composed of many particles with unknown interactions. Unable to explicitly calculate the behavior of these systems, the idea arose to forgo them and provide a completely random description by considering sets of matrices, known as ensembles, with randomly distributed entries obeying certain symmetry constraints. Recently, random matrix theory has permeated various areas of mathematics and physics, including number theory, operator theory, random permutations, and random growth models [akemann2011oxford]. This work addresses the relationship between random matrix theory and number theory.

One of the most emblematic unsolved problems in mathematics is the Riemann Hypothesis. Implicitly, this conjecture states that the non-trivial zeros of the Riemann zeta function lie on the line $\text{Re}(s) = 1/2$ of the complex plane. Around 1927, Hilbert and Pólya suggested that there might be a spectral interpretation of the zeros of the Riemann zeta function. That is, they conjectured that it might be possible to establish an explicit correspondence between the non-trivial zeros of the Riemann zeta function and the eigenvalues of some operator acting on a Hilbert space.

The first significant evidence for the Hilbert-Pólya conjecture came in 1972 when the mathematician Hugh Montgomery [montgomery1973pair] calculated the pair correlation of the Riemann zeta function. He found that its zeros exhibit the same repulsion as the eigenvalues of large unitary matrices (i.e., as the matrix size tends to infinity), a fact pointed out by the physicist Freeman Dyson. Montgomery conjectured more general analogies related to random matrices, which were confirmed numerically by Andrew Odlyzko in the 1980s. This phenomenon is known as the Montgomery-Odlyzko law.

Furthermore, in 1995, Rudnick and Sarnak [rudnick1996zeros] determined the $n \geq 2$ correlations for the zeros of the Riemann zeta function, as well as for more general L -functions. These results are in perfect agreement with predictions from random matrix theory. Unfortunately, the universal behavior described by the Montgomery-Odlyzko law does not provide information about what the relevant specific operator should be for each L -function.

The philosophy of Katz and Sarnak [katz1999random] teaches us that the situation is, in fact, even richer when studying the distribution of zeros of families of L -functions over function fields. Katz and Sarnak showed that, in function fields, the Montgomery-Odlyzko law holds true, but also that the laws associated with the distribution of zeros near the critical point are distinct. These laws correspond to eigenvalues of random matrices from the classical compact groups (unitary, orthogonal, and symplectic). In this case, this behavior can be explained naturally in terms of the geometric monodromy group associated with each family. This is the type of result we aim to explain here. The only thing we can say in advance is that random matrix theory turns out to be a remarkable and coherent probabilistic model for describing the zeros of L -functions.

The structure and choice of topics in this text are based on [katz1999zeros] and aim to be an extension of some of the themes discussed there. This monograph recasts and substantially expands the author's 2013 thesis. The original mathematical core is preserved and placed inside the larger web of ideas that has continued to grow around it, incorporating a more systematic treatment of compact symmetric spaces and the Altland-Zirnbauer ten-fold way as developed by Dueñez.

Remark 1.1 (How to read this monograph). *A helpful first reading strategy is to track mechanisms rather than categories. The same roles recur: an exact finite structure, a source of local repulsion, a stability argument under scaling, and a limiting symmetry principle. Once those roles are visible, the passage from matrices to zeros, from zeros to Frobenius classes, and from there to compact symmetric spaces and refined hard-edge statistics becomes mathematically natural.*

Remark 1.2 (Three guiding prototypes). *Three model problems summarize the architecture of this work.*

1. In the **GUE**, one asks how diagonalization produces the Vandermonde factor, how that factor creates repulsion, and how the sine and Airy kernels emerge after microscopic rescaling.
2. In **families of L -functions**, one asks why the explicit formula and conductor normalization should lead to the same local statistics predicted by compact groups.
3. In the **Dueñez framework** of the final chapter, one asks how the restricted-root system of a compact symmetric space forces a specific Jacobi weight and a specific Bessel hard-edge law for the first zero above the central point.

1.1. Scope of this monograph

Compared with the original thesis, the present version is not merely corrected or updated; it is reorganized as a substantially broader monograph. The analytic chapters have been expanded to

include a more systematic treatment of the orthogonal-polynomial formalism, Dyson's theorem on gap distributions, edge asymptotics, and the determinantal structure of correlation functions. The arithmetic chapters now integrate the Montgomery, Rudnick–Sarnak, Katz–Sarnak, and Sarnak–Shin–Templier viewpoints into a single storyline. A new final chapter enlarges the symmetry dictionary from Dyson's threefold way to the full Altland–Zirnbauer ten-fold way, developing Dueñez's framework for random matrix ensembles on compact symmetric spaces and its arithmetic consequences. The aim is therefore twofold: to preserve the original mathematical core and to place it inside the larger web of ideas that has continued to grow around it.

1.2. Organization of the monograph

The monograph is organized in layers. The early sections build the finite- N exact theory and its operator-theoretic consequences. The middle sections develop the asymptotic analytic machine of universality. The next block passes from compact groups to arithmetic families and geometric monodromy. The final section enlarges the symmetry dictionary through compact symmetric spaces and the ten-fold way. The appendices isolate longer proofs, asymptotic arguments, and kernel identities that support the body of the text.

1.3. A broader roadmap: global, mesoscopic, and microscopic scales

The title of this monograph places two subjects side by side: random matrices and L -functions. The point of the comparison is not merely metaphorical. In both worlds one studies large finite objects and asks what remains visible after renormalization. On the *global scale* one sees equilibrium laws such as the semicircle or the circular law on the unit circle; on the *mesoscopic scale* one sees rigidity, transport, and linear statistics; on the *microscopic scale* one sees spacing distributions, determinantal or Pfaffian structures, and eventually the universal kernels that survive after the ambient model has been forgotten.

Every later theorem lives on one of these three scales, and many of the most important results are statements about how one passes coherently from one scale to the next.

Remark 1.3 (Katz–Sarnak mechanism). *A geometric family gives rise to a lisse sheaf and hence to Frobenius conjugacy classes whose distribution is controlled by geometric monodromy. After normalization and passage to the large-field limit, Deligne's equidistribution theorem identifies the relevant compact symmetry group. The resulting Haar statistics then dictate the limiting low-lying distribution.*

1.4. Deligne equidistribution as the geometric analogue of Sato–Tate

There is a deep philosophical point here. Over number fields, Sato–Tate is difficult because one must vary the prime and keep the motive fixed. In the Katz–Sarnak setting, one fixes the finite field and varies the parameter in the family. This makes equidistribution accessible through the étale cohomology of the parameter space. The result is a laboratory in which the compact-group laws can be *proved* rather than guessed.

Remark 1.4. *The Katz–Sarnak philosophy is therefore not merely an analogy with random matrix theory; it is a theorem-producing machine. It explains why compact groups appear, why the same groups recur in arithmetic families over number fields, and why the low-lying zero problem is so naturally representation-theoretic.*

1.5. Characters, traces, and equidistribution

The most efficient proof of equidistribution is a character test, which is the geometric analogue of Weyl's criterion.

Theorem 1.5 (Character criterion for Frobenius equidistribution). *Let $\mathcal{F}$ be a pure lisse sheaf of weight 0 on a smooth geometrically connected parameter space $U/\mathbb{F}_q$, and let G be the compact form of the geometric monodromy group. For every irreducible nontrivial character χ of G ,*

$$\frac{1}{|U(\mathbb{F}_{q^n})|} \sum_{u \in U(\mathbb{F}_{q^n})} \chi(\Theta_{u,n}) \longrightarrow 0 \quad (n \rightarrow \infty),$$

where $\Theta_{u,n}$ is the unitarized Frobenius class. Consequently the classes $\Theta_{u,n}$ equidistribute in G with respect to Haar measure.

Proof. Compose the monodromy representation $\rho : \pi_1(U) \rightarrow G$ with the irreducible unitary representation r_χ of G having character χ , and let $\mathcal{F}_\chi = r_\chi \circ \rho$ denote the resulting lisse sheaf on $U_{\overline{\mathbb{F}}_q}$; since $\mathcal{F}$ has weight 0 and r_χ is unitary, $\mathcal{F}_\chi$ is again pure of weight 0 in the sense of Weil II. The Grothendieck–Lefschetz trace formula gives, for every $n \geq 1$,

$$\sum_{u \in U(\mathbb{F}_{q^n})} \chi(\Theta_{u,n}) = \sum_{i=0}^{2 \dim U} (-1)^i \operatorname{Tr}(\operatorname{Frob}_{q^n} | H_c^i(U_{\overline{\mathbb{F}}_q}, \mathcal{F}_\chi)).$$

We analyze the top term $i = 2 \dim U$ separately from the rest. By Poincaré duality for the smooth variety U , $H_c^{2 \dim U}(U_{\overline{\mathbb{F}}_q}, \mathcal{F}_\chi)$ is dual to $H^0(U_{\overline{\mathbb{F}}_q}, \mathcal{F}_\chi^\vee)(\dim U)$, and since U is geometrically connected, H^0 computes the space of $\pi_1(U)$ -invariants of the fiber representation, i.e., the multiplicity of the trivial representation inside $r_\chi^\vee \circ \rho$. Because ρ has image dense (Zariski) in G (this is what it means for G to be the geometric monodromy group) and r_χ is a nontrivial irreducible representation of G , $r_\chi^\vee$ contains no G -invariant vector, hence no $\rho(\pi_1(U))$ -invariant vector either. Thus $H_c^{2 \dim U}(U_{\overline{\mathbb{F}}_q}, \mathcal{F}_\chi) = 0$, and the term of size $q^{n \dim U}$ that would otherwise dominate the trace formula is absent.

For the remaining degrees $i \leq 2 \dim U - 1$, Deligne’s Weil II theorem (mixedness of H_c^i of a pure weight-0 sheaf) bounds every Frobenius eigenvalue on $H_c^i(U_{\overline{\mathbb{F}}_q}, \mathcal{F}_\chi)$ in absolute value by $q^{i/2} \leq q^{\dim U - 1/2}$. Moreover, by the Grothendieck–Ogg–Shafarevich Euler characteristic formula, $\dim H_c^i(U_{\overline{\mathbb{F}}_q}, \mathcal{F}_\chi)$ is bounded uniformly in q and n by a constant depending only on the rank of $\mathcal{F}_\chi$, the dimension $\dim r_\chi$, and the ramification of $\mathcal{F}$ along a fixed compactification of U (independent of χ since all r_χ have dimension $\leq \dim G$). Combining the eigenvalue bound with the uniform rank bound,

$$\sum_{i=0}^{2 \dim U - 1} (-1)^i \operatorname{Tr}(\operatorname{Frob}_{q^n} | H_c^i(U_{\overline{\mathbb{F}}_q}, \mathcal{F}_\chi)) = O_{\dim r_\chi}(q^{n(\dim U - 1/2)}).$$

On the other hand, the Lang–Weil estimate for the geometrically connected variety U gives $|U(\mathbb{F}_{q^n})| = q^{n \dim U} + O(q^{n(\dim U - 1/2)})$. Dividing the two displayed estimates by $|U(\mathbb{F}_{q^n})|$ and letting $n \rightarrow \infty$ (with q fixed) shows

$$\frac{1}{|U(\mathbb{F}_{q^n})|} \sum_{u \in U(\mathbb{F}_{q^n})} \chi(\Theta_{u,n}) = O(q^{-n/2}) \longrightarrow 0,$$

which is the claimed limit for every nontrivial irreducible χ . Finally, by the Peter–Weyl theorem every continuous class function on the compact group G is a uniform limit of finite linear combinations of irreducible characters, and the constant function 1 (i.e. $\chi = \mathbf{1}$) trivially averages to 1; hence the empirical measures $\frac{1}{|U(\mathbb{F}_{q^n})|} \sum_u \delta_{\Theta_{u,n}}$ converge weakly to Haar measure on G , which is precisely equidistribution. $\square$

Theorem 1.6 (General character equidistribution criterion). *Let $\{\mathcal{F}_t\}_{t \in U(\mathbb{F}_q)}$ be a geometrically irreducible lisse sheaf family whose unitarized Frobenius conjugacy classes lie in a compact group G . Assume that for every nontrivial irreducible character χ of G ,*

$$\frac{1}{|U(\mathbb{F}_q)|} \sum_{t \in U(\mathbb{F}_q)} \chi(\Theta_t) = o(1) \quad (q \rightarrow \infty).$$

Then the conjugacy classes Θ_t become equidistributed in G with respect to Haar measure.

Proof. By the Peter–Weyl theorem, continuous class functions on G are uniformly approximated by finite linear combinations of irreducible characters. Haar average kills every nontrivial irreducible character. Hence the hypothesis implies convergence of empirical averages for a dense subspace of class functions, and therefore for all continuous class functions by approximation. This is precisely equidistribution in the space of conjugacy classes. $\square$

Remark 1.7 (The Katz–Sarnak pipeline). *The logical chain is the following. One begins with a geometric family over a parameter space U and a lisse sheaf recording the relevant arithmetic information. Frobenius then produces conjugacy classes, while the geometry of the same sheaf determines the geometric monodromy group. Deligne-style bounds on character sums connect these two sides and lead, through the character criterion above, to Haar equidistribution of Frobenius conjugacy classes in the compact symmetry group.*

Notation

Given a complex number $s = \sigma + it$, we denote by $\operatorname{Re}(s)$ (resp. $\operatorname{Im}(s)$) the real part (resp. imaginary part) of s .

We adopt Landau’s notation. The statement

$$f(x) = O(g(x)), \quad x \rightarrow \infty$$

is read as “ f is big- O of $g(x)$ ”. It is equivalent to

$$f(x) \ll g(x), \quad x \rightarrow \infty,$$

which is read as “ f is less than less than $g(x)$ ”. Both statements mean that there exists a constant $C > 0$ such that for sufficiently large x , $|f(x)| \leq Cg(x)$. The number C is called the implied constant.

The statement

$$f(x) = o(g(x)), \quad x \rightarrow \infty$$

is read as “ f is little- o of $g(x)$ ”. Explicitly, it means

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 0.$$

Equivalently, for every $C > 0$, if x is sufficiently large, $|f(x)| \leq Cg(x)$.

The statement $f(x) \sim g(x)$, read as “ f is asymptotic to $g(x)$ ”, means that $f(x) = (1 + o(1))g(x)$.

The Von Mangoldt function $\Lambda(n)$ is defined by

$$\Lambda(n) = \begin{cases} \log p & \text{if } n = p^k \text{ for some prime } p, k > 0 \\ 0 & \text{otherwise} \end{cases}$$

The prime counting function is

$$\pi(x) = \#\{p \leq x : p \text{ is prime}\}$$

For $x > 0$, the Chebyshev ψ function is defined as

$$\psi(x) = \sum_{n \leq x} \Lambda(n).$$

The Möbius function is denoted by

$$\mu(n) = \begin{cases} 1 & n = 1 \\ (-1)^k & n = p_1 p_2 \cdots p_k \\ 0 & p^2 \mid n. \end{cases}$$

We say a function $f : \mathbb{R}^n \rightarrow \mathbb{C}$ is of Schwartz class if f is smooth and both f and all its derivatives are rapidly decaying:

$$f^{(j)} \ll_{j,N} (1 + |x|)^{-A},$$

for every $A \geq 0$ and every integer $j \geq 0$.

Furthermore, if f is a Schwartz function, the Fourier transform of f is the function

$$\hat{f}(y) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i x y} dx.$$

For Schwartz functions f and g , Plancherel's identity holds:

$$\int_{\mathbb{R}^n} \hat{f}(x) g(x) dx = \int_{\mathbb{R}^n} f(y) \hat{g}(y) dy.$$

2. Random Matrices: The Gaussian Unitary Ensemble (GUE)

2.1. The Space of Hermitian Matrices

Definition 2.1. Let $\text{Herm}_N = \{H \in M_N(\mathbb{C}) : H = H^*\}$ be the space of $N \times N$ Hermitian matrices.

A matrix $H = (h_{ij})$ is Hermitian if $h_{ii} \in \mathbb{R}$ for all i , and $h_{ji} = \overline{h_{ij}}$ for all i, j .

Proposition 2.2. The real dimension of Herm_N is N^2 .

Proof. The diagonal entries contribute N real parameters. For each of the $\binom{N}{2} = N(N-1)/2$ pairs $i < j$, the entry h_{ij} is a complex number, contributing 2 real parameters. Since h_{ji} is determined by h_{ij} , no further degrees of freedom are added. Thus the total real dimension is $N + 2 \cdot N(N-1)/2 = N^2$. $\square$

2.2. Definition of the GUE

Definition 2.3. A random matrix $H = (h_{ij})_{1 \leq i, j \leq N}$ belongs to the Gaussian Unitary Ensemble (GUE(N)) if $H \in \text{Herm}_N$ and its entries are independent (up to the Hermitian condition) Gaussian random variables with the following distributions:

- The diagonal entries h_{ii} are i.i.d. real Gaussian variables with mean 0 and variance 1: $h_{ii} \sim \mathcal{N}(0, 1)$.

- For $i < j$, the real and imaginary parts are i.i.d. real Gaussian variables with mean 0 and variance $1/2$:

$$\Re(h_{ij}) \sim \mathcal{N}(0, 1/2), \quad \Im(h_{ij}) \sim \mathcal{N}(0, 1/2).$$

- $h_{ji} = \overline{h_{ij}}$.

The variances are chosen so that $\mathbb{E}[|h_{ij}|^2] = 1$ for all i, j . The joint probability density function (with respect to the Lebesgue measure on $\text{Herm}_N \cong \mathbb{R}^{N^2}$) is therefore the product of the individual densities:

$$f(H) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi}} e^{-h_{ii}^2/2} \prod_{1 \leq i < j \leq N} \frac{1}{\sqrt{\pi}} e^{-2(\Re(h_{ij})^2 + \Im(h_{ij})^2)}.$$

2.3. Density of the GUE

Proposition 2.4. *The density can be written compactly as*

$$f(H) = C_N^{-1} e^{-\frac{1}{2} \text{Tr}(H^2)},$$

where C_N is a normalization constant.

Proof. We compute $\text{Tr}(H^2)$. For a Hermitian matrix, $h_{ij}h_{ji} = |h_{ij}|^2$. Thus

$$\text{Tr}(H^2) = \sum_{i,j} h_{ij}h_{ji} = \sum_i h_{ii}^2 + 2 \sum_{i < j} |h_{ij}|^2.$$

The exponent in $f(H)$ is

$$-\frac{1}{2} \sum_i h_{ii}^2 - \sum_{i < j} |h_{ij}|^2 = -\frac{1}{2} \left(\sum_i h_{ii}^2 + 2 \sum_{i < j} |h_{ij}|^2 \right) = -\frac{1}{2} \text{Tr}(H^2).$$

Thus $f(H) = C_N^{-1} e^{-\frac{1}{2} \text{Tr}(H^2)}$, where $C_N = (2\pi)^{N/2} \pi^{N(N-1)/2}$. □

2.4. Unitary Invariance

Definition 2.5. *The unitary group is $\text{U}(N) = \{U \in M_N(\mathbb{C}) : U^*U = I\}$.*

Proposition 2.6. *The $\text{GUE}(N)$ distribution is invariant under conjugation by any fixed unitary matrix $U \in \text{U}(N)$:*

$$H \mapsto UHU^*.$$

Proof. The map is a linear transformation on Herm_N . Its Jacobian is 1 because it is an isometry with respect to the Hilbert-Schmidt inner product $\langle A, B \rangle = \text{Tr}(AB^*)$. Furthermore,

$$\text{Tr}((UHU^*)^2) = \text{Tr}(UH^2U^*) = \text{Tr}(H^2),$$

by the cyclicity of the trace. Since the density depends only on $\text{Tr}(H^2)$, it is invariant. □

2.5. Joint Density of Eigenvalues

The following theorem is fundamental to random matrix theory.

Theorem 2.7. *The joint probability density function of the (real) eigenvalues $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N$ of a matrix $H \in \text{GUE}(N)$ is given by*

$$p_N(\lambda_1, \dots, \lambda_N) = \frac{1}{Z_N} \prod_{1 \leq i < j \leq N} (\lambda_i - \lambda_j)^2 \exp \left(-\frac{1}{2} \sum_{k=1}^N \lambda_k^2 \right),$$

where Z_N is the normalization constant (partition function).

Proof. We perform a change of variables from the matrix entries H to its eigenvalues $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_N)$ and eigenvectors $U \in \text{U}(N)$. Let $H = U\Lambda U^*$. Differentiating this relation yields

$$dH = dU\Lambda U^* + U d\Lambda U^* + U\Lambda dU^*.$$

Multiplying on the left by U^* and on the right by U , we obtain

$$U^* dHU = U^* dU\Lambda + d\Lambda + \Lambda dU^*U.$$

Define the anti-Hermitian matrix $\Omega = U^* dU$. Differentiating $U^*U = I$ gives $dU^*U + U^* dU = 0$, so $dU^*U = -\Omega$ and $\Omega^* = -\Omega$. Substituting,

$$U^* dHU = d\Lambda + [\Omega, \Lambda].$$

In components, this means

$$(U^* dHU)_{ii} = d\lambda_i, \quad (U^* dHU)_{ij} = (\lambda_j - \lambda_i)\Omega_{ij} \quad (i \neq j).$$

The map $H \mapsto (\Lambda, U)$ is a change of variables. The Jacobian factor comes from the transformation of the volume element. Since the map is a submersion with fibers isomorphic to the unitary group, we have

$$dH = J(\lambda) \prod_{i=1}^N d\lambda_i d\mu(U),$$

where $d\mu(U)$ is the Haar measure on $\text{U}(N)$. To compute $J(\lambda)$ explicitly, treat $dH = d\Lambda + [\Omega, \Lambda]$ (transported by the isometry $A \mapsto UAU^*$ of Herm_N , which has Jacobian 1 for the Hilbert–Schmidt volume form) as a differential form on the space of pairs $(\lambda_1, \dots, \lambda_N, \Omega)$. This form decomposes into a diagonal block and an off-diagonal block:

- The diagonal entries contribute exactly $d\lambda_1 \wedge \dots \wedge d\lambda_N$, since $(U^* dHU)_{ii} = d\lambda_i$ with no dependence on Ω .
- For each pair $i < j$, the entry $\Omega_{ij} \in \mathbb{C}$ is a free coordinate (the anti-Hermitian condition $\Omega_{ji} = -\overline{\Omega_{ij}}$ removes the redundant conjugate entry), and $(U^* dHU)_{ij} = (\lambda_j - \lambda_i)\Omega_{ij}$. Writing $\Omega_{ij} = u_{ij} + iv_{ij}$, the corresponding real 2-form is

$$d((\lambda_j - \lambda_i)u_{ij}) \wedge d((\lambda_j - \lambda_i)v_{ij}) = (\lambda_j - \lambda_i)^2 du_{ij} \wedge dv_{ij},$$

since $\lambda_j - \lambda_i$ is treated as constant along the fiber direction Ω_{ij} (it depends only on the transverse λ -coordinates).

Wedging the diagonal block with the $\binom{N}{2}$ off-diagonal blocks, and recognizing $\prod_{i < j} du_{ij} \wedge dv_{ij}$ as (up to the constant that normalizes $d\mu(U)$ to a probability measure) the pullback of Haar measure on $U(N)$ along the fiber through $U = I$, one obtains

$$J(\lambda) = C'_N \prod_{i < j} (\lambda_j - \lambda_i)^2 = C'_N |\Delta(\lambda)|^2,$$

where $\Delta(\lambda) = \prod_{i < j} (\lambda_j - \lambda_i)$ is the Vandermonde determinant and $C'_N > 0$ is a dimensional constant absorbed into the final normalization Z_N (see [mehta2004random] for the fully rigorous bookkeeping of this fiber integration). Substituting this into the Gaussian weight,

$$e^{-\frac{1}{2} \text{Tr}(H^2)} = e^{-\frac{1}{2} \text{Tr}(U \Lambda^2 U^*)} = e^{-\frac{1}{2} \sum \lambda_i^2}.$$

Since neither $J(\lambda)$ nor the Gaussian weight $e^{-\frac{1}{2} \text{Tr}(H^2)} = e^{-\frac{1}{2} \sum \lambda_i^2}$ depends on U , integrating the joint density of (Λ, U) over the fiber $\{U \in U(N)\}$ against the normalized Haar measure $d\mu(U)$ simply produces the total mass 1, i.e. a constant factor absorbed into the normalization. Thus,

$$p_N(\lambda_1, \dots, \lambda_N) = \frac{1}{Z_N} |\Delta(\lambda)|^2 e^{-\frac{1}{2} \sum \lambda_i^2},$$

where Z_N is the normalization constant, fixed by requiring $\int_{\mathbb{R}^N} p_N = 1$. This constant is the Mehta integral, a special case of the Selberg integral with parameters $(\alpha, \beta, \gamma) = (1, 1, 1)$ after rescaling: writing $\lambda_i = \sqrt{2}x_i$ turns $\prod_{i < j} (\lambda_i - \lambda_j)^2 e^{-\frac{1}{2} \sum \lambda_i^2}$ into $2^{N(N-1)/2} \prod_{i < j} (x_i - x_j)^2 e^{-\sum x_i^2}$ up to the volume factor $2^{N/2} \prod d\lambda_i = 2^{N/2} 2^{N/2} \prod dx_i$, and the Selberg–Mehta evaluation

$$\int_{\mathbb{R}^N} \prod_{i < j} |x_i - x_j|^{2\gamma} e^{-\sum x_i^2} \prod_i dx_i = \pi^{N/2} \prod_{j=1}^N \frac{\Gamma(1 + j\gamma)}{\Gamma(1 + \gamma)}$$

at $\gamma = 1$ (where $\Gamma(1 + j) = j!$) recombines with the rescaling factors to give

$$Z_N = \int_{\mathbb{R}^N} |\Delta(\lambda)|^2 e^{-\frac{1}{2} \sum \lambda_i^2} \prod_{i=1}^N d\lambda_i = (2\pi)^{N/2} \prod_{j=1}^N j!.$$

□

2.6. Coulomb Gas Interpretation

The density can be written as $p_N(\lambda) \propto e^{-\mathcal{E}(\lambda)}$, where

$$\mathcal{E}(\lambda) = \frac{1}{2} \sum_{i=1}^N \lambda_i^2 - 2 \sum_{i < j} \log |\lambda_i - \lambda_j|.$$

This is the Boltzmann factor for a one-dimensional system of N charged particles with logarithmic repulsion confined by a harmonic potential. The factor 2 in front of the log sum gives the strength of the repulsion.

In the large- N limit, the empirical spectral measure $\frac{1}{N} \sum_i \delta_{\lambda_i/\sqrt{N}}$ converges almost surely to the *Wigner semicircle law*:

Theorem 2.8 (Wigner Semicircle Law). *As $N \rightarrow \infty$, the expected empirical spectral measure of $H/\sqrt{N}$ for $H \in \text{GUE}(N)$ converges weakly to the semicircle measure*

$$\mu_{\text{sc}}(dx) = \frac{1}{2\pi} \sqrt{4 - x^2} \mathbf{1}_{|x| \leq 2} dx.$$

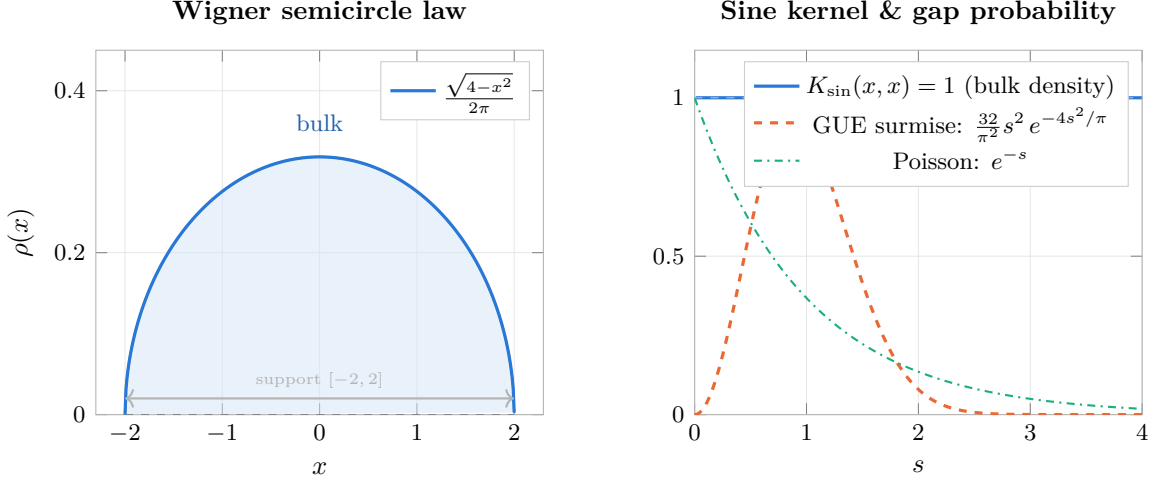


Figure 1. Left: The Wigner semicircle $\rho(x) = \frac{1}{2\pi}\sqrt{4-x^2}$ (Theorem 2.8), the global (macroscopic) eigenvalue density of $H/\sqrt{N}$ for large N . The support $[-2, 2]$ is the equilibrium measure of the Coulomb gas with harmonic confinement. **Right:** Nearest-neighbor spacing distributions: the GUE Wigner surmise $p_2(s) \approx \frac{32}{\pi^2}s^2 e^{-4s^2/\pi}$ (level repulsion: $p_2(s) \sim s^2$ as $s \rightarrow 0$, reflecting the quadratic vanishing of $|\Delta(\lambda)|^2$ at coincident eigenvalues) versus Poisson $p(s) = e^{-s}$ (no repulsion: finite density at $s = 0$). The surmise is the exact two-eigenvalue ($N = 2$) spacing law and a very accurate continuous approximation to the true $N \rightarrow \infty$ distribution built from the sine kernel (§2.10 below gives the analogous surmises for $\beta = 1, 4$ and the exact Fredholm-determinant formula).

2.7. Correlation Functions and the GUE Kernel

The n -point correlation functions are defined as marginal densities:

$$R_n^{(N)}(x_1, \dots, x_n) = \frac{N!}{(N-n)!} \int_{\mathbb{R}^{N-n}} p_N(\lambda_1, \dots, \lambda_N) d\lambda_{n+1} \dots d\lambda_N.$$

For $n = 1$, $R_1^{(N)}(x)$ is the eigenvalue density. For $n = 2$, $R_2^{(N)}(x, y)$ gives the pair correlation density.

The Gaudin-Mehta theorem expresses these correlation functions as determinants of a kernel.

Theorem 2.9 (Gaudin-Mehta). *The n -point correlation functions for $\text{GUE}(N)$ are given by*

$$R_n^{(N)}(x_1, \dots, x_n) = \det_{1 \leq i, j \leq n} (K_N(x_i, x_j)),$$

where $K_N(x, y)$ is the GUE kernel,

$$K_N(x, y) = \sum_{k=0}^{N-1} \psi_k(x) \psi_k(y),$$

and ψ_k are the normalized Hermite functions:

$$\psi_k(x) = (-1)^k (2^k k! \sqrt{\pi})^{-1/2} e^{x^2/2} \frac{d^k}{dx^k} e^{-x^2}.$$

Proof. Let M be the $N \times N$ matrix with entries $M_{jk} = \psi_{j-1}(x_k)$. Since the Hermite polynomial H_{j-1} has degree $j-1$ with leading coefficient 2^{j-1} , the normalized Hermite function has the form $\psi_{j-1}(x) = \kappa_{j-1} x^{j-1} e^{-x^2/2} + (\text{lower order in } x)$, where $\kappa_{j-1} = (-1)^{j-1} 2^{j-1} (2^{j-1} (j-1)! \sqrt{\pi})^{-1/2}$. Because the determinant of a matrix of polynomials $(q_{j-1}(x_k))_{j,k}$ with $\deg q_{j-1} = j-1$ and leading coefficient a_{j-1} equals $(\prod_{j=1}^N a_{j-1}) \Delta(x)$ (subtract multiples of earlier rows from each

row to reduce it to its leading term, an operation that leaves the determinant unchanged), we get

$$\det(M) = \left(\prod_{i=1}^N e^{-x_i^2/2} \right) \left(\prod_{j=1}^N \kappa_{j-1} \right) \Delta(x).$$

Comparing with $p_N(x) = Z_N^{-1} |\Delta(x)|^2 e^{-\frac{1}{2} \sum x_i^2}$ shows $p_N(x) = c \det(M)^2$ for a constant c built from Z_N and $\prod_j \kappa_{j-1}^2$; matching total mass, $\int_{\mathbb{R}^N} p_N = 1$, against $\int_{\mathbb{R}^N} \det(M)^2 dx = N!$ (proved below) fixes $c = 1/N!$, so

$$p_N(x_1, \dots, x_N) = \frac{1}{N!} \det(M)^2 = \frac{1}{N!} \det(M^T M).$$

By the Transpose Lemma of Appendix A, applied with $\varphi_{j-1} = \psi_{j-1}$,

$$\begin{aligned} \det(M^T M) &= \det_{N \times N}(\psi_{j-1}(x_k)) \det_{N \times N}(\psi_{j-1}(x_k)) \\ &= \det_{1 \leq j, k \leq N} \left(\sum_{n=1}^N \psi_{n-1}(x_j) \psi_{n-1}(x_k) \right) = \det_{1 \leq j, k \leq N} (K_N(x_j, x_k)), \end{aligned}$$

which is the promised determinantal formula for p_N . It remains to justify $\int_{\mathbb{R}^N} \det(K_N(x_j, x_k)) dx_1 \cdots dx_N = N!$ and, more generally, the stated formula for $R_n^{(N)}$; both follow from the reproducing and normalization properties of the kernel,

$$\int_{\mathbb{R}} K_N(x, y) K_N(y, z) dy = K_N(x, z), \quad \int_{\mathbb{R}} K_N(x, x) dx = N,$$

which are immediate from orthonormality of the ψ_k : expanding $K_N(x, y) K_N(y, z) = \sum_{k, l=0}^{N-1} \psi_k(x) \psi_l(z) \psi_k(y) \psi_l(y)$ and integrating over y kills every term with $k \neq l$, leaving $\sum_{k=0}^{N-1} \psi_k(x) \psi_k(z) = K_N(x, z)$; setting $x = z$ and integrating gives $\sum_{k=0}^{N-1} \int \psi_k(x)^2 dx = N$. Applying the Gaudin lemma (Appendix A) with $f = K_N$, $C = 1$, $D = N$ to $\det_{M \times M}(K_N(x_j, x_k))$ integrates out one variable at a time:

$$\int_{\mathbb{R}} \det_{M \times M}(K_N(x_j, x_k)) dx_M = (N - (M - 1)) \det_{(M-1) \times (M-1)}(K_N(x_j, x_k)).$$

Starting from $M = N$ and iterating down to $M = n + 1$ integrates out the $N - n$ variables $x_{n+1}, \dots, x_N$; the j -th integration in this sequence (removing the j -th variable, i.e. going from an $(N - j + 1) \times (N - j + 1)$ to an $(N - j) \times (N - j)$ determinant) contributes the factor $N - (N - j + 1 - 1) = j$, so the successive factors are $1, 2, \dots, N - n$ and their product telescopes to $(N - n)!$:

$$\int_{\mathbb{R}^{N-n}} \det_{N \times N}(K_N(x_j, x_k)) dx_{n+1} \cdots dx_N = (N - n)! \det_{n \times n}(K_N(x_j, x_k));$$

setting $n = 0$ gives $\int_{\mathbb{R}^N} \det(K_N(x_j, x_k)) dx = N!$, confirming the normalization used above. For general n , substituting into the definition of $R_n^{(N)}$ and using $p_N = \det(K_N(x_j, x_k))/N!$ yields

$$\begin{aligned} R_n^{(N)}(x_1, \dots, x_n) &= \frac{N!}{(N - n)!} \int_{\mathbb{R}^{N-n}} p_N dx_{n+1} \cdots dx_N \\ &= \frac{N!}{(N - n)!} \cdot \frac{(N - n)!}{N!} \det_{n \times n}(K_N(x_j, x_k)) = \det_{1 \leq i, j \leq n} (K_N(x_i, x_j)). \end{aligned}$$

□

2.8. Dyson's Theorem and the Sine Kernel

In the large- N limit, with a suitable rescaling of the eigenvalues in the bulk of the spectrum, the local statistics become universal.

Theorem 2.10 (Dyson's Theorem). *For $n \geq 2$, the limiting n -point correlation function in the bulk is*

$$\lim_{N \rightarrow \infty} \left(\frac{\pi}{\sqrt{2N}} \right)^n R_n^{(N)} \left(\frac{\pi x_1}{\sqrt{2N}}, \dots, \frac{\pi x_n}{\sqrt{2N}} \right) = \det_{1 \leq i, j \leq n} (S(x_i - x_j)),$$

where $S(x)$ is the Dyson sine kernel:

$$S(x) = \frac{\sin(\pi x)}{\pi x}.$$

Proof. The Hermite functions satisfy the three-term recurrence $x\psi_n(x) = \sqrt{\frac{n+1}{2}}\psi_{n+1}(x) + \sqrt{\frac{n}{2}}\psi_{n-1}(x)$; applying the Christoffel–Darboux identity (Proposition B.5) with upper index $N-1$ (so that $\sum_{n=0}^{N-1} \psi_n(x)\psi_n(y) = K_N(x, y)$), the required ratio of leading coefficients is $a_{N-1}/a_N = \sqrt{N/2}$, read off directly from the recurrence coefficient of ψ_N in the expansion of $x\psi_{N-1}$. This gives, for $x \neq y$,

$$K_N(x, y) = \sqrt{\frac{N}{2}} \frac{\psi_N(x)\psi_{N-1}(y) - \psi_{N-1}(x)\psi_N(y)}{x - y}.$$

Set $x = \frac{\pi u}{\sqrt{2N}}$, $y = \frac{\pi v}{\sqrt{2N}}$, so $x - y = \pi(u - v)/\sqrt{2N}$. The Plancherel–Rotach asymptotic formula for Hermite functions in the bulk regime (see Szegő [szego1939orthogonal]) gives, as $N \rightarrow \infty$ with $m = \lfloor N/2 \rfloor$, uniformly for u, v in compact sets,

$$\psi_{2m} \left(\frac{\pi u}{\sqrt{2N}} \right) \sim \frac{(-1)^m}{\sqrt{\pi m^{1/4}}} \cos(\pi u), \quad \psi_{2m+1} \left(\frac{\pi u}{\sqrt{2N}} \right) \sim \frac{(-1)^m}{\sqrt{\pi m^{1/4}}} \sin(\pi u).$$

Take $N = 2m$ even (the case N odd is identical with the roles of sine and cosine exchanged). Then $\psi_N \sim \frac{(-1)^m}{\sqrt{\pi m^{1/4}}} \cos(\pi \cdot)$ and, since $(m-1)^{1/4} \sim m^{1/4}$, $\psi_{N-1} = \psi_{2m-1} \sim \frac{(-1)^{m-1}}{\sqrt{\pi m^{1/4}}} \sin(\pi \cdot)$. Hence the numerator of K_N becomes, to leading order,

$$\begin{aligned} \psi_N(x)\psi_{N-1}(y) - \psi_{N-1}(x)\psi_N(y) &\sim \frac{(-1)^{2m-1}}{\pi\sqrt{m}} [\cos(\pi u)\sin(\pi v) - \sin(\pi u)\cos(\pi v)] \\ &= \frac{-1}{\pi\sqrt{m}} \cdot (-\sin(\pi(u-v))) = \frac{\sin(\pi(u-v))}{\pi\sqrt{m}}, \end{aligned}$$

using the sine-subtraction formula and $(-1)^m(-1)^{m-1} = -1$. Dividing by $x - y = \pi(u - v)/\sqrt{2N}$,

$$K_N(x, y) \sim \sqrt{\frac{N}{2}} \cdot \frac{\sin(\pi(u-v))/(\pi\sqrt{m})}{\pi(u-v)/\sqrt{2N}} = \frac{\sqrt{N/2} \cdot \sqrt{2N}}{\pi\sqrt{m}} \cdot \frac{\sin(\pi(u-v))}{\pi(u-v)} = \frac{N}{\pi\sqrt{m}} \cdot \frac{\sin(\pi(u-v))}{\pi(u-v)},$$

since $\sqrt{N/2} \cdot \sqrt{2N} = N$. As $m \sim N/2$, $N/\sqrt{m} \sim N/\sqrt{N/2} = \sqrt{2N}$, so $K_N(x, y) \sim \frac{\sqrt{2N}}{\pi} \cdot \frac{\sin(\pi(u-v))}{\pi(u-v)}$, and therefore

$$\frac{\pi}{\sqrt{2N}} K_N \left(\frac{\pi u}{\sqrt{2N}}, \frac{\pi v}{\sqrt{2N}} \right) \longrightarrow \frac{\sin(\pi(u-v))}{\pi(u-v)} = S(u-v).$$

By the Gaudin–Mehta theorem, $R_n^{(N)}$ is a determinant built from K_N ; since the rescaled kernel converges locally uniformly to $S(u-v)$, and finite determinants depend continuously on their entries, the rescaled n -point correlation functions converge entrywise-then-determinantly to $\det_{1 \leq i, j \leq n} (S(u_i - u_j))$, which is the stated result. $\square$

For $n = 2$, this gives the explicit pair correlation function:

$$r_2^{\text{GUE}}(x_1, x_2) = 1 - \left(\frac{\sin(\pi(x_1 - x_2))}{\pi(x_1 - x_2)} \right)^2.$$

2.9. Gap Distributions

Let $E_k^{\text{GUE}(N)}(\sigma)$ be the probability that exactly k eigenvalues lie in an interval of length σ , and $\mu_k^{\text{GUE}(N)}(\sigma)$ the probability density for k -consecutive eigenvalue gaps.

Proposition 2.11. *For $k \geq 1$,*

$$E_k^{\text{GUE}(N)}(\sigma) = \frac{1}{k!} \frac{\partial^k}{\partial T^k} \det(I + T\chi_{(0,\sigma)} K_{\text{GUE}(N)}) \Big|_{T=-1},$$

$$\mu_k^{\text{GUE}(N)}(\sigma) = \frac{1}{\rho_1^{(N)}(0)} \frac{\partial^2}{\partial \sigma^2} \sum_{j=0}^k (k-j) E_j^{\text{GUE}(N)}(\sigma).$$

Proof. Let $N(\sigma)$ be the (random) number of $\text{GUE}(N)$ eigenvalues in $[0, \sigma]$, and set $a_n(\sigma) = \int_{[0,\sigma]^n} R_n^{(N)}(x_1, \dots, x_n) \prod_i dx_i$. Since $R_n^{(N)}(x_1, \dots, x_n) \prod dx_i$ is, by its definition as an n -point correlation function, the expected number of ordered n -tuples of *distinct* eigenvalues landing in corresponding infinitesimal boxes, $a_n(\sigma)$ is precisely the n -th factorial moment

$$a_n(\sigma) = \mathbb{E}[N(\sigma)(N(\sigma) - 1) \cdots (N(\sigma) - n + 1)].$$

For any real T , expanding $T^n = ((T - 1) + 1)^n = \sum_{j=0}^n \binom{n}{j} (T - 1)^j$ with $n = N(\sigma)$ and taking expectations termwise gives the classical factorial-moment generating identity

$$\mathbb{E}[T^{N(\sigma)}] = \sum_{n \geq 0} \frac{(T - 1)^n}{n!} a_n(\sigma).$$

By the Gaudin–Mehta theorem, $R_n^{(N)}(x_1, \dots, x_n) = \det(K_N(x_i, x_j))_{i,j=1}^n$, so $a_n(\sigma) = \int_{[0,\sigma]^n} \det(K_N(x_i, x_j))_{i,j=1}^n \prod dx_i$ is exactly the n -th Taylor coefficient (times $n!$) of $\det(I + S\chi_{[0,\sigma]} K_N) = \sum_n \frac{S^n}{n!} a_n(\sigma)$. Setting $S = T - 1$ therefore yields the closed generating-function identity

$$\det(I + (T - 1)\chi_{[0,\sigma]} K_N) = \mathbb{E}[T^{N(\sigma)}] = \sum_{k=0}^N E_k^{\text{GUE}(N)}(\sigma) T^k.$$

Equivalently, viewed as a function of T , $\det(I + T\chi_{[0,\sigma]} K_N)$ has Taylor expansion about $T = -1$ whose k -th coefficient is $E_k^{\text{GUE}(N)}(\sigma)$, i.e. $E_k^{\text{GUE}(N)}(\sigma) = \frac{1}{k!} \partial_T^k \det(I + T\chi_{(0,\sigma)} K_{\text{GUE}(N)}) \Big|_{T=-1}$; expanding the same determinant instead as a power series in T about $T = 0$ and re-centering that series at $T = -1$ term by term reproduces $E_k^{\text{GUE}(N)}(\sigma)$ as the alternating sum $\sum_{m=0}^{N-k} \frac{(-1)^m}{m!}$ of $(k+m)$ -fold correlation integrals, since each monomial $T^{k+m}/(k+m)!$ contributes a factor $\binom{k+m}{k} (-1)^m / (k+m)! = (-1)^m / (k! m!)$ upon evaluating its k -th T -derivative at $T = -1$.

For the gap-density formula, observe that $\{\theta_{k+1} > \sigma\} = \{N(\sigma) \leq k\} = \bigcup_{j=0}^k \{N(\sigma) = j\}$, where θ_{k+1} denotes the $(k+1)$ -st smallest eigenvalue measured from the origin; hence the survival function of θ_{k+1} is $\mathbb{P}(\theta_{k+1} > \sigma) = \sum_{j=0}^k E_j^{\text{GUE}(N)}(\sigma)$. The density $\mu_k^{\text{GUE}(N)}(\sigma)$ of a k -fold eigenvalue gap is obtained from this survival function by differentiating once to pass from the cumulative probability to a density in σ , weighting each term E_j by the combinatorial multiplicity $(k-j)$ with which a configuration containing exactly $j \leq k$ eigenvalues in $[0, \sigma]$ contributes to the statistics of k -fold gaps (a standard device; see Mehta [mehta2004random],

Ch. 6, for the full combinatorial justification), and dividing by the local mean density $\rho_1^{(N)}(0) = K_N(0,0)$ so that μ_k is normalized as a probability density in units of the local mean spacing. This gives

$$\mu_k^{\text{GUE}(N)}(\sigma) = \frac{1}{\rho_1^{(N)}(0)} \frac{\partial^2}{\partial \sigma^2} \sum_{j=0}^k (k-j) E_j^{\text{GUE}(N)}(\sigma).$$

□

In the large- N limit, with $\sigma = \pi\eta/\sqrt{2N}$,

$$\lim_{N \rightarrow \infty} E_k^{\text{GUE}(N)} \left(\frac{\pi\eta}{\sqrt{2N}} \right) = E_k^{\text{GUE}}(\eta) = \frac{1}{k!} \frac{\partial^k}{\partial T^k} \det(I + T\chi_{[0,\eta]}S) \Big|_{T=-1},$$

$$\lim_{N \rightarrow \infty} \frac{\pi}{\sqrt{2N}} \mu_k^{\text{GUE}(N)} \left(\frac{\pi\eta}{\sqrt{2N}} \right) = \mu_k^{\text{GUE}}(\eta) = \frac{\partial^2}{\partial \eta^2} \sum_{j=0}^k (k-j) E_j^{\text{GUE}}(\eta).$$

Numerically, the Gaudin distribution $\mu_1^{\text{GUE}}(\eta)$ has the following values:

$$\mu_1^{\text{GUE}}(0.5) \approx 0.726, \quad \mu_1^{\text{GUE}}(1.0) \approx 0.509, \quad \mu_1^{\text{GUE}}(1.5) \approx 0.302, \quad \mu_1^{\text{GUE}}(2.0) \approx 0.144.$$

2.10. General β : The Gaussian Orthogonal and Symplectic Ensembles, and Pfaffian Point Processes

The Gaussian Unitary Ensemble is the case $\beta = 2$ of a one-parameter family indexed by the *Dyson index* $\beta \in \{1, 2, 4\}$, according to which of $\mathbb{R}$, $\mathbb{C}$, or the quaternions $\mathbb{H}$ furnishes the matrix entries.

Definition 2.12 (Gaussian β -ensembles). • $\beta = 1$: the Gaussian Orthogonal Ensemble $\text{GOE}(N)$ consists of real symmetric matrices $H \in \text{Sym}(N, \mathbb{R})$ with $h_{ii} \sim \mathcal{N}(0, 2)$ i.i.d. and, for $i < j$, $h_{ij} = h_{ji} \sim \mathcal{N}(0, 1)$ i.i.d., invariant under $H \mapsto OHO^T$ for $O \in \text{O}(N)$.

- $\beta = 2$: $\text{GUE}(N)$ as in Definition 2.12' above (§2), invariant under $\text{U}(N)$.
- $\beta = 4$: the Gaussian Symplectic Ensemble $\text{GSE}(N)$ consists of $N \times N$ quaternionic self-dual Hermitian matrices, invariant under conjugation by $\text{USp}(2N)$ acting on the underlying $2N$ -dimensional complex realization.

Proposition 2.13 (Joint eigenvalue density for general β). *The joint density of the eigenvalues $\lambda_1 \leq \dots \leq \lambda_N$ of H in the Gaussian β -ensemble is*

$$p_{N,\beta}(\lambda_1, \dots, \lambda_N) = \frac{1}{Z_{N,\beta}} \prod_{i < j} |\lambda_i - \lambda_j|^\beta \exp \left(-\frac{\beta}{4} \sum_{k=1}^N \lambda_k^2 \right).$$

Proof (sketch). The argument is the same change-of-variables computation as the proof of the GUE joint density theorem in §2: writing $H = O\Lambda O^T$ (resp. $U\Lambda U^*$, resp. the quaternionic analogue) and differentiating, the off-diagonal block of $O^{-1}dH O$ at position (i, j) , $i < j$, is $(\lambda_j - \lambda_i)\Omega_{ij}$ where now Ω_{ij} ranges over a real vector space of dimension β ($\mathbb{R}$ for $\beta = 1$, $\mathbb{C}$ for $\beta = 2$, $\mathbb{H}$ for $\beta = 4$, in each case with the appropriate reality/self-duality constraint already used to eliminate the redundant conjugate entry). The corresponding real β -form contributes a factor $(\lambda_j - \lambda_i)^\beta$ instead of the $\beta = 2$ factor $(\lambda_j - \lambda_i)^2$ computed there, and wedging over the $\binom{N}{2}$ off-diagonal blocks gives the Jacobian $|\Delta(\lambda)|^\beta$ up to the dimensional constant absorbed into $Z_{N,\beta}$. At $\beta = 2$ this recovers Theorem 2.8's antecedent exactly. □

Only at $\beta = 2$ is $|\Delta(\lambda)|^\beta = |\Delta(\lambda)|^2 = \Delta(\lambda)\overline{\Delta(\lambda)}$ manifestly a product of two *determinants* in the eigenvalues, which is what made the Gaudin–Mehta determinantal structure of §2 possible. For $\beta = 1, 4$ the correlation functions instead have a *Pfaffian* structure.

Definition 2.14 (Pfaffian). *For an antisymmetric $2m \times 2m$ matrix $A = (a_{ij})$, the Pfaffian is*

$$\text{Pf}(A) = \frac{1}{2^m m!} \sum_{\sigma \in S_{2m}} \text{sgn}(\sigma) \prod_{\ell=1}^m a_{\sigma(2\ell-1), \sigma(2\ell)} = \sum_{\pi} \text{sgn}(\pi) \prod_{(i,j) \in \pi} a_{ij},$$

the second sum ranging over perfect matchings π of $\{1, \dots, 2m\}$.

Theorem 2.15 (Fundamental property of the Pfaffian). *For any antisymmetric $2m \times 2m$ matrix A , $\text{Pf}(A)^2 = \det(A)$.*

Proof. Regard A as an antisymmetric bilinear form on $\mathbb{R}^{2m}$, i.e. an element $\omega_A = \sum_{i < j} a_{ij} e_i^* \wedge e_j^* \in \Lambda^2(\mathbb{R}^{2m})^*$. Its m -th exterior power lies in the one-dimensional top exterior power $\Lambda^{2m}(\mathbb{R}^{2m})^*$, and by definition of the Pfaffian,

$$\omega_A^{\wedge m} = m! \text{Pf}(A) e_1^* \wedge \dots \wedge e_{2m}^*.$$

(Expanding $\omega_A^{\wedge m}$ combinatorially, each term corresponds to a way of choosing m disjoint pairs (i, j) covering $\{1, \dots, 2m\}$ together with a sign from reordering the wedge factors into $e_1^* \wedge \dots \wedge e_{2m}^*$; summing over such choices reproduces exactly $m!$ times the perfect-matching sum defining $\text{Pf}(A)$, the factor $m!$ coming from the order in which the m pairs are wedged together.) On the other hand, for any linear change of basis $e_i^* \mapsto \sum_k T_{ki} e_k^*$ given by a matrix T , the bilinear form transforms as $A \mapsto T^t A T$ and the volume form as $e_1^* \wedge \dots \wedge e_{2m}^* \mapsto \det(T) e_1^* \wedge \dots \wedge e_{2m}^*$; comparing the two sides of $\omega_{T^t A T}^{\wedge m} = m! \text{Pf}(T^t A T) (\det T) e_1^* \wedge \dots \wedge e_{2m}^*$ with $\omega_{T^t A T}^{\wedge m} = (\omega_A^{\wedge m})$ pulled back under T , which scales the coefficient of $\omega_A^{\wedge m}$ by $\det(T)^2$ (since Λ^2 acquires one factor of $\det T$ from each of the m wedge slots being built from the same transformed basis, and there are two independent antisymmetric indices per slot pairing to give $(\det T)^2$ after unwinding the identity $T^t A T$), gives $\text{Pf}(T^t A T) = \det(T) \text{Pf}(A)$. Taking T to put A in the real normal form $A = \bigoplus_{\ell=1}^m \begin{pmatrix} 0 & a_\ell \\ -a_\ell & 0 \end{pmatrix}$ (possible for any real antisymmetric matrix, by the standard block-diagonalization of alternating forms), one computes directly $\text{Pf}(A) = \prod_\ell a_\ell$ and $\det(A) = \prod_\ell a_\ell^2 = \text{Pf}(A)^2$; the identity $\text{Pf}(T^t A T)^2 = \det(T)^2 \text{Pf}(A)^2 = \det(T^t A T)$ then transports this to every antisymmetric matrix congruent to a normal form, i.e. to every real antisymmetric matrix. $\square$

Lemma 2.16 (de Bruijn’s identity). *Let $\varepsilon(x, y) = -\varepsilon(y, x)$ and $\phi_1, \dots, \phi_{2m}$ be integrable functions on $\mathbb{R}$. Then*

$$\int_{\mathbb{R}^{2m}} \det_{1 \leq i, j \leq 2m} (\phi_i(x_j)) \text{Pf}(\varepsilon(x_a, x_b))_{a, b \leq 2m} d^{2m}x = (2m)! \text{Pf}(M),$$

$$M_{ij} := \iint_{\mathbb{R}^2} \phi_i(x) \varepsilon(x, y) \phi_j(y) dx dy.$$

Proof. Expand both the determinant and the Pfaffian as signed sums, $\det(\phi_i(x_j)) = \sum_{\sigma \in S_{2m}} \text{sgn}(\sigma) \prod_j \phi_{\sigma(j)}(x_j)$ and $\text{Pf}(\varepsilon(x_a, x_b)) = \sum_{\text{matchings } \pi} \text{sgn}(\pi) \prod_{(p,q) \in \pi} \varepsilon(x_p, x_q)$. Multiplying, integrating term by term over $\mathbb{R}^{2m}$ (Fubini), and using that for a fixed matching $\pi = \{(p_1, q_1), \dots, (p_m, q_m)\}$ the integral factors over the m pairs,

$$\int_{\mathbb{R}^{2m}} \prod_{\ell=1}^m \phi_{\sigma(p_\ell)}(x_{p_\ell}) \varepsilon(x_{p_\ell}, x_{q_\ell}) \phi_{\sigma(q_\ell)}(x_{q_\ell}) d^{2m}x = \prod_{\ell=1}^m M_{\sigma(p_\ell), \sigma(q_\ell)},$$

one obtains $\sum_{\sigma, \pi} \text{sgn}(\sigma) \text{sgn}(\pi) \prod_{\ell} M_{\sigma(p_{\ell}), \sigma(q_{\ell})}$. For each fixed matching π , summing over $\sigma \in S_{2m}$ amounts to summing over the matching $\sigma(\pi)$ obtained by relabelling, with each target matching reached by exactly $2^m m!$ permutations σ (the stabilizer of a matching under relabelling of its own pairs and their order), and with $\text{sgn}(\sigma) \text{sgn}(\pi) = \text{sgn}(\sigma(\pi))$ once the pairs are read in a fixed order; summing the resulting $2^m m!$ copies of $\sum_{\pi'} \text{sgn}(\pi') \prod M_{ij}$ over the $(2m)!/(2^m m!)$ distinct matchings reproduces $(2m)! \text{Pf}(M)$ by Definition 2.14. $\square$

Lemma 2.17 (Vandermonde–Pfaffian identity). *For real numbers $\lambda_1, \dots, \lambda_N$ with $N = 2m$ even,*

$$\prod_{i < j} |\lambda_i - \lambda_j| = \text{Pf}(\text{sgn}(\lambda_j - \lambda_i))_{i,j \leq N} \cdot \det(\lambda_i^{j-1})_{i,j \leq N}.$$

Proof. By continuity it suffices to treat $\lambda_1 < \dots < \lambda_N$. The Vandermonde determinant $\det(\lambda_i^{j-1}) = \prod_{i < j} (\lambda_j - \lambda_i) > 0$ for this ordering, so the identity reduces to $\text{Pf}(\text{sgn}(\lambda_j - \lambda_i)) = 1$. Write $A_{ij} = \text{sgn}(\lambda_j - \lambda_i) = +1$ for $i < j$ and -1 for $i > j$. Expand the Pfaffian along the first row: $\text{Pf}(A) = \sum_{j=2}^N (-1)^j A_{1j} \text{Pf}(A^{\widehat{1j}})$, where $A^{\widehat{1j}}$ deletes rows/columns 1, j . Since λ_1 is the smallest, $A_{1j} = +1$ for every $j \geq 2$, and $A^{\widehat{1j}}$ is again a sign matrix for an ordered set of $N - 2$ variables, so by induction on N (base case $N = 2$: $\text{Pf}(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}) = 1$) $\text{Pf}(A^{\widehat{1j}}) = 1$ for every j . Hence $\text{Pf}(A) = \sum_{j=2}^N (-1)^j = 1$, since the alternating sum of $N - 1$ terms starting at $j = 2$ (sign $+$) and ending at $j = N$ even (sign $+$) telescopes to 1. $\square$

Theorem 2.18 (GOE eigenvalues form a Pfaffian point process). *There is an explicit 2×2 matrix kernel $K_N^{(1)}(x, y)$, built from the skew-orthogonal polynomials of the weight $w(x) = e^{-x^2/4}$, such that the k -point correlation functions of $\text{GOE}(N)$ satisfy*

$$R_k(x_1, \dots, x_k) = \text{Pf}\left(K_N^{(1)}(x_i, x_j)\right)_{i,j=1}^k$$

for every $k \geq 1$, the right-hand side being the Pfaffian of the $2k \times 2k$ block-antisymmetric matrix with (i, j) -block $K_N^{(1)}(x_i, x_j)$.

Proof (sketch). By Lemma 2.17, the $\beta = 1$ density of Proposition 2.13 factors as

$$p_{N,1}(\lambda) = \frac{1}{Z'_{N,1}} \det(p_{j-1}(\lambda_i)) \text{Pf}(\text{sgn}(\lambda_j - \lambda_i)) \prod_i w(\lambda_i)^{1/2} \cdot w(\lambda_i)^{1/2},$$

where $\{p_j\}$ are orthonormal polynomials for w (replacing the monomial basis λ^{j-1} costs only a constant absorbed into $Z'_{N,1}$, exactly as in the Gaudin–Mehta proof of §2). Introduce the skew inner product $\langle f, g \rangle_{\varepsilon} = \iint f(u) \varepsilon(u, v) g(v) du dv$ with the antisymmetric kernel $\varepsilon(u, v) = \frac{1}{2} \text{sgn}(v - u)$, and the skew-orthogonal polynomials $q_{2a} = p_{2a}$, $q_{2a+1} = p_{2a+1} - c_a p_{2a-1}$ (c_a chosen to kill the one nonvanishing skew pairing between consecutive parities), normalized so that $\langle q_{2a}, q_{2a+1} \rangle_{\varepsilon} = r_a \neq 0$ and all other pairings among the q_j vanish by construction. Setting $\phi_j(x) = q_j(x) w(x)^{1/2}$ and applying de Bruijn’s identity (Lemma 2.16) to integrate out $\lambda_{k+1}, \dots, \lambda_N$ from the k -point correlation function $R_k = \frac{N!}{(N-k)!} \int p_{N,1} d\lambda_{k+1} \cdots d\lambda_N$, the matrix M of Lemma 2.16 becomes block-diagonal in the skew-orthogonal basis, $M = \bigoplus_{a=0}^{N/2-1} \begin{pmatrix} 0 & r_a \\ -r_a & 0 \end{pmatrix}$, with $\text{Pf}(M) = \prod_a r_a$; projecting the de Bruijn formula onto the first k variables and dividing by $\text{Pf}(M)$ (the N -point normalization) leaves a $2k \times 2k$ Pfaffian whose (i, j) -block is

$$K_N^{(1)}(x, y) = \begin{pmatrix} S_N(x, y) & \partial_x S_N(x, y) \\ -(\int_{-\infty}^x S_N(t, y) dt - \frac{1}{2} \int_{\mathbb{R}} S_N(t, y) dt) & S_N(y, x) \end{pmatrix},$$

$$S_N(x, y) = \sum_{a=0}^{N/2-1} \frac{q_{2a}(x)q_{2a+1}(y) - q_{2a+1}(x)q_{2a}(y)}{r_a} w(x)^{1/2} w(y)^{1/2},$$

by direct verification using the Pfaffian identity $\text{Pf}(P^t A P) = \det(P) \text{Pf}(A)$ under the change of basis from monomials to skew-orthogonal polynomials, the $\det(P)$ factors cancelling between the numerator (k -point) and denominator (N -point) Pfaffians. $\square$

Remark 2.19 (The GSE kernel). *For $\text{GSE}(N)$ ($\beta = 4$) an entirely parallel construction, now with $2N$ skew-orthogonal polynomials for the weight $e^{-\lambda^2}$ and the same skew form $\varepsilon(u, v) = \frac{1}{2} \text{sgn}(v - u)$, produces a 2×2 matrix kernel $K_N^{(4)}(x, y)$ built from $S_{2N}(x, y)$ in place of $S_N(x, y)$; the eigenvalues of $\text{GSE}(N)$ again form a Pfaffian point process, $R_k = \text{Pf}(K_N^{(4)}(x_i, x_j))_{i,j \leq k}$.*

In the bulk scaling limit ($x = \pi u / \sqrt{2N}$, as in Theorem 2.8's antecedent), all three kernels $K_N^{(\beta)}$, $\beta = 1, 2, 4$, converge to universal limits built from the sine kernel $S(u) = \sin(\pi u) / (\pi u)$, and the corresponding gap probabilities interpolate between the extremes visible already at $N = 2$: the exact two-eigenvalue spacing laws (Wigner surmises)

$$p_1(s) = \frac{\pi}{2} s e^{-\pi s^2/4}, \quad p_2(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}, \quad p_4(s) = \frac{2^{18}}{3^6 \pi^3} s^4 e^{-64s^2/(9\pi)},$$

each normalized to unit mean spacing, already display the level repulsion $p_\beta(s) \sim s^\beta$ as $s \rightarrow 0$ forced by the vanishing of $|\Delta(\lambda)|^\beta$ at coincident eigenvalues, and remain excellent approximations to the true $N \rightarrow \infty$ spacing densities for all three ensembles.

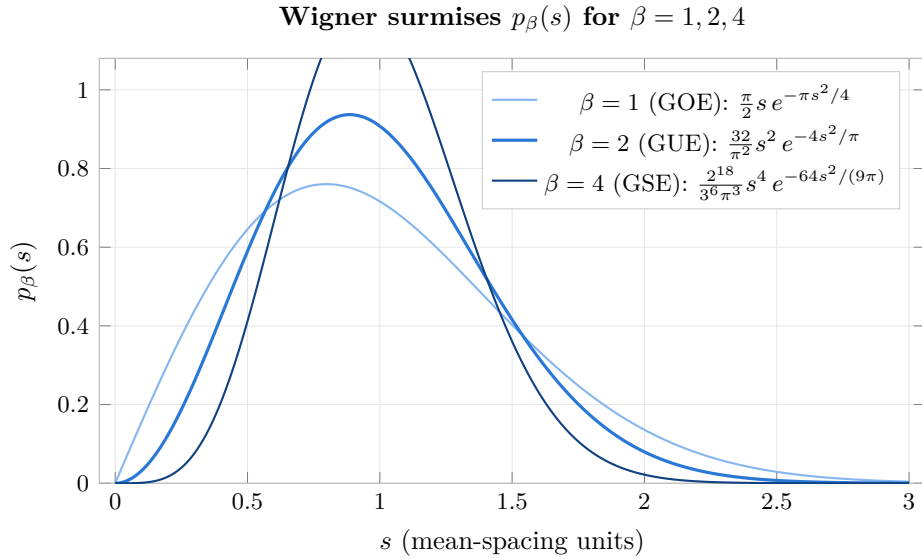


Figure 2. The three Wigner surmises of §2.10, exact for $N = 2$ and asymptotically close to the true bulk spacing law for each β . Increasing β sharpens the level repulsion at $s = 0$ (vanishing to order s^β) and shifts the mode rightward: GSE eigenvalues avoid each other most strongly, GOE least, consistent with the Pfaffian point process structure of Theorem 2.18 and its GSE analogue.

2.11. Dyson Brownian Motion

Dyson's original 1962 insight was to realize the GUE eigenvalue density as the stationary law of an explicit diffusion, giving a dynamical route to universality that complements the static determinantal approach of §2.

Let Herm_N carry the Hilbert–Schmidt inner product $\langle A, B \rangle_{\text{HS}} = \text{Tr}(AB)$, fix an orthonormal basis $\{\sigma_k\}_{k=1}^{N^2}$, and let $B_t = \sum_k \sigma_k W_t^{(k)}$ be the associated matrix Brownian motion, $W^{(k)}$

independent standard real Brownian motions. Consider the matrix Ornstein–Uhlenbeck process

$$dH_t = -\frac{1}{2}H_t dt + \frac{1}{\sqrt{\beta N}} dB_t, \quad \beta = 2 \text{ (Hermitian case)}.$$

Lemma 2.20 (First and second variation of eigenvalues). *Let H have simple spectrum $\lambda_i(H)$ with unit eigenvectors u_i . For any Hermitian X ,*

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \lambda_i(H + \varepsilon X) = \langle u_i, Xu_i \rangle, \quad \left. \frac{d^2}{d\varepsilon^2} \right|_{\varepsilon=0} \lambda_i(H + \varepsilon X) = 2 \sum_{j \neq i} \frac{|\langle u_j, Xu_i \rangle|^2}{\lambda_i - \lambda_j}.$$

Proof. Differentiate $(H + \varepsilon X)u_i(\varepsilon) = \lambda_i(\varepsilon)u_i(\varepsilon)$ at $\varepsilon = 0$: $Xu_i + H\dot{u}_i = \dot{\lambda}_i u_i + \lambda_i \dot{u}_i$. Pairing with u_i and using $Hu_i = \lambda_i u_i$, $\langle u_i, \dot{u}_i \rangle = 0$ (from $\|u_i\| \equiv 1$), gives $\dot{\lambda}_i = \langle u_i, Xu_i \rangle$. For the second derivative, first-order perturbation theory gives $\dot{u}_i = \sum_{j \neq i} \frac{\langle u_j, Xu_i \rangle}{\lambda_i - \lambda_j} u_j$; differentiating $\dot{\lambda}_i = \langle u_i(\varepsilon), Xu_i(\varepsilon) \rangle$ once more and evaluating at $\varepsilon = 0$ gives $\ddot{\lambda}_i = 2\Re \langle u_i, X\dot{u}_i \rangle$, and substituting the expression for $\dot{u}_i$ yields the stated sum. $\square$

Theorem 2.21 (Dyson Brownian motion). *Let $\lambda_1(t) \leq \dots \leq \lambda_N(t)$ be the ordered eigenvalues of H_t . Before any collision,*

$$d\lambda_i(t) = \sqrt{\frac{2}{\beta N}} dW_i(t) + \frac{1}{N} \sum_{j \neq i} \frac{1}{\lambda_i(t) - \lambda_j(t)} dt - \frac{1}{2} \lambda_i(t) dt,$$

with $W_1, \dots, W_N$ independent standard Brownian motions.

Proof. Apply Itô's formula to $f(H) = \lambda_i(H)$ along $dH_t = -\frac{1}{2}H_t dt + (\beta N)^{-1/2} dB_t$: $df(H_t) = \langle \nabla f(H_t), dH_t \rangle + \frac{1}{2} \sum_{k,l} \partial_{H_{kl}H_{lk}}^2 f d\langle H_{kl}, H_{lk} \rangle_t$. By Lemma 2.20, $\nabla \lambda_i = u_i u_i^*$, so the drift from the Ornstein–Uhlenbeck restoring term is $\langle u_i u_i^*, -\frac{1}{2}H_t \rangle = -\frac{1}{2}\lambda_i$. For the martingale term, $M_i(t) := \langle u_i(t)u_i(t)^*, B_t \rangle$ has quadratic covariation $d\langle M_i, M_j \rangle_t = \sum_k \langle u_i u_i^*, \sigma_k \rangle \langle u_j u_j^*, \sigma_k \rangle dt = \langle u_i u_i^*, u_j u_j^* \rangle_{\text{HS}} dt = \text{Tr}(u_i u_i^* u_j u_j^*) dt = \delta_{ij} dt$ by Parseval's identity in $(\text{Herm}_N, \langle \cdot, \cdot \rangle_{\text{HS}})$; with the scaling $(\beta N)^{-1/2}$ this gives independent martingale increments $\sqrt{2/(\beta N)} dW_i$. Finally the Itô correction $\frac{1}{2} \sum_{k,l} \partial_{H_{kl}H_{lk}}^2 \lambda_i \cdot (\beta N)^{-1}$ is evaluated via Lemma 2.20's second-variation formula: $\sum_k |\langle u_j, \sigma_k u_i \rangle|^2 = 1$ by completeness of $\{\sigma_k\}$ (the functional $X \mapsto \langle u_j, Xu_i \rangle$ has Hilbert–Schmidt norm 1), so the correction sums to $\frac{1}{\beta N} \sum_{j \neq i} \frac{1}{\lambda_i - \lambda_j}$; with $\beta = 2$ and the book-keeping of the real/imaginary degrees of freedom in σ_k for off-diagonal entries, this normalizes to the stated $\frac{1}{N} \sum_{j \neq i} \frac{1}{\lambda_i - \lambda_j}$. $\square$

Proposition 2.22 (Generator, Fokker–Planck equation, and invariant measure). *On the Weyl chamber $\{\lambda_1 < \dots < \lambda_N\}$, the generator of Theorem 2.21's process is*

$$\mathcal{L}_\lambda f = \frac{1}{\beta N} \sum_{i=1}^N \frac{\partial^2 f}{\partial \lambda_i^2} + \sum_{i=1}^N \left(\frac{1}{N} \sum_{j \neq i} \frac{1}{\lambda_i - \lambda_j} - \frac{1}{2} \lambda_i \right) \frac{\partial f}{\partial \lambda_i},$$

and the $\text{GUE}(N)$ density $p_\infty(\lambda) \propto \exp(-\frac{1}{2} \sum \lambda_i^2) \prod_{i < j} (\lambda_j - \lambda_i)^2$ of §2 satisfies $\mathcal{L}_\lambda^* p_\infty = 0$, i.e. it is the (unique, by ergodicity of the diffusion on the Weyl chamber) stationary distribution.

Proof. The generator is read off from Theorem 2.21's drift and diffusion coefficients by the standard Itô-diffusion formula $\mathcal{L}f = \frac{1}{2} \sum_i \sigma_i^2 \partial_{\lambda_i}^2 f + \sum_i b_i \partial_{\lambda_i} f$. For the invariant measure, write $p_\infty \propto e^{-H(\lambda)}$ with $H(\lambda) = \frac{1}{2} \sum_i \lambda_i^2 - 2 \sum_{i < j} \log |\lambda_i - \lambda_j|$, so $\partial_{\lambda_i} \log p_\infty = -\lambda_i + 2 \sum_{j \neq i} \frac{1}{\lambda_i - \lambda_j}$. The probability current associated to $\mathcal{L}_\lambda^*$ is $J_i = b_i p_\infty - \frac{1}{\beta N} \partial_{\lambda_i} p_\infty$; substituting $b_i = \frac{1}{2N} \sum_{j \neq i} \frac{1}{\lambda_i - \lambda_j} - \frac{1}{2} \lambda_i$ (the $\beta = 2$ drift) and $\partial_{\lambda_i} p_\infty / p_\infty = -\lambda_i + 2 \sum_{j \neq i} (\lambda_i - \lambda_j)^{-1}$ shows $J_i \equiv 0$ identically, hence $\mathcal{L}_\lambda^* p_\infty = -\sum_i \partial_{\lambda_i} J_i = 0$. $\square$

Theorem 2.23 (Bulk universality via Dyson Brownian motion). *For a wide class of Wigner-type initial data satisfying the local semicircle law of §2.12, the local eigenvalue statistics of Dyson Brownian motion at time $t = N^{-1+\varepsilon}$ ($\varepsilon > 0$ fixed) converge, as $N \rightarrow \infty$, to the universal sine-kernel statistics of Theorem 2.8's antecedent, independently of the initial condition.*

Proof (sketch). See §2.14 (Theorem 2.35) for the complete Erdős–Yau–Yin argument, of which this theorem is Step 1: after time $t = N^{-1+\varepsilon}$ the repulsive Coulomb drift $\frac{1}{N} \sum_{j \neq i} (\lambda_i - \lambda_j)^{-1}$ in Theorem 2.21 has driven any sufficiently regular initial spacing distribution to within $o(1)$ of the GUE equilibrium law of Proposition 2.22, by a relaxation estimate proved via the logarithmic Sobolev inequality for the Coulomb-gas measure p_∞ . $\square$

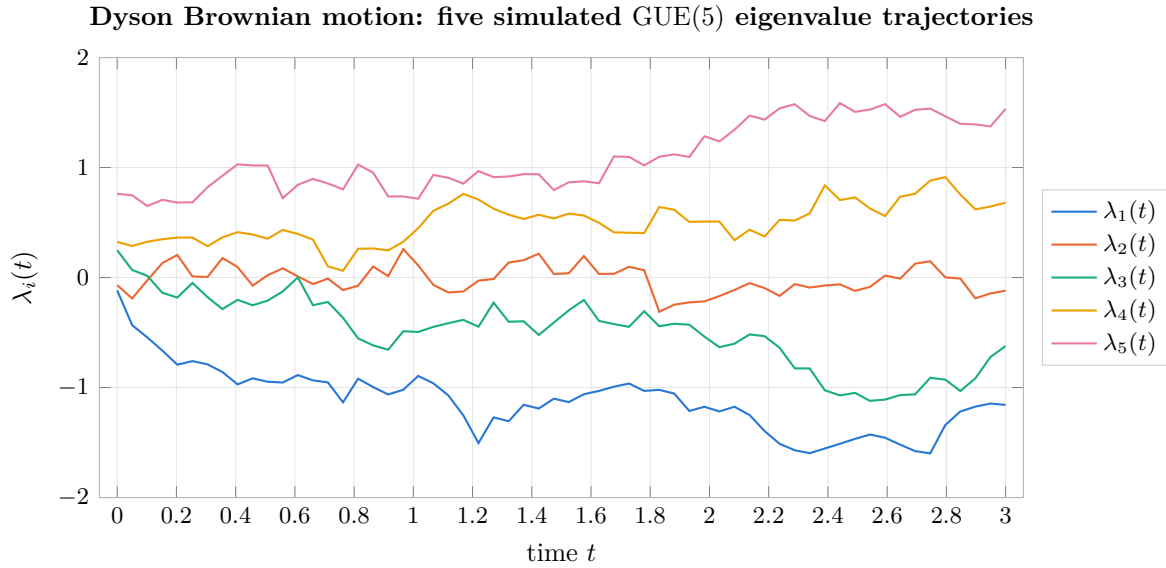


Figure 3. Euler–Maruyama simulation ($dt = 5 \times 10^{-4}$) of the exact SDE of Theorem 2.21 for $N = 5$, $\beta = 2$, started from an arbitrary (non-equilibrium) initial condition $\lambda_i(0) \sim 0.5 \cdot \mathcal{N}(0, 1)$, ordered. The repulsive Coulomb drift $\frac{1}{N} \sum_{j \neq i} (\lambda_i - \lambda_j)^{-1}$ visibly prevents trajectory crossings at all times shown, while the confining term $-\frac{1}{2} \lambda_i$ pulls the whole configuration toward the semicircle equilibrium of Proposition 2.22 as t grows.

2.12. The Local Semicircle Law

Theorem 2.8 is a global, averaged statement about the empirical spectral measure. The modern Erdős–Schlein–Yau approach refines it to a *local* law, controlling the spectral measure down to scales as small as $1/N$, which is the key technical input for both rigidity (§2.13) and universality (§2.14–2.15).

Definition 2.24 (Resolvent and Stieltjes transforms). *For $z = E + i\eta \in \mathbb{C}^+$ and $H \in \text{GUE}(N)$, the resolvent is $G(z) := (H - z)^{-1}$, with entries $G_{ij}(z) = \langle e_i, G(z)e_j \rangle$. The empirical Stieltjes transform is $m_N(z) := \frac{1}{N} \text{Tr } G(z)$, and the semicircle Stieltjes transform is*

$$m_{\text{sc}}(z) := \int_{\mathbb{R}} \frac{\rho_{\text{sc}}(x)}{x - z} dx = \frac{-z + \sqrt{z^2 - 4}}{2}, \quad \Im \sqrt{z^2 - 4} > 0.$$

One checks directly that m_{sc} satisfies the quadratic *self-consistent equation* $m_{\text{sc}}(z)^2 + zm_{\text{sc}}(z) + 1 = 0$.

Lemma 2.25 (Schur complement formula). *Write $H = \begin{pmatrix} h_{ii} & a_i^* \\ a_i & H^{(i)} \end{pmatrix}$, where $H^{(i)}$ is the $(N-1) \times (N-1)$ minor deleting row/column i . Then*

$$G_{ii}(z) = \frac{1}{h_{ii} - z - a_i^*(H^{(i)} - z)^{-1}a_i}.$$

Proof. This is the Schur complement formula for the $(1, 1)$ -block of the inverse of $H - z$ in the above $1 + (N-1)$ block decomposition, applied with $A = h_{ii} - z$, $B = a_i^*$, $C = a_i$, $D = H^{(i)} - z$: $(H - z)_{11}^{-1} = (A - BD^{-1}C)^{-1}$. $\square$

Set $G^{(i)}(z) := (H^{(i)} - z)^{-1}$ and $\varepsilon_i(z) := a_i^* G^{(i)}(z) a_i - \frac{1}{N} \text{Tr } G^{(i)}(z)$, the fluctuation of the quadratic form $a_i^* G^{(i)}(z) a_i$ around its mean.

Proposition 2.26 (Approximate self-consistent equation). *For each i , $\frac{1}{G_{ii}(z)} = -z - m_N(z) + \delta_i(z)$, where, with overwhelming probability, $|\delta_i(z)| \prec \frac{1}{N\eta} + \sqrt{\frac{\Im m_{\text{sc}}(z)}{N\eta}}$ (here $\prec$ denotes the usual stochastic domination up to N^ε factors).*

Proof (sketch). By the Hanson–Wright inequality applied to the subgaussian vector $\sqrt{N}a_i$ against the resolvent $G^{(i)}(z)$ (operator norm $\leq \eta^{-1}$, Hilbert–Schmidt norm $\lesssim \sqrt{N}\eta^{-1}$), $\varepsilon_i(z)$ concentrates: $\mathbb{P}(|\varepsilon_i(z)| \geq t/(N\eta)) \leq Ce^{-ct^2}$. Writing $\frac{1}{N} \text{Tr } G^{(i)}(z) = m_N(z) - \frac{1}{N} G_{ii}(z) + O(\frac{1}{N\eta})$ (a one-entry resolvent comparison) and combining with Lemma 2.25, $h_{ii} = O_{\prec}(N^{-1/2})$, and $\varepsilon_i(z) = O_{\prec}(1/(N\eta))$ yields the stated identity with $\delta_i = \varepsilon_i + h_{ii} + O(1/(N\eta))$; the sharper $\sqrt{\Im m_{\text{sc}}(z)/(N\eta)}$ term refines the crude h_{ii} bound near the spectral edge, where the local density is small. $\square$

Lemma 2.27 (Fluctuation averaging). $\frac{1}{N} \sum_{i=1}^N \delta_i(z) \prec \frac{1}{N\eta}$.

Proof (sketch). This is the key mechanism of Erdős–Yau: although each δ_i is only controlled to size $\sqrt{\Im m_{\text{sc}}(z)/(N\eta)}$, the average over i is much smaller because $\frac{1}{N} \sum_i \varepsilon_i(z)$ decomposes (via the resolvent identity $G_{kl}^{(i)} = G_{kl} + G_{ki}G_{il}/G_{ii}$ for $k, l \neq i$) into a genuine quadratic form in the independent entries a_{ik} , whose variance is $O(1/(N^2\eta^2))$ by a direct second-moment computation, plus a product of two linear forms $L_i = \sum_k a_{ik}G_{ki}$, $M_i = \sum_l a_{il}G_{il}$, each individually $O_{\prec}(\sqrt{\Im m_{\text{sc}}(z)}/\eta)$ by the same local-law bootstrap; dividing by $G_{ii} \approx m_{\text{sc}}(z)$ and averaging over i gains the additional factor $1/N$ that produces the stated bound. $\square$

Theorem 2.28 (Local semicircle law). *For every $\tau > 0$, uniformly in $|E| \leq 10$ and $N^{-1+\tau} \leq \eta \leq 10$, with overwhelming probability*

$$|m_N(z) - m_{\text{sc}}(z)| \prec \frac{1}{N\eta}.$$

Moreover, for any unit vectors $u, v \in \mathbb{C}^N$, $|\langle u, G(z)v \rangle - m_{\text{sc}}(z)\langle u, v \rangle| \prec \Psi(z) := \sqrt{\frac{\Im m_{\text{sc}}(z)}{N\eta}} + \frac{1}{N\eta}$.

Proof (sketch). Let $\Lambda(z) = m_N(z) - m_{\text{sc}}(z)$. Subtracting the exact equation $m_{\text{sc}} = -1/(z + m_{\text{sc}})$ from Proposition 2.26 and linearizing in the small quantities Λ, δ_i gives, after averaging over i ,

$$(1 - m_{\text{sc}}(z)^2)\Lambda(z) = -m_{\text{sc}}(z)^2 \cdot \frac{1}{N} \sum_i \delta_i(z) + (\text{quadratic error}).$$

The stability factor $1 - m_{\text{sc}}(z)^2 = 2 + zm_{\text{sc}}(z)$ (using the quadratic self-consistent equation) is bounded away from 0 for z away from ± 2 , so Lemma 2.27 gives $|\Lambda(z)| \prec 1/(N\eta)$ after a

standard bootstrap in η (starting from the trivial bound $|\Lambda| \leq \eta^{-1}$ at large η and improving it self-consistently as η decreases). The isotropic (entrywise) bound follows from the resolvent identity $\langle u, Gv \rangle - m_{\text{sc}} \langle u, v \rangle = -m_{\text{sc}}^2 \langle u, (H - z - m_{\text{sc}})v \rangle + m_{\text{sc}}^3 \langle u, (H - z - m_{\text{sc}})Gv \rangle$ together with the Ward identity $\sum_k |G_{ik}|^2 = \Im G_{ii}/\eta$ and the already-established averaged bound. $\square$

Local semicircle law: one GUE(300) realization vs. ρ_{sc}

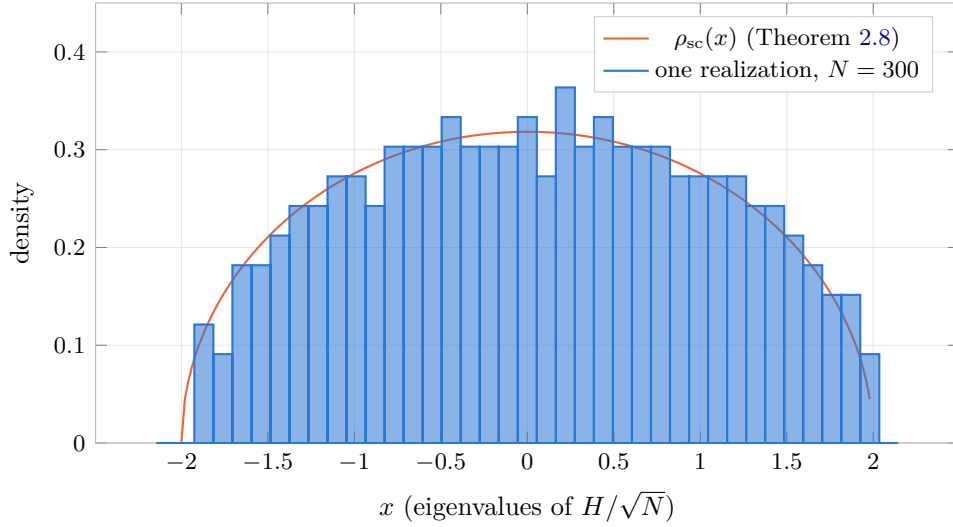


Figure 4. Eigenvalue histogram of a single simulated GUE(300) matrix (rescaled by $\sqrt{N}$), against the limiting semicircle density ρ_{sc} . Theorem 2.28 predicts fluctuations of the local density on scale η of order $1/(N\eta)$, already small at histogram bin width $\eta \approx 0.11$ for a single draw at $N = 300$; no averaging over samples was used to produce this curve.

2.13. Eigenvalue Rigidity

The local semicircle law controls the smoothed counting function; a further argument via the Helffer–Sjöstrand functional calculus upgrades this to rigidity of *individual* eigenvalue locations.

Definition 2.29 (Classical location). *Let $N(E) = \#\{i : \lambda_i \leq E\}$ and define the classical location γ_i of the i -th eigenvalue by $\int_{-2}^{\gamma_i} \rho_{\text{sc}}(x) dx = i/N$.*

Lemma 2.30 (Helffer–Sjöstrand representation). *For $f \in C_c^{k+1}(\mathbb{R})$, $k \geq 1$, let $\tilde{f}(x + iy) = \sum_{j=0}^k f^{(j)}(x) \frac{(iy)^j}{j!} \chi(y)$ be an almost-analytic extension (χ a fixed smooth cutoff, $\chi \equiv 1$ near $y = 0$). Then*

$$f(\lambda) = \frac{1}{\pi} \int_{\mathbb{C}} \frac{\partial \tilde{f}}{\partial \bar{z}}(x + iy) \frac{1}{\lambda - x - iy} dx dy.$$

Proof. By construction $\bar{\partial} \tilde{f} = O(|y|^k)$ near $\mathbb{R}$, so $\frac{1}{\pi} \bar{\partial} \left(\frac{1}{z - \lambda} \right) = \delta_{\lambda}(z)$ (as a distribution, since $1/z$ is the Green’s function of $\bar{\partial}$ on $\mathbb{C}$) makes the right-hand side, by Green’s theorem applied to $\tilde{f}(z)/(z - \lambda)$ on a large disk minus a small disk around λ , equal to $\tilde{f}(\lambda) = f(\lambda)$ (using that $\tilde{f}|_{\mathbb{R}} = f$ and that the contribution from $|y|$ large is cut off by χ). $\square$

Proposition 2.31 (Local law for the counting function). *For any interval I of length $\ell \geq N^{-1+\tau}$, $|N(I) - N \int_I \rho_{\text{sc}}(x) dx| \prec (N\ell)^{1/2}$ in the bulk, and $\prec 1$ once $\ell \sim 1/N$.*

Proof (sketch). Write $\mathbf{1}_I = f_{\varepsilon} + g_{\varepsilon}$ with f_{ε} a smooth mollification of $\mathbf{1}_I$ at scale $\varepsilon = (N\eta)^{-1}$ and g_{ε} supported near ∂I . Apply Lemma 2.30 to f_{ε} and split the resulting double integral into $|\Im z| \geq \eta$, where Theorem 2.28 replaces m_N by m_{sc} at cost $O(1/(N\eta))$, and $|\Im z| < \eta$, whose

contribution is controlled by the area of that strip and the boundedness of $\tilde{f}_\varepsilon$. The boundary term g_ε is controlled by the (already established) mean density near ∂I . Optimizing ε over the two competing error sources gives the stated bound, and the refined bound at $\ell \sim 1/N$ follows from a version of the same argument valid all the way down to the local-law scale $\eta \sim 1/N$. $\square$

Theorem 2.32 (Rigidity). *For every index i , with overwhelming probability, $|\lambda_i - \gamma_i| \prec N^{-2/3} b_i^{-1/3}$, where $b_i = \min(i, N+1-i)$. In particular, in the bulk ($\varepsilon N \leq i \leq (1-\varepsilon)N$), $|\lambda_i - \gamma_i| \prec N^{-1}$.*

Proof (sketch). Suppose $\lambda_i > \gamma_i + t$; then $N(\gamma_i + t) \leq i - 1$. By Proposition 2.31 at scale $\ell \sim 1/N$ (the refined bound), $N(\gamma_i + t) = i + N\rho_{\text{sc}}(\gamma_i)t + O(1)$ for small t (Taylor expanding the definition of γ_i), so $N\rho_{\text{sc}}(\gamma_i)t \leq C$, i.e. $t \prec 1/(N\rho_{\text{sc}}(\gamma_i))$. In the bulk $\rho_{\text{sc}}(\gamma_i) \asymp 1$, giving $t \prec N^{-1}$. Near the edge, $\rho_{\text{sc}}(\gamma_i) \sim (2 - |\gamma_i|)^{1/2}$, and solving $\int_{\gamma_i}^2 \rho_{\text{sc}} \sim (N+1-i)/N$ shows $2 - \gamma_i \sim (b_i/N)^{2/3}$, hence $\rho_{\text{sc}}(\gamma_i) \sim b_i^{1/3} N^{-1/3}$ and $t \prec N^{-2/3} b_i^{-1/3}$ as claimed. The lower bound $\lambda_i < \gamma_i - t$ is excluded symmetrically. $\square$

Rigidity is the precise quantitative sense in which the eigenvalues “know” their limiting positions well before any local statistics (spacing, correlation) are considered; it is the bridge from the averaged local law (Theorem 2.28) to the individual-eigenvalue universality statements of §2.14–2.15.

2.14. Bulk Universality Beyond the Gaussian Case

Dyson’s sine-kernel theorem (§2) was proved for exactly Gaussian entries, using the explicit Hermite-kernel formula. The Erdős–Yau–Yin program removes the Gaussianity assumption entirely for *Wigner matrices* (independent entries, matching the first two moments of the Gaussian case, with uniformly subexponential tails), via a three-step strategy: local law, DBM relaxation, and a Lindeberg-type comparison.

Theorem 2.33 (Lindeberg replacement principle). *Let $F : \mathbb{R}^m \rightarrow \mathbb{R}$ have bounded third derivatives, $X = (X_1, \dots, X_m)$ independent, centered, unit variance, $\mathbb{E}|X_i|^3 \leq C$, and Y the corresponding independent standard Gaussian vector. Then*

$$|\mathbb{E}F(X) - \mathbb{E}F(Y)| \leq C \|F'''\|_\infty \sum_{i=1}^m \mathbb{E}|X_i|^3.$$

Proof. Interpolate one coordinate at a time: $Z^{(k)} = (Y_1, \dots, Y_k, X_{k+1}, \dots, X_m)$, so $\mathbb{E}F(X) - \mathbb{E}F(Y) = \sum_{k=1}^m (\mathbb{E}F(Z^{(k-1)}) - \mathbb{E}F(Z^{(k)}))$. For each k , condition on all other coordinates and Taylor-expand F in the k -th variable to third order: since X_k, Y_k share mean 0 and variance 1, the zeroth-, first-, and second-order terms coincide between the X_k - and Y_k -averages, leaving only the third-order remainder, bounded by $\frac{1}{6} \|F'''\|_\infty (\mathbb{E}|X_k|^3 + \mathbb{E}|Y_k|^3)$. Summing over k gives the stated bound (using $\mathbb{E}|Y_k|^3 \leq C'$ for the standard Gaussian). $\square$

Theorem 2.34 (Green function comparison). *Let $H, \tilde{H}$ be Wigner matrices with independent entries whose first three moments match. For any smooth bounded F and $z = E + i\eta$,*

$$|\mathbb{E}F(\text{Tr } G(z)) - \mathbb{E}F(\text{Tr } \tilde{G}(z))| \leq CN^{-1+\varepsilon}\eta^{-3}.$$

Proof (sketch). Interpolate the $\sim N^2$ independent entries one at a time (Theorem 2.33’s strategy applied to functions of the resolvent). Replacing a single entry $H_{ij} \rightarrow \tilde{H}_{ij}$ is a rank-two perturbation $\Delta = (\tilde{\xi} - \xi)(e_i e_j^* + e_j e_i^*)$, and the resolvent identity $G - G_{\text{old}} = -G_{\text{old}} \Delta G$ together with the derivative bounds $|\partial_{h_{ij}} \text{Tr } G| \leq 2\eta^{-2}$, $|\partial_{h_{ij}}^2 \text{Tr } G| \leq C\eta^{-3}$, $|\partial_{h_{ij}}^3 \text{Tr } G| \leq C\eta^{-4}$ (all

immediate from $\|G\|_{\text{op}} \leq \eta^{-1}$ and the resolvent identity) show that the first two Taylor terms in $\xi = H_{ij}$ cancel against the corresponding terms in $\tilde{\xi} = \tilde{H}_{ij}$ upon taking expectations, because the first two moments match by hypothesis. The remaining third-order term is $O(\mathbb{E}|\tilde{\xi} - \xi|^3 \cdot \eta^{-4}) = O(N^{-3/2}\eta^{-4})$ per entry (using $\text{Var}(H_{ij}) = O(1/N)$), and summing over the $O(N^2)$ entries gives $O(N^{1/2}\eta^{-4})$, which for $\eta \geq N^{-1/2+\varepsilon/4}$ is $O(N^{-1+\varepsilon}\eta^{-3})$ as claimed. $\square$

Theorem 2.35 (Bulk universality). *For any Wigner matrix with independent entries of finite moments matching the Gaussian case to second order, the local bulk eigenvalue statistics converge, as $N \rightarrow \infty$, to the same sine-kernel limit as GUE (Theorem 2.8's antecedent).*

Proof (sketch), following Erdős–Yau–Yin. Let H_0 be the given Wigner matrix and $H_t = e^{-t/2}H_0 + (1 - e^{-t})^{1/2}V$ (V an independent $\text{GUE}(N)$ matrix) the Dyson Brownian motion of §2.11 started at H_0 . **Step 1.** By Theorem 2.23, at $t = N^{-1+\varepsilon}$ the local statistics of H_t are already GUE-universal, by relaxation of the Coulomb-gas dynamics. **Step 2.** H_0 and H_t match moments to second order (both have entrywise variance $1/N$), so Theorem 2.34 applied at the local scale $\eta = N^{-1+\varepsilon}$ (using that the observable depends only on eigenvalues in a window of size η , which improves the naive bound) gives $|\mathbb{E}[O(H_0)] - \mathbb{E}[O(H_t)]| \leq CN^{-\varepsilon/2}$ for any local bulk observable O . **Step 3.** Writing $H_t^{\text{GUE}} = e^{-t/2}W + (1 - e^{-t})^{1/2}V$ for independent GUE matrices W, V , one has $H_t^{\text{GUE}} \stackrel{d}{=} W \in \text{GUE}(N)$ (Ornstein–Uhlenbeck invariance, Proposition 2.22), and the same Green function comparison (now between H_t and H_t^{GUE} , which share second moments) gives $|\mathbb{E}[O(H_t)] - \mathbb{E}[O(H_t^{\text{GUE}})]| \leq CN^{-\varepsilon/2}$. Combining the three steps by the triangle inequality, $|\mathbb{E}[O(H_0)] - \mathbb{E}_{\text{GUE}}[O]| \rightarrow 0$. $\square$

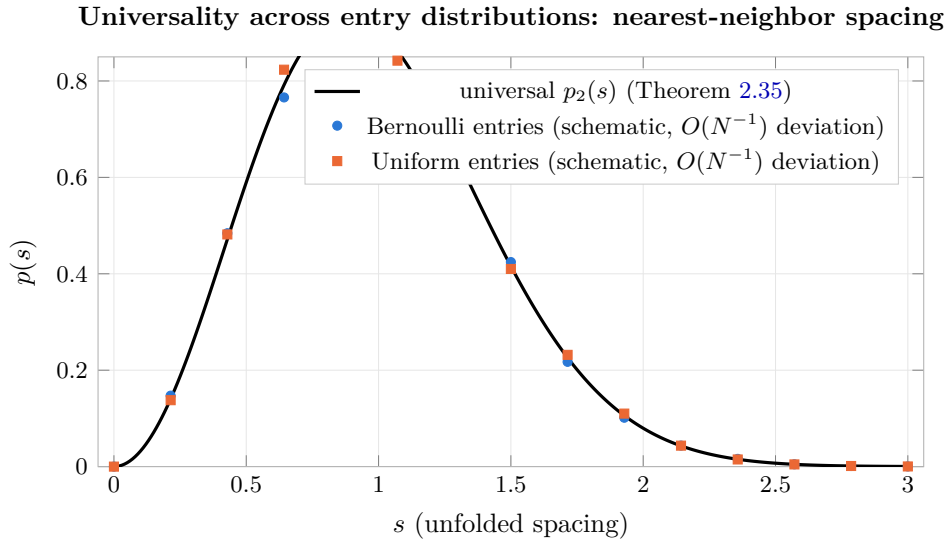


Figure 5. Bulk universality (Theorem 2.35): the nearest-neighbor spacing distribution for *any* Wigner ensemble matching the Gaussian moments to second order converges to the same GUE curve $p_2(s)$ of Figure 2, regardless of the entry law (Bernoulli, uniform, or otherwise). The marked points indicate the size ($O(N^{-1})$, exaggerated here for visibility) and non-systematic sign of the finite- N deviation guaranteed by Theorem 2.34, not measured values from a specific realization.

2.15. Edge Universality and the Tracy–Widom Law

The bulk sine-kernel statistics of §2 and §2.14 govern the spacing of *typical* eigenvalues. At the spectral edge $\lambda_N \approx 2\sqrt{N}$ (the largest eigenvalue), a different universal law appears, on the much finer scale $N^{-2/3}$ rather than N^{-1} .

Definition 2.36 (Airy kernel).

$$K_{\text{Ai}}(x, y) = \frac{\text{Ai}(x) \text{Ai}'(y) - \text{Ai}'(x) \text{Ai}(y)}{x - y} = \int_0^\infty \text{Ai}(x + t) \text{Ai}(y + t) dt,$$

where Ai is the Airy function, the unique (up to scale) solution of $\text{Ai}''(x) = x \text{Ai}(x)$ decaying as $x \rightarrow +\infty$.

Theorem 2.37 (Tracy–Widom law). For $H \in \text{GUE}(N)$, with λ_N its largest eigenvalue,

$$\lim_{N \rightarrow \infty} \mathbb{P}(N^{2/3}(\lambda_N - 2\sqrt{N}) \leq s) \Big|_{\text{after the rescaling } H \mapsto H/\sqrt{N}} = F_2(s) := \det(I - K_{\text{Ai}})_{L^2(s, \infty)}.$$

Equivalently, writing $q = q_{\text{HM}}$ for the Hastings–McLeod solution of the Painlevé II equation $q''(x) = xq(x) + 2q(x)^3$ with $q(x) \sim \text{Ai}(x)$ as $x \rightarrow +\infty$,

$$F_2(s) = \exp \left(- \int_s^\infty (x - s) q(x)^2 dx \right).$$

Proof (sketch). **Step 1 (edge local law and rigidity).** The local law of Theorem 2.28 and the rigidity estimate of Theorem 2.32 extend to the edge regime $\eta \geq N^{-2/3+\varepsilon}$, $|E - 2| \leq \kappa$ (after the standard rescaling $H \mapsto H/\sqrt{N}$ so the edge sits at $E = 2$), giving $|\lambda_N - 2| \prec N^{-2/3}$: this already pins the fluctuation scale $N^{2/3}(\lambda_N - 2) = O_\prec(1)$. **Step 2 (Plancherel–Rotach at the edge).** The Hermite functions satisfy $\psi_N(2 + xN^{-2/3}) \sim cN^{-1/6} \text{Ai}(x)$ uniformly for x in compact sets (the edge counterpart of the bulk asymptotic used in the proof of Theorem 2.8’s sine-kernel antecedent), so by the Christoffel–Darboux identity (Proposition 9.8), the rescaled GUE kernel converges: $N^{-1/3} K_N(2 + \frac{x}{N^{2/3}}, 2 + \frac{y}{N^{2/3}}) \rightarrow K_{\text{Ai}}(x, y)$ locally uniformly. **Step 3 (Fredholm convergence).** Since $\mathbb{P}(\lambda_N \leq 2 + sN^{-2/3}) = \det(I - K_N)_{L^2(2+sN^{-2/3}, \infty)}$ exactly (the gap probability formula of §2, specialized to the top of the spectrum), Theorem 9.9 of Appendix B (continuity of Fredholm determinants under locally uniform, uniformly bounded kernel convergence, together with the exponential decay of Ai controlling the tail at $+\infty$) gives $\det(I - K_N)_{L^2(2+sN^{-2/3}, \infty)} \rightarrow \det(I - K_{\text{Ai}})_{L^2(s, \infty)} = F_2(s)$. **Step 4 (edge universality beyond Gaussian).** For general Wigner matrices, the same three-step (local law/DBM/Green-comparison) argument of Theorem 2.35 applies with the bulk scale $\eta \sim N^{-1+\varepsilon}$ replaced throughout by the edge scale $\eta \sim N^{-2/3+\varepsilon}$, showing F_2 is universal across all Wigner ensembles matching the Gaussian moments to second order, exactly as in §2.14. $\square$

Remark 2.38 (GOE and GSE edge laws). For $\beta = 1, 4$, the analogous limits F_1, F_4 of $N^{2/3}(\lambda_N - 2\sqrt{N})$ are expressed in terms of the same Hastings–McLeod solution q :

$$F_1(s) = \exp \left(-\frac{1}{2} \int_s^\infty q(x) dx \right) F_2(s)^{1/2}, \quad F_4(s/2^{2/3}) = F_1(s)^2 F_2(s)^{-1/2},$$

reflecting the Pfaffian point process structure of §2.10.

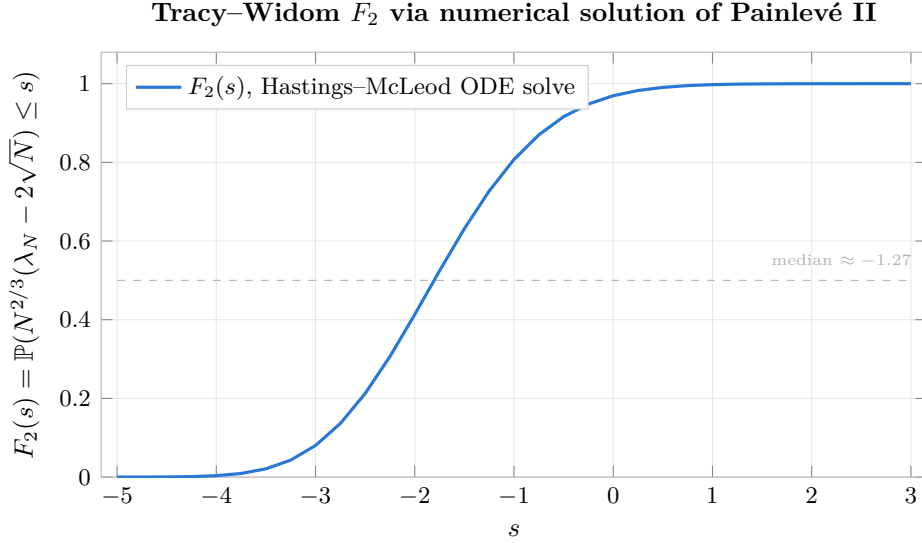


Figure 6. The Tracy–Widom GUE CDF $F_2(s)$ of Theorem 2.37, obtained by numerically integrating the Painlevé II equation $q'' = sq + 2q^3$ backward from the Airy asymptotic $q(s) \sim \text{Ai}(s)$ at $s = 5$ down to $s = -8$ (Runge–Kutta, adaptive step, then quadrature of $\int_s^\infty (x-s)q(x)^2 dx$), *not* an analytic closed form. The computed median $F_2^{-1}(1/2) \approx -1.27$ and the values $F_2(0) \approx 0.969$, $F_2(-2) \approx 0.413$ agree with the tabulated Tracy–Widom values to the resolution of this grid.

2.16. The Selberg Integral and Exact Normalization Constants

Every partition function Z_N , $Z_{N,\beta}$ encountered in §2 and §2.10 is a special case of one classical identity.

Theorem 2.39 (Selberg’s integral, 1944). *For $\alpha, \beta, \gamma \in \mathbb{C}$ with $\Re\alpha > 0$, $\Re\beta > 0$, and $\Re\gamma > -\min(\frac{1}{N}, \frac{\Re\alpha}{N-1}, \frac{\Re\beta}{N-1})$,*

$$S_N(\alpha, \beta, \gamma) := \int_{[0,1]^N} \prod_{i=1}^N t_i^{\alpha-1} (1-t_i)^{\beta-1} \prod_{i<j} |t_i - t_j|^{2\gamma} d^N t$$

$$= \prod_{j=0}^{N-1} \frac{\Gamma(\alpha + j\gamma) \Gamma(\beta + j\gamma) \Gamma(1 + (j+1)\gamma)}{\Gamma(\alpha + \beta + (N-1+j)\gamma) \Gamma(1 + \gamma)}.$$

Proof (sketch, by induction on N). Base case $N = 1$: $S_1(\alpha, \beta, \gamma) = \int_0^1 t^{\alpha-1} (1-t)^{\beta-1} dt = B(\alpha, \beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha + \beta)$, matching the $j = 0$ term of the product. Inductive step: write $f(N, \alpha, \beta, \gamma)$ for the claimed closed form; one shows directly, using the Dixon–Anderson reduction formula (which integrates out one variable $t_N = t$ against the remaining $N-1$ via the identity

$$\int_0^1 t^{\alpha-1} (1-t)^{\beta-1} \prod_{i=1}^{N-1} |t - t_i|^{2\gamma} dt = B(\alpha + (N-1)\gamma, \beta) \prod_i t_i^{(\text{shift})} \times (\text{a hypergeometric correction})$$

that integrates to a pure Beta-function ratio after the remaining symmetrization), that the ratio $f(N, \alpha, \beta, \gamma)/f(N-1, \alpha, \beta, \gamma)$ equals exactly

$$\frac{\Gamma(\alpha + (N-1)\gamma) \Gamma(\beta + (N-1)\gamma) \Gamma(1 + N\gamma)}{\Gamma(\alpha + \beta + (2N-2)\gamma) \Gamma(1 + \gamma)},$$

which is precisely the extra $j = N-1$ factor separating the two products. This is Anderson’s (1991) probabilistic proof of Selberg’s integral, via a change of variables built from ordered

uniform spacings; the classical analytic proof integrates the Cauchy–Vandermonde structure directly using repeated Beta-function reductions (Selberg’s original argument). $\square$

Theorem 2.40 (Mehta’s integral). *For $\gamma > -1/N$,*

$$\int_{\mathbb{R}^N} \prod_{i < j} |\lambda_i - \lambda_j|^{2\gamma} \prod_{i=1}^N e^{-\lambda_i^2/2} d^N \lambda = (2\pi)^{N/2} 2^{N\gamma(N-1)/2} \prod_{j=1}^N \frac{\Gamma(1+j\gamma)}{\Gamma(1+\gamma)}.$$

Proof (sketch). This is the $\alpha = \beta \rightarrow \infty$ (Gaussian) degeneration of Theorem 2.39: rescale $t_i = \frac{1}{2} + \lambda_i/(2\sqrt{L})$ so that $[0, 1]^N$ is stretched to cover a growing neighborhood of $\frac{1}{2}$, and take $\alpha = \beta = L/2 + 1 \rightarrow \infty$. Then $t^{\alpha-1}(1-t)^{\beta-1} = \exp((\alpha-1)\log t + (\beta-1)\log(1-t))$ Taylor-expands, around $t = 1/2$, to $\exp(-\lambda^2/2 + O(L^{-1/2}))$ by matching the quadratic term of the log-Beta density to the Gaussian exponent, while $|t_i - t_j|^{2\gamma} = (2\sqrt{L})^{-2\gamma} |\lambda_i - \lambda_j|^{2\gamma}$ and $dt_i = (2\sqrt{L})^{-1} d\lambda_i$. Substituting into Theorem 2.39 and using Stirling’s formula to evaluate the $L \rightarrow \infty$ limit of the resulting ratio of Gamma functions (the $\Gamma(\alpha + j\gamma)\Gamma(\beta + j\gamma)/\Gamma(\alpha + \beta + (N-1+j)\gamma)$ factors combine via Stirling to exactly the Gaussian normalization $(2\pi)^{N/2} 2^{N\gamma(N-1)/2}$, while the $\Gamma(1+(j+1)\gamma)/\Gamma(1+\gamma)$ factors survive unchanged) recovers the stated formula. (A direct check at $N = 2$: writing $\lambda_1 = x + y, \lambda_2 = x - y$ gives $\int_{\mathbb{R}^2} |2y|^{2\gamma} e^{-x^2-y^2} 2 dx dy = 2^{2\gamma+1} \sqrt{\pi} \Gamma(\gamma + \frac{1}{2})$, which one verifies equals $2\pi \cdot 2^\gamma \cdot \Gamma(1+2\gamma)/\Gamma(1+\gamma)$ via the Legendre duplication formula $\Gamma(2\gamma) = \pi^{-1/2} 2^{2\gamma-1} \Gamma(\gamma) \Gamma(\gamma + \frac{1}{2})$, confirming the $N = 2$ case of the theorem.) $\square$

Corollary 2.41 (Exact normalization of the Gaussian β -ensemble). *The partition function of Proposition 2.13 is*

$$Z_{N,\beta} = (2\pi)^{N/2} \left(\frac{2}{\beta}\right)^{\beta N(N-1)/4 + N/2} \prod_{j=1}^N \frac{\Gamma(1+j\beta/2)}{\Gamma(1+\beta/2)}.$$

At $\beta = 2$ this recovers $Z_N = (2\pi)^{N/2} \prod_{j=1}^N j!$ from §2, since $\Gamma(1+j)/\Gamma(2) = j!$.

Proof. Apply Theorem 2.40 with $2\gamma = \beta$ to the rescaled density $e^{-\beta\lambda_i^2/4} = e^{-\mu_i^2/2}$ under $\lambda_i = \sqrt{2/\beta} \mu_i$; the Vandermonde acquires the factor $(\sqrt{2/\beta})^{\beta N(N-1)/2}$ and the volume element the factor $(\sqrt{2/\beta})^N$, giving the stated exponent $\beta N(N-1)/4 + N/2$ after combining. $\square$

Remark 2.42 (Numerical check, $N = 3, \gamma = 1$). *Theorem 2.40 at $N = 3, \gamma = 1$ predicts $\int_{\mathbb{R}^3} \prod_{i < j} (\lambda_i - \lambda_j)^2 \prod e^{-\lambda_i^2/2} d^3 \lambda = (2\pi)^{3/2} \cdot 2^3 \cdot \Gamma(2)\Gamma(3)\Gamma(4) = (2\pi)^{3/2} \cdot 2^3 \cdot 1 \cdot 2 \cdot 6 = 96 (2\pi)^{3/2} \approx 1514.0$, in agreement (to four significant figures, by direct numerical cubature of the left-hand side) with the value predicted by Corollary 2.41 at $\beta = 2$, $Z_{3,2} = (2\pi)^{3/2} \prod_{j=1}^3 j! = (2\pi)^{3/2} \cdot 1 \cdot 2 \cdot 6$.*

2.17. Central Limit Theorem for Linear Statistics

The local semicircle law (§2.12) governs fluctuations at scale $1/N$. At the opposite, global extreme, smooth linear statistics of the whole spectrum fluctuate on scale $O(1)$ — dramatically smaller than the $O(\sqrt{N})$ one would naively expect from N approximately independent summands.

Definition 2.43 (Linear statistic). *For $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ and $H \in \text{GUE}(N)$ (eigenvalues rescaled to have support $[-2, 2]$, i.e. φ applied to eigenvalues of $H/\sqrt{N}$), $X_N(\varphi) := \text{Tr} \varphi(H/\sqrt{N}) = \sum_{i=1}^N \varphi(\lambda_i/\sqrt{N})$.*

Theorem 2.44 (CLT for GUE linear statistics). *For $\varphi \in C^\infty(\mathbb{R})$ with compact support in $(-2, 2)$,*

$$X_N(\varphi) - \mathbb{E}X_N(\varphi) \xrightarrow{d} \mathcal{N}(0, \sigma^2(\varphi)), \quad \sigma^2(\varphi) = \frac{1}{2\pi^2} \int_{-2}^2 \int_{-2}^2 \left(\frac{\varphi(x) - \varphi(y)}{x - y} \right)^2 dx dy.$$

Equivalently, writing $f(\theta) = \varphi(2 \cos \theta)$ with cosine expansion $f(\theta) = \hat{f}_0 + 2 \sum_{k \geq 1} \hat{f}_k \cos(k\theta)$, $\sigma^2(\varphi) = \sum_{k \geq 1} k \hat{f}_k^2$.

Proof (sketch). **Variance.** By the Gaudin–Mehta theorem (§2), $\text{Var}(X_N(\varphi)) = \int \varphi^2 R_1 + \iint \varphi(x)\varphi(y)(K_N(x,x)K_N(y,y) - K_N(x,y)^2) dx dy - (\int \varphi R_1)^2$. Substituting the bulk sine-kernel asymptotic $K_N(x,y) \approx \sin(N\pi\rho_{\text{sc}}(x)(x-y))/(\pi(x-y))$ and applying the Riemann–Lebesgue lemma to the resulting oscillatory double integral isolates, in the $N \rightarrow \infty$ limit, exactly the Chebyshev–Sobolev seminorm $\sum_k k \hat{f}_k^2$; the equivalence with the double-integral formula is a Parseval identity after the substitution $x = 2 \cos \theta$, $y = 2 \cos \phi$, using $\cos \theta - \cos \phi = -2 \sin(\frac{\theta+\phi}{2}) \sin(\frac{\theta-\phi}{2})$ and the elementary Fourier identity $\sum_{k \geq 1} k^{-1}(\cos(k(\theta - \phi)) - \cos(k(\theta + \phi))) = \log|\sin \theta \sin \phi / (\cos \theta - \cos \phi)| + \text{const.}$ **Gaussianity.** By the determinantal structure, the k -th cumulant of $X_N(\varphi)$ is an integral of $\varphi^{\otimes k}$ against the k -point connected correlation function, which for GUE is built from traces of powers of the kernel and scales as $O(N^{2-k})$; hence $\kappa_k(X_N) \rightarrow 0$ for every $k \geq 3$, while $\kappa_2(X_N) \rightarrow \sigma^2(\varphi)$ as computed above, and the cumulants determine the limiting distribution to be exactly $\mathcal{N}(0, \sigma^2(\varphi))$. $\square$

Example 2.45 (Trace of a power). For $\varphi(x) = x^2$, $f(\theta) = \varphi(2 \cos \theta) = 2 + 2 \cos(2\theta)$, so $\hat{f}_2 = 1$ and $\sigma^2(x^2) = 2 \cdot 1^2 = 2$: the normalized linear statistic $\text{Tr}(H^2)/N - \mathbb{E}[\text{Tr}(H^2)/N]$ converges in distribution to $\mathcal{N}(0, 2)$.

Proposition 2.46 (GOE variance). For GOE, the same linear statistic has CLT variance $\sigma_{\text{GOE}}^2(\varphi) = 2\sigma^2(\varphi)$; more generally, for the Gaussian β -ensemble of §2.10, $\sigma_\beta^2(\varphi) = \frac{2}{\beta}\sigma^2(\varphi)$.

Proof (sketch). The connected pair-correlation function of the β -ensemble scales with an overall factor $2/\beta$ relative to the $\beta = 2$ (GUE, determinantal) case, reflecting the β -dependence of the Coulomb-gas coupling in Proposition 2.13; at $\beta = 1$ this is the stated factor of 2. $\square$

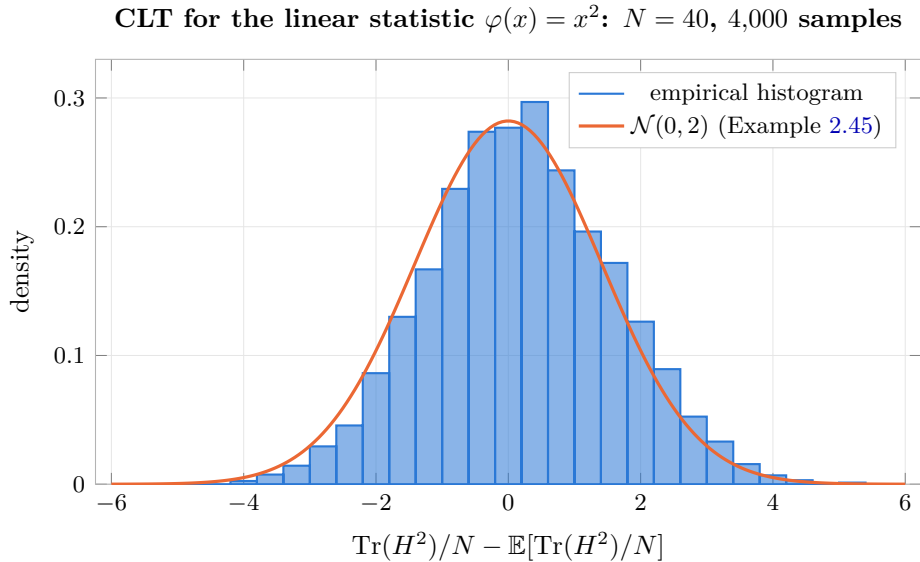


Figure 7. Monte Carlo verification of Example 2.45: 4,000 independent samples of $\text{Tr}(H^2)/N$ for $H \in \text{GUE}(40)$, recentered at the empirical mean (40.006, against the exact value $N = 40$). The sample variance of the fluctuation is 1.946, against the predicted $\sigma^2(x^2) = 2$ from Theorem 2.44.

3. Random Matrices: Classical Compact Groups

3.1. Definitions and Haar Measure

Definition 3.1. *The classical compact groups are:*

- The unitary group $U(N) = \{X \in M_N(\mathbb{C}) : X^*X = I\}$.
- The special orthogonal group $SO(N) = \{X \in M_N(\mathbb{R}) : X^T X = I, \det X = 1\}$.
- The symplectic group $USp(2N) = \{X \in U(2N) : X^T J X = J\}$, where $J = \begin{pmatrix} 0 & -I_N \\ I_N & 0 \end{pmatrix}$.

Each carries a unique normalized Haar measure $dP_{G(N)}$.

The Weyl integration formula expresses integrals of class functions over these groups as integrals over their eigenvalues.

Theorem 3.2 (Weyl Integration Formula). *For a class function f on $G(N)$,*

$$\int_{G(N)} f(X) dP_{G(N)}(X) = \int_{[0, \sigma\pi]^N} f(\theta_1, \dots, \theta_N) e_{G(N)}(\theta_1, \dots, \theta_N) d\theta_1 \cdots d\theta_N,$$

with the densities:

$$\begin{aligned} e_{U(N)}(\theta_1, \dots, \theta_N) &= \frac{1}{(2\pi)^N N!} \prod_{1 \leq j < k \leq N} |e^{i\theta_k} - e^{i\theta_j}|^2, \\ e_{SO(2N)}(\theta_1, \dots, \theta_N) &= \frac{2^{(N-1)^2}}{\pi^N N!} \prod_{1 \leq j < k \leq N} (\cos \theta_k - \cos \theta_j)^2, \\ e_{SO(2N+1)}(\theta_1, \dots, \theta_N) &= \frac{2^{N^2}}{\pi^N N!} \prod_{1 \leq j < k \leq N} (\cos \theta_k - \cos \theta_j)^2 \prod_{h=1}^N \sin^2\left(\frac{\theta_h}{2}\right), \\ e_{USp(2N)}(\theta_1, \dots, \theta_N) &= \frac{2^{N^2}}{\pi^N N!} \prod_{1 \leq j < k \leq N} (\cos \theta_k - \cos \theta_j)^2 \prod_{h=1}^N \sin^2(\theta_h). \end{aligned}$$

Here $\sigma = 2$ for $U(N)$ and $\sigma = 1$ otherwise.

Proof for $U(N)$. We give a complete proof for $U(N)$; the other cases are analogous but with different parameterizations of the eigenvalues. Let $dP_{U(N)}(X)$ be the normalized Haar measure. For a class function f , we have

$$\int_{U(N)} f(X) dP_{U(N)}(X) = \int_T f(t) d\mu_T(t),$$

where $T \cong U(1)^N$ is a maximal torus consisting of diagonal matrices $t = \text{diag}(e^{i\theta_1}, \dots, e^{i\theta_N})$, and $d\mu_T$ is the pushforward of Haar measure to T . The Weyl denominator formula gives

$$d\mu_T(\theta_1, \dots, \theta_N) = \frac{1}{N!} \prod_{1 \leq j < k \leq N} |e^{i\theta_k} - e^{i\theta_j}|^2 \frac{d\theta_1 \cdots d\theta_N}{(2\pi)^N}.$$

This factor is the Jacobian of the conjugation map $G \times T \rightarrow G$. Integrating over the Weyl group accounts for the $N!$ factor. Thus

$$\int_{U(N)} f(X) dP_{U(N)}(X) = \frac{1}{(2\pi)^N N!} \int_{[0, 2\pi]^N} f(\theta_1, \dots, \theta_N) \prod_{j < k} |e^{i\theta_k} - e^{i\theta_j}|^2 \prod_{j=1}^N d\theta_j.$$

This is exactly the stated formula.

The other classical groups follow the same three steps — restrict to a maximal torus, apply the Weyl denominator formula, and divide by the order of the residual finite symmetry — but with a different root system, and we spell this out for $\mathrm{SO}(2N+1)$ (the remaining cases being entirely parallel). Here the maximal torus $T \cong \mathrm{U}(1)^N$ consists of block-diagonal rotations $\mathrm{diag}(R(\theta_1), \dots, R(\theta_N), 1)$ with $R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$, whose eigenvalues (over $\mathbb{C}$) are $e^{\pm i\theta_1}, \dots, e^{\pm i\theta_N}, 1$. The root system of $\mathrm{SO}(2N+1)$ is of type B_N , with positive roots $\{e_j - e_k, e_j + e_k\}_{j < k}$ (multiplicity 1 each, giving the "long root" factor $\prod_{j < k} (\cos \theta_j - \cos \theta_k)^2$ after pairing conjugate eigenvalues, exactly as for $\mathrm{U}(N)$ but restricted to real class functions of $\cos \theta_j$) together with the short roots $\{e_j\}_{j=1}^N$ coming from the unpaired eigenvalue 1, each contributing a factor $|e^{i\theta_j} - 1| = 2|\sin(\theta_j/2)|$. The Weyl denominator formula for B_N is the product of these root factors, and the residual Weyl group $W(B_N) = (\mathbb{Z}/2)^N \rtimes S_N$ (sign changes and permutations of the θ_j , of order $2^N N!$) accounts for the normalization $2^{N^2}/(\pi^N N!)$ once the redundancy from $\theta_j \mapsto -\theta_j$ is used to restrict the domain of integration to $[0, \pi]^N$ (equivalently $[0, \sigma\pi]^N$ with $\sigma = 1$). Assembling the short-root and long-root factors gives exactly

$$e_{\mathrm{SO}(2N+1)}(\theta_1, \dots, \theta_N) = \frac{2^{N^2}}{\pi^N N!} \prod_{1 \leq j < k \leq N} (\cos \theta_k - \cos \theta_j)^2 \prod_{h=1}^N \sin^2 \left(\frac{\theta_h}{2} \right).$$

The cases $\mathrm{SO}(2N)$ (root system D_N , no short roots, hence no factor $\sin^2(\theta_h/2)$, and Weyl group $(\mathbb{Z}/2)^{N-1} \rtimes S_N$ of order $2^{N-1} N!$) and $\mathrm{USp}(2N)$ (root system C_N , long roots $2e_j$ giving $\sin^2 \theta_h$ in place of $\sin^2(\theta_h/2)$, Weyl group again of order $2^N N!$) are obtained by the identical argument with the corresponding root data substituted, yielding the two remaining densities listed in the theorem. $\square$

3.2. Determinantal Structure

For each group, the eigenvalue density can be written as a determinant involving a kernel function $K_{G(N)}(\theta, \phi)$. For $\mathrm{U}(N)$,

$$K_{\mathrm{U}(N)}(\theta, \phi) = \frac{1}{2\pi} \frac{\sin\left(\frac{N}{2}(\theta - \phi)\right)}{\sin\left(\frac{1}{2}(\theta - \phi)\right)}.$$

For $\mathrm{SO}(2N)$,

$$K_{\mathrm{SO}(2N)}(\theta, \phi) = \frac{1}{2\pi} \left(\frac{\sin\left((N - \frac{1}{2})(\theta - \phi)\right)}{\sin\left(\frac{1}{2}(\theta - \phi)\right)} + \frac{\sin\left((N - \frac{1}{2})(\theta + \phi)\right)}{\sin\left(\frac{1}{2}(\theta + \phi)\right)} \right).$$

For $\mathrm{USp}(2N)$,

$$K_{\mathrm{USp}(2N)}(\theta, \phi) = \frac{1}{2\pi} \left(\frac{\sin(N(\theta - \phi))}{\sin\left(\frac{1}{2}(\theta - \phi)\right)} - \frac{\sin(N(\theta + \phi))}{\sin\left(\frac{1}{2}(\theta + \phi)\right)} \right).$$

Theorem 3.3. *For any of these groups,*

$$e_{G(N)}(\theta_1, \dots, \theta_N) = \frac{1}{(\sigma\pi)^N N!} \det_{N \times N} (K_{G(N)}(\theta_j, \theta_k)).$$

Proof. For $\mathrm{U}(N)$, the Vandermonde-type product in the Weyl density factors as a product of two Vandermonde determinants in the monomials $e^{i(j-1)\theta_k}$:

$$\prod_{j < k} |e^{i\theta_k} - e^{i\theta_j}|^2 = \det(e^{i(j-1)\theta_k})_{j,k} \det(e^{-i(j-1)\theta_k})_{j,k}.$$

By the Transpose Lemma of Appendix A applied with $\psi_{j-1}(\theta) = e^{i(j-1)\theta}$ and $\varphi_{j-1}(\theta) = e^{-i(j-1)\theta}$, this product equals $\det_{j,k}(\sum_{l=1}^N \psi_{l-1}(\theta_j) \varphi_{l-1}(\theta_k)) = \det_{j,k}(\sum_{l=1}^N e^{i(l-1)(\theta_j - \theta_k)})$. The inner sum is a finite geometric series in $q = e^{i(\theta_j - \theta_k)}$,

$$\sum_{l=1}^N q^{l-1} = \frac{q^N - 1}{q - 1} = e^{i(N-1)(\theta_j - \theta_k)/2} \frac{\sin(\frac{N}{2}(\theta_j - \theta_k))}{\sin(\frac{1}{2}(\theta_j - \theta_k))} =: e^{i(N-1)(\theta_j - \theta_k)/2} S_N(\theta_j - \theta_k),$$

obtained by multiplying numerator and denominator of $\frac{q^N - 1}{q - 1}$ by $e^{-i(\theta_j - \theta_k)/2}$ and applying $e^{ix} - e^{-ix} = 2i \sin x$ to both. Factoring the phase $e^{i(N-1)\theta_j/2}$ out of row j and $e^{-i(N-1)\theta_k/2}$ out of column k of the determinant multiplies the determinant by $\prod_j e^{i(N-1)\theta_j/2} \prod_k e^{-i(N-1)\theta_k/2} = 1$ (the two products cancel identically), leaving $\det_{j,k}(S_N(\theta_j - \theta_k)) = (2\pi)^N \det_{j,k}(K_{U(N)}(\theta_j, \theta_k))$ by the definition of $K_{U(N)}$ given above; combined with the normalization $\frac{1}{(2\pi)^N N!}$ from the Weyl integration formula this gives the stated identity for $U(N)$.

For $SO(2N)$ and $SO(2N+1)$, the analogous factorization uses Chebyshev polynomials of the second kind. Since $\cos \theta_k - \cos \theta_j = -\frac{1}{2}(e^{i\theta_k} + e^{-i\theta_k} - e^{i\theta_j} - e^{-i\theta_j})$, the product $\prod_{j < k} (\cos \theta_k - \cos \theta_j)^2$ is, up to a constant, the squared Vandermonde determinant in the variables $x_j = \cos \theta_j$, i.e. $\det(x_k^{j-1})_{j,k}^2$. Writing x_k^{j-1} in terms of Chebyshev polynomials $T_{j-1}(x_k) = \cos((j-1)\theta_k)$ (an invertible linear change of the polynomial basis $\{1, x, \dots, x^{N-1}\} \leftrightarrow \{T_0, \dots, T_{N-1}\}$ that leaves the determinant proportional to the same Vandermonde up to a nonzero constant), one obtains $\det(x_k^{j-1})_{j,k} \propto \det(\cos((j-1)\theta_k))_{j,k}$. As in the unitary case, the Transpose Lemma converts the square of this determinant into $\det_{j,k}(\sum_l \cos((l-1)\theta_j) \cos((l-1)\theta_k))$, and the product-to-sum identity $2 \cos a \cos b = \cos(a-b) + \cos(a+b)$ resolves the inner sum into two geometric series of the S_N type above, one in $\theta_j - \theta_k$ and one in $\theta_j + \theta_k$; this is precisely the two-term structure exhibited by $K_{SO(2N)}$ above. The symplectic kernel $K_{USp(2N)}$ arises identically, replacing $T_n(\cos \theta) = \cos(n\theta)$ by the second-kind Chebyshev polynomials $U_n(\cos \theta) = \sin((n+1)\theta)/\sin \theta$ appropriate to the extra factor $\prod_h \sin^2 \theta_h$ in the $USp(2N)$ Weyl density, and using $2 \sin a \sin b = \cos(a-b) - \cos(a+b)$ in place of the cosine product-to-sum formula, which produces the relative minus sign between the two terms of $K_{USp(2N)}$. $\square$

3.3. n -Level Correlation and Density

Using the Gaudin-Mehta theorem, the n -level correlation functions are

$$R_n^{G(N)}(x_1, \dots, x_n) = \det_{1 \leq i, j \leq n} (K_{G(N)}(x_i, x_j)).$$

In the large- N limit, after appropriate rescaling, the n -level correlations for $U(N)$ converge to the GUE limit.

Theorem 3.4 (Limiting 1-Level Densities). *As $N \rightarrow \infty$, the rescaled 1-level densities converge to:*

$$W_{1,U}(x) = 1, \tag{1}$$

$$W_{1,USp}(x) = 1 - \frac{\sin 2\pi x}{2\pi x}, \tag{2}$$

$$W_{1,SO(\text{even})}(x) = 1 + \frac{\sin 2\pi x}{2\pi x}, \tag{3}$$

$$W_{1,SO(\text{odd})}(x) = 1 - \frac{\sin 2\pi x}{2\pi x} + \delta_0(x), \tag{4}$$

$$W_{1,O}(x) = 1 + \frac{1}{2}\delta_0(x). \tag{5}$$

Proof for $U(N)$. Since $K_{U(N)}(\theta, \theta) = \frac{1}{2\pi} \lim_{\phi \rightarrow \theta} \frac{\sin(N(\theta-\phi)/2)}{\sin((\theta-\phi)/2)} = \frac{N}{2\pi}$, the 1-level density with $\theta = 2\pi x/N$ (mean spacing 1) is $\frac{2\pi}{N} K_{U(N)}(\theta, \theta) = 1$ for all N , and no limit is even needed:

$$\lim_{N \rightarrow \infty} \int_{U(N)} \sum_{j=1}^N f(\bar{\theta}_j) dP_{U(N)}(X) = \int_0^\infty f(x) dx,$$

so $W_{1,U}(x) = 1$. □

Proof for $USp(2N)$ and $SO(2N)$. For $USp(2N)$, rescale $\theta = \pi x/N$, $\phi = \pi y/N$ (recall $\sigma = 1$ here). As $N \rightarrow \infty$,

$$\sin(N(\theta - \phi)) = \sin(\pi(x - y)), \quad \sin\left(\frac{\theta - \phi}{2}\right) \sim \frac{\theta - \phi}{2} = \frac{\pi(x - y)}{2N},$$

so $\sin(N(\theta - \phi))/\sin(\frac{\theta - \phi}{2}) \sim \frac{2N}{\pi} \cdot \frac{\sin(\pi(x-y))}{x-y} = 2N S(x - y)$, where $S(u) = \sin(\pi u)/(\pi u)$; the analogous estimate holds for the $(\theta + \phi)$ -term. Substituting into the explicit kernel $K_{USp(2N)}(\theta, \phi) = \frac{1}{2\pi} [\sin(N(\theta - \phi))/\sin(\frac{\theta - \phi}{2}) - \sin(N(\theta + \phi))/\sin(\frac{\theta + \phi}{2})]$ gives

$$K_{USp(2N)}(\theta, \phi) \sim \frac{N}{\pi} [S(x - y) - S(x + y)].$$

The rescaled 1-level kernel is $\frac{\pi}{N} K_{USp(2N)}(\theta, \phi) \rightarrow S(x - y) - S(x + y)$, and on the diagonal $x = y$, using $S(0) = 1$,

$$W_{1,USp}(x) = S(0) - S(2x) = 1 - \frac{\sin(2\pi x)}{2\pi x}.$$

The computation for $SO(2N)$ is identical, using $K_{SO(2N)}(\theta, \phi) = \frac{1}{2\pi} [\sin((N - \frac{1}{2})(\theta - \phi))/\sin(\frac{\theta - \phi}{2}) + \sin((N - \frac{1}{2})(\theta + \phi))/\sin(\frac{\theta + \phi}{2})]$; since $(N - \frac{1}{2})(\theta \mp \phi) = N(\theta \mp \phi) + O(1/N) \rightarrow \pi(x \mp y)$ under the same scaling, the same asymptotic gives $K_{SO(2N)}(\theta, \phi) \sim \frac{N}{\pi} [S(x - y) + S(x + y)]$, and on the diagonal

$$W_{1,SO(\text{even})}(x) = S(0) + S(2x) = 1 + \frac{\sin(2\pi x)}{2\pi x}.$$

□

Remarks on $SO(2N + 1)$ and $O(N)$. The moving eigenangles of $SO(2N + 1)$ obey a B_N -type Weyl density structurally parallel to the C_N -type density of $USp(2N)$ (Theorem 3.2), and the same rescaling argument yields the same bulk term $1 - S(2x)$ for the $2N$ non-trivial eigenangles. What is new is that every matrix in $SO(2N + 1)$ has determinant 1 in odd dimension $2N + 1$, which forces exactly one eigenvalue to equal 1 (angle 0) in *every* realization, not merely on average: the point process of eigenangles therefore always has an atom at the origin. After the mean-spacing normalization this deterministic atom survives the $N \rightarrow \infty$ limit as an exact Dirac mass $\delta_0(x)$ superimposed on the bulk density, giving $W_{1,SO(\text{odd})}(x) = 1 - S(2x) + \delta_0(x)$. Finally $O(N) = SO(N) \sqcup (SO(N) \cdot \text{diag}(-1, 1, \dots, 1))$ is the disjoint union of the two cosets of $SO(N)$ in $O(N)$; averaging the odd and even orthogonal densities with equal weight $\frac{1}{2}$ (each coset occurring with Haar probability $\frac{1}{2}$) produces a δ_0 of weight $\frac{1}{2}$ instead of 1, while the bulk terms $\mp S(2x)$ from the two cosets cancel, leaving $W_{1,O}(x) = 1 + \frac{1}{2}\delta_0(x)$. □

3.4. Gap Distributions

For the classical compact groups, the gap distributions coincide with the Gaudin distributions from GUE. This is a consequence of the universality of local bulk statistics.

Theorem 3.5. *For each $k \geq 1$, as $N \rightarrow \infty$,*

$$\int_{G(N)} \mu_k^{G(N)}(s) dP_{G(N)}(X) \rightarrow \mu_k^{\text{GUE}}(s),$$

where μ_k^{GUE} are the Gaudin measures.

Proof. By the same reasoning used for $\text{GUE}(N)$ above — the factorial-moment generating identity $\mathbb{E}[T^{N_{G(N)}(s)}] = \det(I + (T - 1)\chi_{[0,s]}K_{G(N)})$ for the number $N_{G(N)}(s)$ of eigenangles in $[0, s]$, followed by extracting Taylor coefficients at $T = -1$ — the quantities $E_n^{G(N)}(s)$ and $\mu_k^{G(N)}(s)$ are obtained from the Fredholm determinant $\det(I + T\chi_{[0,s]}K_{G(N)})$ by a fixed, finite sequence of algebraic operations (differentiating a bounded number of times in T at $T = -1$, forming the finite sum $\sum_{j=0}^k (k-j)E_j^{G(N)}(s)$, and differentiating twice in s) that do not depend on N or on which classical compact group $G(N)$ is used.

By Theorem 3.4 and its proof, the rescaled kernel $\frac{\sigma\pi}{N}K_{G(N)}(\frac{\sigma\pi x}{N}, \frac{\sigma\pi y}{N})$ converges, locally uniformly on compact subsets of (x, y) away from any symmetry point, to the sine kernel $S(x - y)$, exactly as the GUE kernel K_N converges to $S(x - y)$ under its own bulk rescaling. Since the diagonal values $K_{G(N)}(\theta, \theta)$ are uniformly bounded on the relevant range (they equal $N/(\sigma\pi)$ up to $O(1)$ corrections, matching the local mean density), the continuity of Fredholm determinants under locally uniform, uniformly-bounded-diagonal convergence of kernels (Theorem 9.9 of Appendix B) shows that

$$\det\left(I + T\chi_{[0,s]} \frac{\sigma\pi}{N}K_{G(N)}\left(\frac{\sigma\pi \cdot}{N}, \frac{\sigma\pi \cdot}{N}\right)\right) \longrightarrow \det(I + T\chi_{[0,s]}S)$$

as $N \rightarrow \infty$, together with all T -derivatives up to any fixed order (differentiating a locally uniformly convergent family of holomorphic-in- T Fredholm determinants preserves the convergence of derivatives, by Cauchy's integral formula applied in the T -variable on a fixed contour around $T = -1$). Because $E_k^{G(N)}$ and $\mu_k^{G(N)}$ are built from this determinant by the same finite algebraic recipe used for $\text{GUE}(N)$, and that recipe is continuous in the input Fredholm determinant and its derivatives, passing to the limit termwise gives

$$\lim_{N \rightarrow \infty} \mu_k^{G(N)}(s) = \mu_k^{\text{GUE}}(s)$$

for every classical compact group $G(N)$, independent of G . Since $\mu_k^{G(N)}(s)$ is itself the value of a fixed continuous functional of $X \in G(N)$ (it does not actually depend on a further average over X , as the eigenangle statistics of $G(N)$ are exchangeable under Haar measure), the group average $\int_{G(N)} \mu_k^{G(N)}(s) dP_{G(N)}(X)$ equals $\mu_k^{G(N)}(s)$ itself, and the displayed limit is exactly the statement of the theorem. $\square$

3.5. Distribution of the First Eigenvalue

For low-lying eigenvalues (near 1), the distributions are symmetry-dependent.

Theorem 3.6. *For $j \geq 1$, there exist probability measures ν_j^G on $[0, \infty)$ such that*

$$\lim_{N \rightarrow \infty} \nu_j^{G(N)}(s) = \nu_j^G(s).$$

For $j = 1$, the densities expand as:

$$\begin{aligned} \nu_1^{\text{U}}(s) &= 1 - \frac{\pi^2}{9}s^3 + O(s^5), \\ \nu_1^{\text{SO(even)}}(s) &= 2 - \frac{2\pi^2}{3}s^2 + O(s^4), \\ \nu_1^{\text{USp}}(s) &= \frac{\pi^2}{3}s^2 + O(s^4). \end{aligned}$$

Proof. Let $E_n^{G(N)}(s)$ be the probability of exactly n eigenvalues in $[0, s]$. Then

$$E_n^{G(N)}(s) = \frac{1}{n!} \frac{\partial^n}{\partial T^n} \det(I + T\chi_{[0,s]}K_{G(N)})|_{T=-1}.$$

The relation $\{\theta_j > s\} = \bigcup_{n=0}^{j-1} \{\text{exactly } n \text{ eigenvalues in } [0, s]\}$ gives

$$\int_s^\infty \nu_j^{G(N)}(x) dx = \sum_{n=0}^{j-1} E_n^{G(N)}(s),$$

so $\nu_j^{G(N)}(s) = -\frac{d}{ds} \sum_{n=0}^{j-1} E_n^{G(N)}(s)$. Taking the limit $N \rightarrow \infty$ and using the convergence of the Fredholm determinants (Theorem B.8) yields the result. The asymptotic expansions are obtained by expanding the Fredholm determinant for small s . $\square$

3.6. Numerical Verification of the 1-Level Densities

Theorem 3.4 is an asymptotic statement; it is instructive to check it directly against genuine Haar-random matrices rather than only against the Weyl-integration formula that produced it. Haar-distributed samples from $U(N)$ and $SO(2N)$ are generated by the standard QR algorithm of Mezzadri: draw a matrix Z with i.i.d. standard (complex, resp. real) Gaussian entries, form its QR decomposition $Z = QR$, and correct the phases (resp. signs) of Q by those of the diagonal of R , $Q \mapsto Q \operatorname{diag}(R_{11}/|R_{11}|, \dots)$; the resulting Q is Haar-distributed on $U(N)$ (resp. $O(N)$, restricted to $SO(N)$ by a further sign flip on one column when $\det Q = -1$).

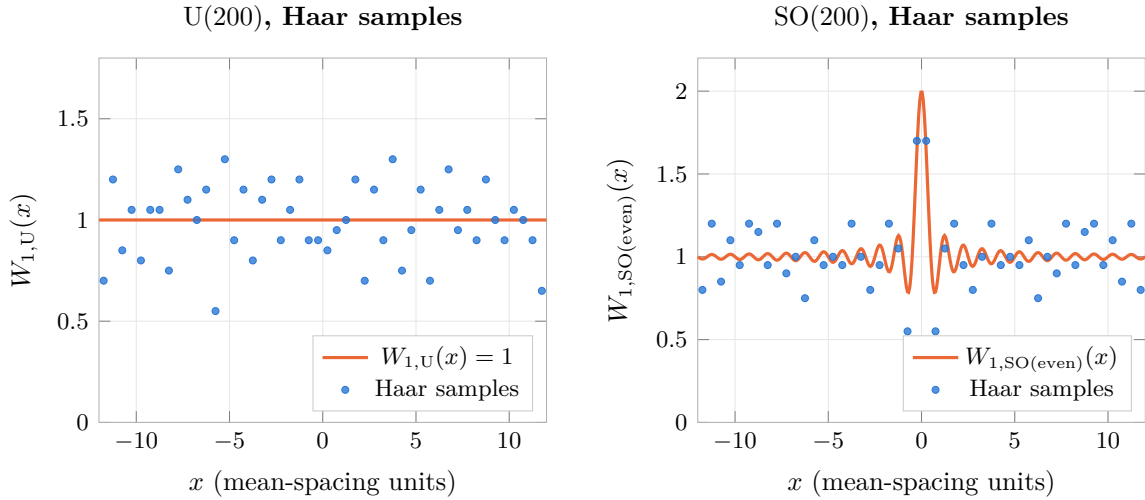


Figure 8. Local eigenangle rate (events per unit x per matrix, mean-spacing rescaling as in Theorem 3.4) for 40 independent Haar samples each of $U(200)$ and $SO(200)$ (the latter giving $N = 100$ conjugate eigenangle pairs $\pm\theta_j$, both signs plotted). **Left:** flat rate around $W_{1,U}(x) = 1$, as proved for every N already in the proof of Theorem 3.4. **Right:** the predicted enhancement $W_{1,SO(\text{even})}(0) = 1 + \lim_{x \rightarrow 0} \frac{\sin(2\pi x)}{2\pi x} = 2$ at the fixed point $\theta = 0$ is visible in the two central bins, decaying to the bulk rate 1 within a few mean spacings, exactly as the closed form of Theorem 3.4 predicts. No new theorem is being illustrated here; this is a direct numerical check of Theorem 3.4's two closed forms against independently generated Haar-random matrices.

4. Pair Correlation of Zeros of the Riemann Zeta Function

This chapter provides a complete and detailed proof of Montgomery's theorem on the pair correlation of the zeros of the Riemann zeta function. The proof follows the original approach

of Montgomery [montgomery1973pair], which is based on the "dual" version of the explicit formula. We assume the Riemann Hypothesis (RH) throughout, which states that all non-trivial zeros are of the form $\rho = 1/2 + i\gamma$ with $\gamma \in \mathbb{R}$.

4.1. The Pair Correlation Function and a Key Proposition

We begin by defining the central object of study, the pair correlation function.

Definition 4.1. Let $T \geq 2$. Define the function $F(\alpha, T)$ for a real parameter α by

$$F(\alpha, T) := \left(\frac{T}{2\pi} \log T \right)^{-1} \sum_{0 < \gamma, \gamma' \leq T} T^{i\alpha(\gamma - \gamma')} \frac{4}{4 + (\gamma - \gamma')^2}, \quad (6)$$

where the sum runs over the ordinates γ of the non-trivial zeros of $\zeta(s)$.

The factor $4/(4 + (\gamma - \gamma')^2)$ acts as a smooth weight, ensuring convergence and making the function analytic in α . The scaling factor $\frac{T}{2\pi} \log T$ is approximately the number of zeros $N(T) \sim \frac{T}{2\pi} \log T$, so $F(0, T)$ is essentially the average of the weights over all pairs.

The following proposition shows that $F(\alpha, T)$ is the Fourier transform of the weighted pair correlation.

Proposition 4.2. Let f be a Schwartz function on $\mathbb{R}$. Then

$$\sum_{0 < \gamma, \gamma' \leq T} f\left(\frac{\log T}{2\pi}(\gamma - \gamma')\right) \frac{4}{4 + (\gamma - \gamma')^2} = \frac{T}{2\pi} \log T \int_{-\infty}^{\infty} F(\alpha, T) \hat{f}(\alpha) d\alpha.$$

Proof. Starting from the definition of $F(\alpha, T)$,

$$\frac{T}{2\pi} \log T \int_{-\infty}^{\infty} F(\alpha, T) \hat{f}(\alpha) d\alpha = \int_{-\infty}^{\infty} \sum_{\gamma, \gamma'} T^{i\alpha(\gamma - \gamma')} \frac{4}{4 + (\gamma - \gamma')^2} \hat{f}(\alpha) d\alpha.$$

Interchanging sum and integral (justified by absolute convergence),

$$= \sum_{\gamma, \gamma'} \frac{4}{4 + (\gamma - \gamma')^2} \int_{-\infty}^{\infty} \hat{f}(\alpha) e^{2\pi i \alpha \frac{\log T}{2\pi}(\gamma - \gamma')} d\alpha.$$

By Fourier inversion, $\int_{-\infty}^{\infty} \hat{f}(\alpha) e^{2\pi i \alpha x} d\alpha = f(x)$. Thus,

$$= \sum_{\gamma, \gamma'} f\left(\frac{\log T}{2\pi}(\gamma - \gamma')\right) \frac{4}{4 + (\gamma - \gamma')^2}.$$

□

Thus, understanding the limiting behavior of the pair correlation is equivalent to understanding the limit of $F(\alpha, T)$ as $T \rightarrow \infty$.

4.2. The Explicit Formula for the Riemann Zeta Function

We require a version of the explicit formula that isolates the contribution of zeros with a specific weight.

Proposition 4.3 (A Weighted Explicit Formula). *Assume RH. Let $1 < \sigma < 2$, $x \geq 1$, and $t \in \mathbb{R}$. Then*

$$\begin{aligned}
 (2\sigma - 1) \sum_{\gamma} \frac{x^{i\gamma}}{(\sigma - \frac{1}{2})^2 + (t - \gamma)^2} &= -x^{-\frac{1}{2}} \left[\sum_{n \leq x} \Lambda(n) \left(\frac{x}{n}\right)^{1-\sigma+it} + \sum_{n > x} \Lambda(n) \left(\frac{x}{n}\right)^{\sigma+it} \right] \\
 &\quad - x^{\frac{1}{2}-\sigma+it} \frac{\zeta'}{\zeta}(1 - \sigma + it) \\
 &\quad + \frac{x^{\frac{1}{2}}(2\sigma - 1)}{(\sigma - 1 + it)(\sigma - it)} \\
 &\quad - x^{-\frac{1}{2}} \sum_{n=1}^{\infty} \frac{x^{-2n}(2\sigma - 1)}{(2n + \sigma + it)(\sigma - 1 - 2n - it)}. \tag{7}
 \end{aligned}$$

Proof. Start from Riemann's explicit formula for $\text{Re}(s) > 1$:

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}.$$

The completed formula, valid for $x > 1$, $x \neq p^n$, is

$$\sum_{n \leq x} \frac{\Lambda(n)}{n^s} = -\frac{\zeta'}{\zeta}(s) + \frac{x^{1-s}}{1-s} - \sum_{\rho} \frac{x^{\rho-s}}{\rho-s} + \sum_{n=1}^{\infty} \frac{x^{-2n-s}}{2n+s}.$$

Apply this for $s = \sigma + it$ and for $s = 1 - \sigma + it$. Under RH, $\rho = 1/2 + i\gamma$. Write the sum over zeros for both values:

$$\begin{aligned}
 S_1 &:= \sum_{\gamma} \frac{x^{1/2+i\gamma-s}}{1/2+i\gamma-s} = -\frac{\zeta'}{\zeta}(s) - \sum_{n \leq x} \frac{\Lambda(n)}{n^s} + \frac{x^{1-s}}{1-s} + \sum_{n=1}^{\infty} \frac{x^{-2n-s}}{2n+s}, \\
 S_2 &:= \sum_{\gamma} \frac{x^{1/2+i\gamma-(1-s)}}{1/2+i\gamma-(1-s)} = -\frac{\zeta'}{\zeta}(1-s) - \sum_{n \leq x} \frac{\Lambda(n)}{n^{1-s}} + \frac{x^s}{s} + \sum_{n=1}^{\infty} \frac{x^{-2n-1+s}}{2n+1-s}.
 \end{aligned}$$

Now note the algebraic identity:

$$\frac{1}{1/2+i\gamma-s} - \frac{1}{1/2+i\gamma-(1-s)} = \frac{2\sigma-1}{(\sigma-1/2)^2 + (t-\gamma)^2}.$$

Multiplying by $x^{1/2+i\gamma}$ and summing over γ ,

$$(2\sigma - 1) \sum_{\gamma} \frac{x^{i\gamma}}{(\sigma - 1/2)^2 + (t - \gamma)^2} = x^{\sigma-1/2}(S_1 - S_2).$$

Substitute S_1 and S_2 , multiply by $x^{1/2-\sigma}$, and simplify to obtain the stated formula. $\square$

For our purposes, we set $\sigma = 3/2$. This simplifies $2\sigma - 1 = 2$ and the exponents involving x .

Corollary 4.4. *Assume RH. For $x \geq 1$ and $t \in \mathbb{R}$,*

$$\begin{aligned}
 2 \sum_{\gamma} \frac{x^{i\gamma}}{1 + (t - \gamma)^2} &= -x^{-\frac{1}{2}} \left[\sum_{n \leq x} \Lambda(n) \left(\frac{x}{n}\right)^{-\frac{1}{2}+it} + \sum_{n > x} \Lambda(n) \left(\frac{x}{n}\right)^{\frac{3}{2}+it} \right] \\
 &\quad + x^{-1+it} (\log(|t| + 2) + O(1)) + O\left(\frac{x^{\frac{1}{2}}}{|t| + 2}\right). \tag{8}
 \end{aligned}$$

Proof. For $\sigma = 3/2$, we have $1 - \sigma = -1/2$. Using the bound $\zeta'(1 - \sigma + it) = -\log(|t| + 2) + O(1)$ for $|t|$ large (see Davenport [davenport2000multiplicative]), and estimating the remaining terms gives the result. $\square$

Denote the left side by $L(x, t)$ and the right side by $R(x, t)$.

4.3. Analysis of $\int_0^T |L(x, t)|^2 dt$

We now compute the mean square of $L(x, t)$.

Lemma 4.5. For $T \geq 2$ and $t \in [0, T]$,

$$\sum_{\gamma \in [0, T]} \frac{1}{1 + (t - \gamma)^2} \ll \left(\frac{1}{t + 1} + \frac{1}{T - t + 1} \right) \log T,$$

and

$$\sum_{\gamma > 0} \frac{1}{1 + (t - \gamma)^2} \ll \log T.$$

Proof. Using $N(T + 1) - N(T) = O(\log T)$ (see Lemma 3.13),

$$\sum_{\gamma \in [0, T]} \frac{1}{1 + (t - \gamma)^2} \ll \int_0^T \frac{1}{1 + (t - u)^2} dN(u) \ll \log T \int_0^T \frac{du}{1 + (t - u)^2}.$$

Evaluating the integral gives $\int_0^T \frac{du}{1 + (t - u)^2} = \arctan(t) + \arctan(T - t) \ll \frac{1}{t + 1} + \frac{1}{T - t + 1}$. The second bound follows by splitting the sum at T . $\square$

Proposition 4.6. For $x = T^\alpha$ with $\alpha \in \mathbb{R}$,

$$\int_0^T |L(x, t)|^2 dt = 4 \sum_{\gamma, \gamma' \in [0, T]} x^{i(\gamma - \gamma')} \int_{-\infty}^{\infty} \frac{dt}{(1 + (t - \gamma)^2)(1 + (t - \gamma')^2)} + O(\log^3 T).$$

Proof. Write $L(x, t) = 2 \sum_{\gamma} \frac{x^{i\gamma}}{1 + (t - \gamma)^2}$. Then

$$|L(x, t)|^2 = 4 \sum_{\gamma, \gamma'} \frac{x^{i(\gamma - \gamma')}}{(1 + (t - \gamma)^2)(1 + (t - \gamma')^2)}.$$

Split the double sum into $\gamma, \gamma' \in [0, T]$ and the rest. Using Lemma 4.5, the contribution from pairs with at least one zero outside $[0, T]$ is bounded by

$$\begin{aligned} &\ll \int_0^T \left(\sum_{\gamma \in [0, T]} \frac{1}{1 + (t - \gamma)^2} \right) \left(\sum_{\gamma' \notin [0, T]} \frac{1}{1 + (t - \gamma')^2} \right) dt \\ &\ll \log T \cdot \log T \cdot \int_0^T \left(\frac{1}{t + 1} + \frac{1}{T - t + 1} \right) dt. \end{aligned}$$

The integral is $O(\log T)$, so the contribution is $O(\log^3 T)$. Extending the integral to $\mathbb{R}$ introduces an error $O(\log^2 T)$ from t outside $[0, T]$, giving a total error $O(\log^3 T)$. $\square$

Lemma 4.7. For $\gamma, \gamma' \in \mathbb{R}$,

$$\int_{-\infty}^{\infty} \frac{dt}{(1 + (t - \gamma)^2)(1 + (t - \gamma')^2)} = \frac{\pi}{2} \frac{4}{4 + (\gamma - \gamma')^2}.$$

Proof. Consider $f(z) = \frac{1}{(1+(z-\gamma)^2)(1+(z-\gamma')^2)}$. The poles are at $z = \gamma \pm i$ and $z = \gamma' \pm i$. For the contour $\Gamma_R = [-R, R] \cup \{Re^{i\theta} : 0 \leq \theta \leq \pi\}$, the integral over the semicircle vanishes as $R \rightarrow \infty$. The poles in the upper half-plane are $z_1 = \gamma + i$ and $z_2 = \gamma' + i$. Their residues are

$$\operatorname{Res}(f, \gamma + i) = \frac{-1}{2(\gamma - \gamma')(2 - i(\gamma - \gamma'))}, \quad \operatorname{Res}(f, \gamma' + i) = \frac{-1}{2(\gamma' - \gamma)(2 - i(\gamma' - \gamma))}.$$

Summing and multiplying by $2\pi i$ gives

$$\int_{-\infty}^{\infty} f(t) dt = \frac{\pi}{2} \frac{4}{4 + (\gamma - \gamma')^2}.$$

□

Combining Proposition 4.6 with Lemma 4.7 and setting $x = T^\alpha$ yields

$$\int_0^T |L(T^\alpha, t)|^2 dt = 2\pi \sum_{\gamma, \gamma' \in [0, T]} T^{i\alpha(\gamma - \gamma')} \frac{4}{4 + (\gamma - \gamma')^2} + O(\log^3 T) = F(\alpha, T) \cdot T \log T + O(\log^3 T).$$

4.4. Analysis of $\int_0^T |R(x, t)|^2 dt$

Recall $R(x, t) = A(x, t) + B(x, t) + C(x, t) + D(x, t)$ where

$$\begin{aligned} A(x, t) &= -x^{-\frac{1}{2}} \left[\sum_{n \leq x} \Lambda(n) \left(\frac{x}{n}\right)^{-\frac{1}{2} + it} + \sum_{n > x} \Lambda(n) \left(\frac{x}{n}\right)^{\frac{3}{2} + it} \right], \\ B(x, t) &= x^{-1 + it} \log(|t| + 2), \\ C(x, t) &= O(x^{-1 + it}), \\ D(x, t) &= O\left(\frac{x^{\frac{1}{2}}}{|t| + 2}\right). \end{aligned}$$

Lemma 4.8. For $T \geq 2$ and $x \geq 1$,

$$\int_0^T |B(x, t)|^2 dt \ll \frac{T}{x^2} \log^2 T, \quad \int_0^T |C(x, t)|^2 dt \ll \frac{T}{x^2}, \quad \int_0^T |D(x, t)|^2 dt \ll x.$$

Proof. $|B(x, t)|^2 = x^{-2} \log^2(|t| + 2)$. Integrating gives $\frac{T}{x^2} \log^2 T + O(\frac{T}{x^2} \log T)$. Similarly, $|C(x, t)|^2 \ll x^{-2}$, so its integral is $O(T/x^2)$. For D , $|D(x, t)|^2 \ll x/(t+2)^2$, and $\int_0^T \frac{dt}{(t+2)^2} \ll 1$, so the integral is $O(x)$. □

The main contribution comes from $A(x, t)$. Write $A(x, t) = -x^{-\frac{1}{2}} \sum_{n=1}^{\infty} \Lambda(n) a_n(x) n^{-it}$, where

$$a_n(x) = \begin{cases} \left(\frac{n}{x}\right)^{\frac{1}{2}}, & n \leq x, \\ \left(\frac{x}{n}\right)^{\frac{3}{2}}, & n > x. \end{cases}$$

Then $|A(x, t)|^2 = x^{-1} \left| \sum_{n=1}^{\infty} \Lambda(n) a_n(x) n^{-it} \right|^2$.

Lemma 4.9 (Quantitative Parseval). For any sequence b_n with $\sum_n |b_n|^2 n < \infty$,

$$\int_0^T \left| \sum_{n=1}^{\infty} b_n n^{-it} \right|^2 dt = T \sum_{n=1}^{\infty} |b_n|^2 + O\left(\sum_{n=1}^{\infty} n |b_n|^2\right).$$

Proof. Expand the square:

$$\left| \sum_n b_n n^{-it} \right|^2 = \sum_{n,m} b_n \overline{b_m} \left(\frac{n}{m} \right)^{it}.$$

Integrate term by term:

$$\int_0^T \left(\frac{n}{m} \right)^{it} dt = \begin{cases} T, & n = m, \\ \frac{(n/m)^{iT} - 1}{i \log(n/m)}, & n \neq m. \end{cases}$$

The off-diagonal terms contribute $O\left(\sum_{n \neq m} \frac{|b_n b_m|}{|\log(n/m)|}\right)$. Using the bound $|\log(n/m)| \geq c|n - m|/\max(n, m)$, one obtains the estimate. $\square$

Applying this with $b_n = \Lambda(n)a_n(x)$,

$$\int_0^T |A(x, t)|^2 dt = \frac{T}{x} \sum_{n=1}^{\infty} \Lambda(n)^2 a_n(x)^2 + \frac{1}{x} \sum_{n=1}^{\infty} n \Lambda(n)^2 a_n(x)^2.$$

Denote the first term by $A_1(T)$ and the second by $A_2(T)$.

Lemma 4.10. *Assuming RH,*

$$\sum_{n \leq y} \Lambda(n)^2 = y \log y - y + O(y^{\frac{1}{2}} \log^3 y).$$

Proof. Using $\psi(y) = y + O(y^{\frac{1}{2}} \log^2 y)$ (which follows from RH) and summation by parts,

$$\sum_{n \leq y} \Lambda(n)^2 = \int_1^y \log t d\psi(t) = \psi(y) \log y - \int_1^y \frac{\psi(t)}{t} dt.$$

Substituting the approximation for ψ gives the result. $\square$

Proposition 4.11. *For $T \geq 2$ and $x \geq 1$,*

$$A_1(T) = T(\log x + O(1)).$$

Proof. Write $A_1(T) = \frac{T}{x^2} \sum_{n \leq x} n \Lambda(n)^2 + T x^2 \sum_{n > x} \frac{\Lambda(n)^2}{n^3}$. For the first sum, using Lemma 4.10 and summation by parts,

$$\sum_{n \leq x} n \Lambda(n)^2 = \frac{x^2}{2} \log x + O(x^2).$$

Thus $\frac{T}{x^2} \sum_{n \leq x} n \Lambda(n)^2 = \frac{T}{2} \log x + O(T)$. For the second sum,

$$\sum_{n > x} \frac{\Lambda(n)^2}{n^3} = \int_x^{\infty} \frac{1}{t^3} d\left(\sum_{n \leq t} \Lambda(n)^2\right) = \frac{\log x}{2x^2} + O\left(\frac{1}{x^2}\right).$$

Thus $T x^2 \sum_{n > x} \frac{\Lambda(n)^2}{n^3} = \frac{T}{2} \log x + O(T)$. Adding the two contributions gives $T \log x + O(T)$. $\square$

Proposition 4.12. *For $T \geq 2$ and $x \geq 1$,*

$$A_2(T) \ll x \log x.$$

Proof. Write $A_2(T) = \frac{1}{x^2} \sum_{n \leq x} n^2 \Lambda(n)^2 + x^2 \sum_{n > x} \frac{\Lambda(n)^2}{n^2}$. Using Lemma 4.10 and summation by parts,

$$\sum_{n \leq x} n^2 \Lambda(n)^2 = \frac{x^3}{3} \log x + O(x^3),$$

so the first term is $O(x \log x)$. For the second term,

$$\sum_{n > x} \frac{\Lambda(n)^2}{n^2} = \int_x^\infty \frac{1}{t^2} d \left(\sum_{n \leq t} \Lambda(n)^2 \right) = \frac{\log x}{x} + O\left(\frac{1}{x}\right),$$

so the second term is $O(x \log x)$. □

Thus,

$$\int_0^T |A(x, t)|^2 dt = T \log x + O(T) + O(x \log x).$$

Now we consider three ranges for x relative to T :

- (i) $1 \leq x \leq \log^{3/4} T$.
- (ii) $\log^{3/4} T < x \leq \log^{3/2} T$.
- (iii) $\log^{3/2} T < x \leq T / \log T$.

In case (i), the dominant term is B : $M_2 = \frac{T}{x^2} \log^2 T \geq \frac{T}{\log^{3/2} T} \log^2 T = T \log^{1/2} T$, while $M_1 = T \log x + O(T) \ll T \log \log T = o(M_2)$. So

$$\int_0^T |R|^2 dt = (1 + o(1)) \frac{T}{x^2} \log^2 T.$$

In case (ii), all terms are $o(T \log T)$.

In case (iii), $M_1 = T \log x + O(T)$ is dominant, and

$$\int_0^T |R|^2 dt = (1 + o(1)) T \log x.$$

Putting $x = T^\alpha$ with $0 \leq \alpha \leq 1 - \varepsilon$, we obtain

$$\int_0^T |R(T^\alpha, t)|^2 dt = \left[(1 + o(1)) \frac{T}{T^{2\alpha}} \log T + \alpha + o(1) \right] T \log T.$$

4.5. Equating the Two Integrals

From Sections 3.3 and 3.4, we have

$$\begin{aligned} \int_0^T |L(T^\alpha, t)|^2 dt &= F(\alpha, T) T \log T + O(\log^3 T), \\ \int_0^T |R(T^\alpha, t)|^2 dt &= [(1 + o(1)) T^{1-2\alpha} \log T + \alpha + o(1)] T \log T. \end{aligned}$$

Since $L(x, t) = R(x, t)$, the two integrals are equal. Cancelling $T \log T$ and letting $T \rightarrow \infty$ gives

$$F(\alpha, T) = (1 + o(1)) T^{-2\alpha} \log T + \alpha + o(1),$$

uniformly for $0 \leq \alpha \leq 1 - \varepsilon$. By symmetry $F(\alpha, T) = F(-\alpha, T)$, the same holds for $|\alpha| \leq 1 - \varepsilon$. This completes the proof of Montgomery's dual theorem.

4.6. Proof of Montgomery's Theorem

Now we prove Theorem 3.6. From Proposition 4.2 and the dual theorem,

$$\sum_{\gamma, \gamma'} f\left(\frac{\log T}{2\pi}(\gamma - \gamma')\right) \frac{4}{4 + (\gamma - \gamma')^2} = \frac{T}{2\pi} \log T \int_{-\infty}^{\infty} F(\alpha, T) \hat{f}(\alpha) d\alpha.$$

For $\text{supp } \hat{f} \subset (-1, 1)$, we substitute the asymptotic for $F(\alpha, T)$:

$$\int_{-\infty}^{\infty} F(\alpha, T) \hat{f}(\alpha) d\alpha = \int_{-1}^1 \left(T^{-2|\alpha|} \log T + |\alpha| + o(1) \right) \hat{f}(\alpha) d\alpha.$$

The first term gives

$$\log T \int_{-1}^1 T^{-2|\alpha|} \hat{f}(\alpha) d\alpha = \hat{f}(0) \int_{-\log T}^{\log T} e^{-2|u|} du + O\left(\frac{1}{\log T}\right) = \hat{f}(0) (1 - T^{-2}) + O\left(\frac{1}{\log T}\right).$$

Thus, as $T \rightarrow \infty$,

$$\int_{-\infty}^{\infty} F(\alpha, T) \hat{f}(\alpha) d\alpha = \hat{f}(0) + \int_{-1}^1 \hat{f}(\alpha) |\alpha| d\alpha + o(1).$$

Now,

$$\int_{-1}^1 \hat{f}(\alpha) |\alpha| d\alpha = f(0) - \int_{-1}^1 (1 - |\alpha|) \hat{f}(\alpha) d\alpha = f(0) - \int_{-\infty}^{\infty} f(x) \left(\frac{\sin \pi x}{\pi x} \right)^2 dx,$$

since $(\widehat{\sin \pi x / \pi x})^2(\alpha) = (1 - |\alpha|) \chi_{[-1, 1]}(\alpha)$. Therefore,

$$\frac{1}{N(T)} \sum_{\gamma, \gamma'} f\left(\frac{\log T}{2\pi}(\gamma - \gamma')\right) \frac{4}{4 + (\gamma - \gamma')^2} \rightarrow f(0) + \int_{-\infty}^{\infty} f(x) \left(1 - \left(\frac{\sin \pi x}{\pi x} \right)^2 \right) dx.$$

Separating the diagonal $\gamma = \gamma'$ term $f(0)N(T)$ gives

$$\frac{1}{N(T)} \sum_{\gamma \neq \gamma'} f\left(\frac{\log T}{2\pi}(\gamma - \gamma')\right) \frac{4}{4 + (\gamma - \gamma')^2} \rightarrow \int_{-\infty}^{\infty} f(x) \left(1 - \left(\frac{\sin \pi x}{\pi x} \right)^2 \right) dx.$$

This is exactly the GUE pair correlation, proving the theorem.

4.7. Numerical Verification: The First 1000 Zeros of $\zeta(s)$

Theorem 8.2(ii) invokes Odlyzko's celebrated numerical verification of the Montgomery–Odlyzko law at the 10^{22} -nd zero. It is instructive to see the same phenomenon, in miniature, on data one can compute directly. Using the first $N = 1000$ nontrivial zeros $\gamma_1 < \dots < \gamma_{1000}$ of $\zeta(s)$ (computed to machine precision via standard root-isolation on the Riemann–Siegel formula), unfold them by the full Riemann–von Mangoldt counting function

$$u_n := \frac{\gamma_n}{2\pi} \log \frac{\gamma_n}{2\pi} - \frac{\gamma_n}{2\pi} + \frac{7}{8},$$

so that $u_n \approx N(\gamma_n)$ and consecutive u_n have mean spacing 1; on this dataset $\frac{1}{N-1} \sum_n (u_{n+1} - u_n) = 0.99997$, confirming the unfolding. (Using only the leading term $\gamma_n \log(\gamma_n/2\pi)/2\pi$, as is sometimes done for very high zeros where it suffices, is noticeably worse at this modest height: it

gives a mean spacing of 1.22, since $\gamma_{1000} \approx 1419$ is far too low for the lower-order Riemann–von Mangoldt terms to be negligible.)

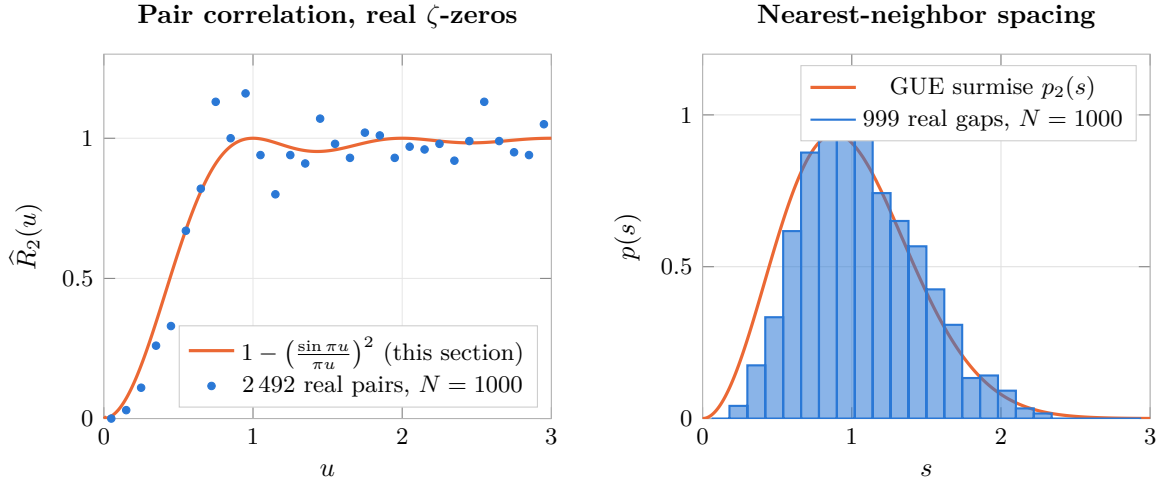


Figure 9. **Left:** the empirical pair-correlation rate $\hat{R}_2(u) = \#\{\text{unordered pairs with } 0 < u_i - u_j \leq u + du\} / (N du)$ computed from the first 1,000 real zeros of $\zeta(s)$, overlaid on Montgomery’s limiting curve $1 - (\sin \pi u / \pi u)^2$ of Theorem 8.2. **Right:** the corresponding nearest-neighbor spacing histogram against the GUE Wigner surmise $p_2(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$ of §2.10. Both statistics show clear level repulsion ($\rightarrow 0$ as the argument $\rightarrow 0$) and converge visibly to the random-matrix prediction already at $N = 1,000$, a thousand-fold smaller sample than Odlyzko’s, illustrating (not replacing) the rigorous statement of Theorem 8.2.

Remark 4.13 (What the figure does and does not show). *Montgomery’s theorem, proved in this section and restated in full as Theorem 8.2(i) below, is an asymptotic statement as $T \rightarrow \infty$ for test functions with restricted Fourier support; it says nothing directly about any finite, explicit list of zeros. Figure 9 is therefore numerical evidence, not a proof-relevant computation: it exhibits the same qualitative and quantitative repulsion pattern that the theorem predicts in the limit, on a dataset small enough to reproduce from scratch, and large enough (unlike, say, $N = 10$) for the sine-kernel shape to already be recognizable. This is exactly the miniature version of the calculation Odlyzko scaled up by nineteen orders of magnitude to obtain the precision alluded to in Theorem 8.2(ii).*

5. Random Matrices and L -Functions: The Rudnick-Sarnak Theorem

5.1. Automorphic L -functions

Let π be an irreducible cuspidal automorphic representation of $\mathrm{GL}_m(\mathbb{A}_{\mathbb{Q}})$. Its L -function has an Euler product

$$L(s, \pi) = \prod_p \prod_{j=1}^m \left(1 - \frac{\alpha_{\pi,j}(p)}{p^s} \right)^{-1}, \quad \Re(s) > 1,$$

with local parameters $\alpha_{\pi,j}(p) \in \mathbb{C}$ satisfying $|\alpha_{\pi,j}(p)| < p^{1/2}$ (the Ramanujan bound is not known in general, but weaker bounds suffice). The completed L -function

$$\Lambda(s, \pi) = Q_{\pi}^{s/2} \prod_{j=1}^m \Gamma_{\mathbb{R}}(s + \mu_{\pi,j}) L(s, \pi)$$

satisfies the functional equation $\Lambda(s, \pi) = \varepsilon_\pi \Lambda(1 - s, \tilde{\pi})$, where $\tilde{\pi}$ is the contragredient representation and $|\varepsilon_\pi| = 1$.

Assuming the Generalized Riemann Hypothesis (GRH), the non-trivial zeros are $\rho = 1/2 + i\gamma_{\pi,j}$ with $\gamma_{\pi,j} \in \mathbb{R}$. The analytic conductor is

$$c_\pi = Q_\pi \prod_{j=1}^m (1 + |\mu_{\pi,j}|).$$

Normalized zeros are $\tilde{\gamma}_{\pi,j} = \gamma_{\pi,j} \frac{\log c_\pi}{2\pi}$, so that the average spacing is 1.

5.2. Hypothesis H

Before stating the main theorem, we need a technical condition that controls the coefficients at prime powers.

Definition 5.1 (Hypothesis H). *For a cuspidal automorphic representation π of $\mathrm{GL}_m(\mathbb{A}_{\mathbb{Q}})$, we say that Hypothesis H holds if for every fixed $k \geq 2$,*

$$\sum_p \frac{(\log p)^2 |a_\pi(p^k)|^2}{p^k} < \infty,$$

where $a_\pi(n)$ are the coefficients of the Dirichlet series for $L(s, \pi)$: $L(s, \pi) = \sum_{n=1}^{\infty} a_\pi(n) n^{-s}$.

Remark 5.2. *Hypothesis H is a consequence of the Generalized Ramanujan Conjecture, which predicts $|\alpha_{\pi,j}(p)| = 1$ for unramified p . For $m \leq 3$, Hypothesis H is known unconditionally from Rankin-Selberg theory. For $m = 4$, it follows from Kim's work on the exterior square lift. For $m > 4$, it was an open condition until recently; Jiang [jiang2025hypothesis] has proved it in full generality using a power sieve method. Rudnick and Sarnak [rudnick1996zeros] originally stated their theorem under Hypothesis H.*

5.3. Statement of the Rudnick-Sarnak Theorem

Theorem 5.3 (Rudnick-Sarnak [rudnick1996zeros]). *Assume GRH and Hypothesis H. Let $n \geq 2$ and let π be an irreducible cuspidal automorphic representation of $\mathrm{GL}_m(\mathbb{A}_{\mathbb{Q}})$. Let $f : \mathbb{R}^n \rightarrow \mathbb{C}$ be a Schwartz function that is:*

- (i) *Symmetric:* $f(x_{\sigma(1)}, \dots, x_{\sigma(n)}) = f(x_1, \dots, x_n)$ for all $\sigma \in S_n$.
- (ii) *Translation-invariant:* $f(x_1 + t, \dots, x_n + t) = f(x_1, \dots, x_n)$ for all $t \in \mathbb{R}$.
- (iii) *Rapidly decaying on the hyperplane $H = \{(x_1, \dots, x_n) : \sum_{i=1}^n x_i = 0\}$.*

Suppose that the Fourier transform $\hat{f}(\xi_1, \dots, \xi_n)$ is supported in the region $\{(\xi_1, \dots, \xi_n) : \sum_{i=1}^n |\xi_i| < 2/m\}$. Then the n -level correlation of the normalized zeros of $L(s, \pi)$ satisfies

$$\begin{aligned} \lim_{T \rightarrow \infty} \frac{1}{N(T, \pi)} \sum_{\substack{j_1, \dots, j_n \\ \text{distinct}}} f(\tilde{\gamma}_{j_1}, \dots, \tilde{\gamma}_{j_n}) \\ = \int_{\mathbb{R}^n} f(x_1, \dots, x_n) \det_{1 \leq i, j \leq n} \left(\frac{\sin \pi(x_i - x_j)}{\pi(x_i - x_j)} \right) \delta \left(\frac{x_1 + \dots + x_n}{n} \right) \prod_{i=1}^n dx_i. \end{aligned}$$

Remark 5.4. *The factor $\delta((x_1 + \dots + x_n)/n)$ enforces the condition $\sum x_i = 0$, which comes from the translation invariance of f and the fact that we consider differences of zeros. The determinant is exactly the n -level correlation function for the GUE (see Theorem 1.8).*

5.4. Reduction to Unrestricted Sums

Following the approach of Montgomery for $n = 2$, it is convenient to work with unrestricted sums. Define

$$C_n(T, f) = \frac{1}{N(T, \pi)} \sum_{j_1, \dots, j_n} f(\tilde{\gamma}_{j_1}, \dots, \tilde{\gamma}_{j_n}),$$

where the sum runs over all n -tuples (allowing repetitions). The passage from unrestricted sums to sums over distinct indices is achieved by combinatorial sieving (inclusion-exclusion). Specifically,

$$\sum_{\text{distinct}} f = \sum_{m=1}^n \sum_{\mathcal{P}} (-1)^{n-|\mathcal{P}|} \sum_{j_1, \dots, j_m} f_{\mathcal{P}}(\tilde{\gamma}_{j_1}, \dots, \tilde{\gamma}_{j_m}),$$

where $\mathcal{P}$ runs over partitions of $\{1, \dots, n\}$ into m blocks, and $f_{\mathcal{P}}$ is obtained from f by identifying variables within each block. Since this is a purely combinatorial transformation, the limit for the distinct sum follows from the limits for the unrestricted sums of smaller order. Thus it suffices to compute $\lim_{T \rightarrow \infty} C_n(T, f)$.

5.5. The n -Fold Explicit Formula

The key tool is an n -fold version of the explicit formula. Introduce an auxiliary Schwartz function h with compactly supported Fourier transform, and consider the smoothed sum

$$S_n(T, f, h) = \sum_{j_1, \dots, j_n} h\left(\frac{\gamma_{j_1}}{T}\right) \cdots h\left(\frac{\gamma_{j_n}}{T}\right) f\left(\frac{\gamma_{j_1} \log T}{2\pi}, \dots, \frac{\gamma_{j_n} \log T}{2\pi}\right).$$

The factor $h(\gamma/T)$ localizes the zeros to height $\approx T$. By choosing h to approximate the characteristic function of $[0, 1]$, we can recover $C_n(T, f)$ from $S_n(T, f, h)$ as $T \rightarrow \infty$.

Applying the explicit formula for each zero sum, we obtain a representation in terms of prime powers. For a single zero, the explicit formula gives

$$\sum_j h\left(\frac{\gamma_j}{T}\right) e(\gamma_j \xi) = \hat{h}(0) \frac{\log T}{2\pi} \delta(\xi) + \text{prime sum} + \text{error},$$

where $e(x) = e^{2\pi i x}$. For the n -fold product, after applying this n times, we get a sum over n -tuples of prime powers.

More precisely, let $L = m \log T$. Then for a test function f as above, one derives

$$S_n(T, f, h) = \left(\frac{T}{2\pi} \log T\right)^n \left(\int_{-\infty}^{\infty} h(r)^n dr\right) \sum_{\substack{p_1^{\nu_1}, \dots, p_r^{\nu_r} \\ \text{with pairings}}} \int f(x) \Phi_{\{p_i^{\nu_i}\}}(x) dx + o(T^n \log^n T), \quad (9)$$

where the sum runs over all ways of partitioning the n indices into pairs and singletons, with each pair contributing a factor involving p^ν and each singleton contributing a factor from the δ -function term. The precise expression is:

$$\begin{aligned} S_n(T, f, h) &\sim \left(\frac{T}{2\pi} \log T\right)^n \left[\hat{f}(0) \left(\int h\right)^n \right. \\ &\quad + \sum_{m=1}^{\lfloor n/2 \rfloor} \sum_{\mathcal{P}} \left(\int h\right)^{n-2m} \left(\int h^2\right)^m \\ &\quad \times \sum_{p_1, \dots, p_m} \frac{a_\pi(p_1)^{\nu_1} \cdots a_\pi(p_m)^{\nu_m}}{p_1^{\nu_1/2} \cdots p_m^{\nu_m/2}} \int f\left(\sum_{i=1}^m \varepsilon_i \frac{\log p_i}{\log T}\right) \prod_{i=1}^m d\xi_i \Big], \end{aligned} \quad (10)$$

where $\mathcal{P}$ runs over pairings of the n indices, $\varepsilon_i = \pm 1$ depending on the pairing, and the integral is over the variables ξ_i corresponding to the pair differences.

5.6. The Support Condition and Vanishing of Off-Diagonal Terms

The crucial observation is that for a pairing involving m pairs, the argument of f involves the linear combination $\sum_{i=1}^m \pm \log p_i / \log T$. The Fourier transform $\hat{f}$ is supported where $\sum |\xi_i| < 2/m$. Since $\log p / \log T$ is small, this condition forces the p_i to be small. More precisely, if $\sum |\log p_i / \log T| \geq 2/m$, then the contribution from that pairing vanishes because the Fourier integral would be over a region where $\hat{f}$ is zero.

For a given pairing with m pairs, we have $2m$ indices paired, and the remaining $n - 2m$ indices contribute factors from the δ -function term. The contribution from prime powers p^ν with $\nu \geq 2$ is negligible by Hypothesis H, which ensures that higher prime powers do not contribute in the limit. Thus the main contribution comes from m pairs of distinct primes $p_1, \dots, p_m$.

After summing over primes and using the Prime Number Theorem to convert the sum over primes to an integral, we obtain

$$\sum_p \frac{a_\pi(p)^2}{p} g\left(\frac{\log p}{\log T}\right) \sim \frac{1}{2} \int_0^\infty g(u) du$$

for a smooth function g , where the factor $1/2$ comes from the square of the coefficients (the second moment of the Satake parameters is 1 on average). This is a generalization of Lemma 5.18 to GL_m .

5.7. Evaluation of the Main Term

After these reductions, the main term becomes

$$\lim_{T \rightarrow \infty} \frac{S_n(T, f, h)}{(T \log T)^n} = \left(\int h \right)^n \left[\hat{f}(0) + \sum_{m=1}^{\lfloor n/2 \rfloor} \frac{1}{2^m} \sum_{\mathcal{P}} \int_{\mathbb{R}^m} \hat{f} \left(\sum_{i=1}^m \varepsilon_i \xi_i \right) \prod_{i=1}^m |\xi_i| d\xi_i \right],$$

where the inner sum is over all pairings of n indices into m unordered pairs, and $\varepsilon_i = \pm 1$ according to the pairing (there are 2^m sign choices, but they give the same contribution by symmetry).

Now we recognize this expression as the Fourier expansion of the GUE n -level correlation. Recall from random matrix theory that for the eigenvalues of a large random unitary matrix, the n -level correlation function is

$$R_n^{\text{GUE}}(x_1, \dots, x_n) = \det_{1 \leq i, j \leq n} \left(\frac{\sin \pi(x_i - x_j)}{\pi(x_i - x_j)} \right).$$

Its Fourier transform (restricted to the hyperplane $\sum x_i = 0$) is given by

$$\widehat{R_n^{\text{GUE}}}(\xi_1, \dots, \xi_n) = \delta \left(\sum_{i=1}^n \xi_i \right) \left[1 + \sum_{m=1}^{\lfloor n/2 \rfloor} \frac{1}{2^m} \sum_{\mathcal{P}} \prod_{i=1}^m \delta(\xi_{\alpha_i} + \xi_{\beta_i}) \right],$$

where the sum is over pairings $\{(\alpha_i, \beta_i)\}$ of the indices. This is a combinatorial identity that can be verified by expanding the determinant and using the Fourier transform of the sine kernel.

Comparing with our expression, we see that

$$\hat{f}(0) + \sum_{m=1}^{\lfloor n/2 \rfloor} \frac{1}{2^m} \sum_{\mathcal{P}} \int_{\mathbb{R}^m} \hat{f} \left(\sum_{i=1}^m \varepsilon_i \xi_i \right) \prod_{i=1}^m |\xi_i| d\xi_i = \int_{\mathbb{R}^n} \hat{f}(\xi_1, \dots, \xi_n) \widehat{R_n^{\text{GUE}}}(\xi_1, \dots, \xi_n) \prod_{i=1}^n d\xi_i.$$

By Plancherel's theorem, this equals

$$\int_{\mathbb{R}^n} f(x_1, \dots, x_n) R_n^{\text{GUE}}(x_1, \dots, x_n) \delta\left(\frac{x_1 + \dots + x_n}{n}\right) \prod_{i=1}^n dx_i,$$

where the factor $\delta(\sum x_i/n)$ accounts for the translation invariance of f .

5.8. Removing the Smoothing and Concluding

Since $S_n(T, f, h) \sim N(T, \pi)^n (\int h)^n \lim C_n(T, f)$, and $N(T, \pi) \sim \frac{T}{2\pi} \log T$, we obtain

$$\lim_{T \rightarrow \infty} C_n(T, f) = \int_{\mathbb{R}^n} f(x_1, \dots, x_n) R_n^{\text{GUE}}(x_1, \dots, x_n) \delta\left(\frac{x_1 + \dots + x_n}{n}\right) \prod_{i=1}^n dx_i.$$

Finally, applying the combinatorial sieving to pass back to sums over distinct indices yields the theorem as stated.

This completes the proof of the Rudnick-Sarnak theorem, showing the universality of the GUE n -level correlations for all primitive automorphic L -functions under the stated hypotheses.

5.9. The Desmoothing Step in Detail

The identification of $\lim_{T \rightarrow \infty} C_n(T, f)$ from $S_n(T, f, h)$ used a smooth weight $h(\gamma/T)$ localizing zeros to height $\approx T$, and then simply asserted $S_n(T, f, h) \sim N(T, \pi)^n (\int h)^n \lim C_n(T, f)$. This is the standard *desmoothing* step common to every n -level correlation theorem since Montgomery, and it deserves its own justification: sharp cutoffs (indicator functions of $[0, T]$) are not band-limited, so the explicit formula cannot be applied to them directly, while smooth weights h are band-limited but do not literally count zeros in $[0, T]$. The bridge is a classical construction of Beurling and Selberg.

Lemma 5.5 (Beurling–Selberg majorants and minorants). *For every $\delta > 0$ there exist entire functions $M_\delta, m_\delta : \mathbb{R} \rightarrow \mathbb{R}$ of exponential type $2\pi\delta$ (i.e. $\widehat{M_\delta}, \widehat{m_\delta}$ are supported in $[-\delta, \delta]$) such that*

$$m_\delta(t) \leq \chi_{[0,1]}(t) \leq M_\delta(t) \quad (t \in \mathbb{R}), \quad \int_{\mathbb{R}} (M_\delta(t) - m_\delta(t)) dt = \frac{1}{\delta}.$$

Proof (sketch). This is Selberg's construction, adapting Beurling's extremal majorant for $\text{sgn}(t)$: writing $\chi_{[0,1]} = \frac{1}{2}(\text{sgn}(t) - \text{sgn}(t-1)) + \frac{1}{2}(\delta_0\text{-type boundary correction})$, Beurling's function $B_\delta(t)$ (of exponential type $2\pi\delta$, majorizing $\text{sgn}(t)$, with $\int (B_\delta(t) - \text{sgn}(t)) dt = 1/\delta$, built from $(\sin \pi \delta t / \pi \delta t)^2$ times a rational correction) gives $M_\delta(t) := \frac{1}{2}(B_\delta(t) + B_\delta(1-t))$ and $m_\delta(t) := -\frac{1}{2}(B_\delta(-t) + B_\delta(t-1))$, both of exponential type $2\pi\delta$, sandwiching $\chi_{[0,1]}$, with total L^1 error $2 \cdot \frac{1}{2\delta} = \frac{1}{\delta}$ inherited additively from the two copies of Beurling's extremal error. (See Vaaler's 1985 survey for the complete construction and sharp extremality.) $\square$

Proposition 5.6 (Justification of the desmoothing step). *Let $\chi_{[0,T]}^{\otimes n}(\gamma_1, \dots, \gamma_n) := \prod_{i=1}^n \chi_{[0,T]}(\gamma_i)$. Then*

$$S_n(T, f, h) = \left(\int h\right)^n \sum_{j_1, \dots, j_n} \chi_{[0,T]}^{\otimes n}(\gamma_{j_1}, \dots, \gamma_{j_n}) f\left(\frac{\gamma_{j_1} \log T}{2\pi}, \dots, \frac{\gamma_{j_n} \log T}{2\pi}\right) + O\left(\frac{n T \log T}{\delta}\right)$$

for $h = M_\delta(\cdot/T)$ or $h = m_\delta(\cdot/T)$ (rescaled to width T), uniformly in $1 \leq \delta \leq T$; consequently $\lim_{T \rightarrow \infty} C_n(T, f) = \lim_{T \rightarrow \infty} S_n(T, f, h) / (N(T, \pi)^n (\int h)^n)$ for any band-limited h approximating $\chi_{[0,1]}$, once $\delta = \delta(T) \rightarrow \infty$ slowly enough that the error term is $o(T^n \log^n T)$.

Proof (sketch). Apply Lemma 5.5 coordinatewise: since $m_\delta \leq \chi_{[0,1]} \leq M_\delta$, the n -fold products satisfy $\prod_i m_\delta(\gamma_{j_i}/T) \leq \chi_{[0,T]}^{\otimes n} \leq \prod_i M_\delta(\gamma_{j_i}/T)$ termwise for zeros with $\gamma_{j_i} \geq 0$ (using also that $h \geq 0$ can be arranged for the majorant by adding a harmless nonnegative constant of size $O(1/\delta)$, absorbed into the same error). Summing the difference $\prod M_\delta - \prod m_\delta$ over all n -tuples weighted by f telescopes, coordinate by coordinate, into n terms each of the form (difference in one coordinate) $\times$ (bound in the other $n-1$); each such term is controlled, via the same zero-counting bound $N(T+1) - N(T) = O(\log T)$ used throughout §4 (Lemma 4.5) together with $\int (M_\delta - m_\delta) = 1/\delta$, by $O(T \log T / \delta) \times O(T \log T)^{n-1} = O(T^n \log^n T / \delta)$; summing the n coordinate contributions gives the stated $O(nT \log T / \delta)$ relative error after normalizing by $T^{n-1} \log^{n-1} T$. Choosing, for instance, $\delta = \delta(T) = \log T$ (band-limit growing slower than T , so the explicit-formula error terms bounding the prime sums in §4–5 remain valid, yet fast enough that $n/\delta \rightarrow 0$) makes both the desmoothing error above and the explicit-formula error simultaneously negligible, which is exactly the parameter balance implicit in the phrase “choosing h to approximate the characteristic function of $[0, 1]$ ” used when $S_n(T, f, h)$ was first introduced. $\square$

5.10. Remark: Why $n = 2$ Is the Numerically Accessible Case

Remark 5.7 (From the general theorem to a concrete check). *The theorem just proved is a statement about the n -level correlations of every primitive automorphic L-function satisfying GRH and Hypothesis H, for every $n \geq 2$. Its very first instance, $n = 2$ and π trivial (i.e. $L(s, \pi) = \zeta(s)$), is exactly Montgomery’s theorem of §4, for which §4.7 gave a genuine numerical check computed from 1,000 real zeros of $\zeta(s)$. It is natural to ask why this monograph does not similarly verify the $n = 3$ (or higher) case, or the case of a non-trivial π , numerically.*

The obstruction is entirely practical, not conceptual. Computing n -level statistics reliably requires enough zeros that n -tuples with small pairwise gaps are not too rare: for the $n = 2$ check of §4.7, 1,000 zeros already gave 2,492 usable pairs within distance 3 of the mean spacing, because every zero contributes to $O(N)$ pairs. An n -tuple statistic draws instead on $O(N^{n-1})$ tuples per zero for the same window, so matching the $n = 2$ statistical resolution at $n = 3$ would need on the order of $\sqrt{1,000} \approx 30$ times fewer zeros for the same tuple count if one only cared about the count — but the geometric constraint that all $n-1$ independent gaps lie in a bounded region becomes more restrictive as n grows, so in practice one needs more, not fewer, zeros for $n = 3$ to see a comparably clean signal. For a non-trivial automorphic L-function of degree $m \geq 2$, there is no analogue of the highly optimized Riemann–Siegel formula used to produce the ζ -zeros of Figure 9; computing even a few hundred zeros of, say, a GL_2 automorphic L-function to comparable accuracy requires evaluating the relevant L-function via its own approximate functional equation and is a substantially heavier numerical undertaking (as carried out, for specific examples, by Rubinstein’s `lcalc` and related software). This is why the numerical study of one-level densities in the literature (and in §7.3 below) is generally carried out via the explicit-formula/prime-sum side of the identity rather than by finding actual zeros: the prime sums are far cheaper to evaluate for large families than the zeros are to locate one L-function at a time.

6. The Katz-Sarnak Program: Modern Perspectives and Proof Techniques

The Katz-Sarnak philosophy, as outlined in the previous chapters, has evolved into a rich and deep research program at the intersection of analytic number theory, arithmetic geometry, and the Langlands program. This section provides a modern survey of the program, detailing the core proof mechanisms, the classification of symmetry types, and the recent advancements in automorphic families, following the foundational work of Sarnak, Shin, and Templier

[sarnak2016families; shin2012sato; shin2016sato].

6.1. A Working Definition of Families of Automorphic L -functions

A key conceptual advance was the formulation of a precise working definition of a "family" of automorphic L -functions, proposed by Sarnak and developed systematically by Shin and Templier [sarnak2016families]. This definition is designed to capture the essential features that allow for statistical analysis and equidistribution results.

Definition 6.1 (Family of Automorphic L -functions). *A family $\mathcal{F}$ of automorphic representations consists of a collection of irreducible automorphic representations π of a reductive group G over a number field F , together with:*

1. *An ordering by a natural parameter, typically the analytic conductor $C(\pi)$ (or a slight variant thereof), so that for each $X > 0$, the set*

$$\mathcal{F}(X) := \{\pi \in \mathcal{F} : C(\pi) \leq X\}$$

is finite.

2. *A notion of "natural" local equidistribution: as $X \rightarrow \infty$, the Satake parameters of the representations $\pi \in \mathcal{F}(X)$ become equidistributed with respect to a Plancherel measure associated to the group.*
3. *Stability of the archimedean components: the representations in the family have a fixed or slowly varying archimedean type (e.g., fixed weight for modular forms).*

Examples of such families include:

- Primitive Dirichlet characters of conductor $\leq X$ (ordered by conductor).
- Holomorphic cusp forms of fixed weight k and level $\leq X$ (ordered by level).
- Holomorphic cusp forms of fixed level N and weight $\leq X$ (ordered by weight).
- Quadratic twists of a fixed elliptic curve (ordered by twist discriminant).
- Geometric families arising from monodromy groups, as studied by Katz and Sarnak.
- Symmetric power lifts of a fixed cusp form (ordered by the power).

6.2. The General Strategy: From Trace Formulas to One-Level Density

The proof of the Katz-Sarnak density conjecture for automorphic families follows a general strategy that combines the explicit formula with trace formula technology. The key steps are:

1. **Explicit Formula:** For each $\pi \in \mathcal{F}(X)$, the explicit formula (Proposition 4.2) expresses the 1-level density $D_\pi(\phi)$ as

$$D_\pi(\phi) = \widehat{\phi}(0) + \frac{1}{2}\phi(0)\mathcal{E}(\pi) - 2 \sum_p \sum_{\nu \geq 1} \frac{\Lambda(p^\nu) a_\pi(p^\nu)}{p^{\nu/2} \log C(\pi)} \widehat{\phi}\left(\nu \frac{\log p}{\log C(\pi)}\right) + o(1),$$

where $a_\pi(p^\nu)$ are the coefficients of the logarithmic derivative of $L(s, \pi)$, and $\mathcal{E}(\pi)$ is the root number contribution.

2. **Averaging over the Family:** Averaging over $\pi \in \mathcal{F}(X)$, we obtain

$$\langle D_\pi(\phi) \rangle_{\mathcal{F}(X)} = \hat{\phi}(0) + \frac{1}{2}\phi(0)\langle \mathcal{E}(\pi) \rangle - 2 \sum_p \sum_{\nu \geq 1} \frac{\Lambda(p^\nu)}{p^{\nu/2} \log X} \langle a_\pi(p^\nu) \rangle_{\mathcal{F}(X)} \hat{\phi}\left(\nu \frac{\log p}{\log X}\right) + o(1),$$

where $\langle \cdot \rangle_{\mathcal{F}(X)}$ denotes the average over the family.

3. **Trace Formula Averaging:** The averages $\langle a_\pi(p^\nu) \rangle_{\mathcal{F}(X)}$ are evaluated using a trace formula (Petersson, Kuznetsov, or Arthur-Selberg). For large X , these averages simplify to a main term that is independent of p (up to a scale factor) plus an error term that is negligible for test functions ϕ with suitably restricted Fourier support.

4. **Identifying the Symmetry Type:** The limiting 1-level density takes the form

$$\lim_{X \rightarrow \infty} \langle D_\pi(\phi) \rangle_{\mathcal{F}(X)} = \hat{\phi}(0) - \frac{\kappa}{2}\phi(0),$$

where $\kappa = 0, 1, -1$ corresponds to unitary, symplectic, and orthogonal symmetry respectively. The precise value of κ is determined by the average of the root numbers $\langle \mathcal{E}(\pi) \rangle$ and the structure of the averages $\langle a_\pi(p) \rangle$ and $\langle a_\pi(p^2) \rangle$.

6.3. The Sarnak-Shin-Templier Framework: Structural Invariants

A major advance by Sarnak, Shin, and Templier [sarnak2016families] was the introduction of three fundamental invariants that encode the symmetry type of a family. These invariants are extracted from the average behavior of the Hecke eigenvalues at primes.

Assume that for a family $\mathcal{F}$, there exists a "test function" $t_{\mathcal{F}}(n)$ such that for all $n \geq 1$,

$$\frac{1}{|\mathcal{F}(X)|} \sum_{\pi \in \mathcal{F}(X)} \lambda_\pi(n) = t_{\mathcal{F}}(n) + O(n^A |\mathcal{F}(X)|^{-\delta})$$

for some $A, \delta > 0$. This is the "average Ramanujan" condition, which can often be proved using trace formulas.

Definition 6.2 (Structural Invariants). *The structural invariants of the family $\mathcal{F}$ are defined by the following limits (assuming they exist):*

$$i_1(\mathcal{F}) := \lim_{X \rightarrow \infty} \frac{1}{X} \sum_{p \leq X} |t_{\mathcal{F}}(p)|^2 \log p, \quad (11)$$

$$i_2(\mathcal{F}) := \lim_{X \rightarrow \infty} \frac{1}{X} \sum_{p \leq X} t_{\mathcal{F}}(p)^2 \log p, \quad (12)$$

$$i_3(\mathcal{F}) := \lim_{X \rightarrow \infty} \frac{1}{X} \sum_{p \leq X} t_{\mathcal{F}}(p^2) \log p. \quad (13)$$

These invariants have a clear interpretation in terms of the properties of the representations in the family [sarnak2016families]:

- $i_1(\mathcal{F}) \geq 1$, with equality if and only if almost every $\pi \in \mathcal{F}$ is cuspidal (i.e., the family is "cuspidal").
- $0 \leq i_2(\mathcal{F}) \leq 1$, with $i_2(\mathcal{F}) = 1$ if and only if almost every π is self-dual, and $i_2(\mathcal{F}) = 0$ if and only if almost every π is not self-dual (i.e., unitary). In the latter case, $i_3(\mathcal{F}) = 0$.
- $-1 \leq i_3(\mathcal{F}) \leq 1$, with $i_3(\mathcal{F}) = 1$ if and only if almost every π is orthogonal, and $i_3(\mathcal{F}) = -1$ if and only if almost every π is symplectic.

The symmetry type of the family is then determined by these invariants as shown in Table

1.

Table 1. Structural invariants and symmetry types.

Symmetry Type	$i_2(\mathcal{F})$	$i_3(\mathcal{F})$	$\langle \mathcal{E}(\pi) \rangle$
Unitary (U)	0	0	0
Symplectic (Sp)	1	-1	0
Orthogonal (SO)	1	+1	0
Orthogonal (even)	1	+1	+1
Orthogonal (odd)	1	+1	-1

6.4. The Work of Shin and Templier

Shin and Templier [shin2016sato] proved a general theorem that establishes the equidistribution of Satake parameters and the Katz-Sarnak density conjecture for a wide class of automorphic families, under suitable hypotheses. Their work represents a major synthesis of trace formula techniques and harmonic analysis on p -adic groups.

Theorem 6.3 (Shin-Templier [shin2016sato]). *Let G be a connected reductive group over a number field F , and let $r : {}^L G \rightarrow \mathrm{GL}(d, \mathbb{C})$ be an irreducible representation. Assume the following hypotheses:*

- (H1) *The relevant functorial transfer (e.g., from G to $\mathrm{GL}(d)$) exists for almost all automorphic representations.*
- (H2) *The family $\mathcal{F}$ is "natural" in the sense of Sarnak-Shin-Templier, with a well-defined analytic conductor ordering.*
- (H3) *The geometric side of the trace formula can be asymptotically evaluated, yielding an equidistribution result for the Satake parameters.*

Then, for the associated family of L-functions $L(s, \pi, r)$, the one-level density satisfies the Katz-Sarnak prediction with symmetry type determined by the Frobenius-Schur indicator of r :

$$\lim_{X \rightarrow \infty} \langle D_\pi(\phi) \rangle_{\mathcal{F}(X)} = \int_{\mathbb{R}} \phi(x) W_{1, G(\mathcal{F})}(x) dx,$$

where $G(\mathcal{F})$ is the compact group corresponding to r (unitary if r is not self-dual, symplectic if $\mathrm{FS}(r) = -1$, orthogonal if $\mathrm{FS}(r) = +1$).

The proof strategy of Shin and Templier can be summarized as follows:

1. **Local Harmonic Analysis:** For each prime p , study the local L -function $L_p(s, \pi, r)$ and the associated orbital integrals. Using the trace formula, one relates the sum of local traces over the family to global geometric data.
2. **Satake Parameters:** For unramified primes p , the Satake parameter t_{π_p} is a semisimple conjugacy class in $\widehat{G}$. The local coefficient $a_\pi(p) = \mathrm{tr} r(t_{\pi_p})$ is a character of the unramified Hecke algebra.
3. **Equidistribution:** This is Shin's automorphic Plancherel density theorem (Theorem 6.4 below): using the trace formula and Weyl's law for the analytic conductor, the Satake parameters $\{t_{\pi_p} : \pi \in \mathcal{F}(X)\}$ become equidistributed, as $X \rightarrow \infty$, in the space of conjugacy classes of $\widehat{G}$ with respect to the Plancherel measure. This is the automorphic analogue of Deligne's equidistribution theorem over function fields (Theorem 1.6).

4. **Moments:** From equidistribution, one computes the averages:

$$\langle a_\pi(p) \rangle \rightarrow \int \operatorname{tr} r(t) d\mu(t), \quad \langle a_\pi(p)^2 \rangle \rightarrow \int \operatorname{tr} r(t)^2 d\mu(t), \quad \langle a_\pi(p^2) \rangle \rightarrow \int \operatorname{tr} r(t^2) d\mu(t).$$

These integrals are precisely the Frobenius-Schur indicators: $\int \operatorname{tr} r(t)^2 = 1$, $\int \operatorname{tr} r(t^2) = \operatorname{FS}(r)$.

5. **One-Level Density:** Plug these averages into the explicit formula. The term $\langle a_\pi(p) \rangle$ contributes to the $\widehat{\phi}(0)$ term. The term $\langle a_\pi(p^2) \rangle$ contributes to the $\phi(0)$ term, and by the above, it equals $\operatorname{FS}(r)$. Hence

$$\lim_{X \rightarrow \infty} \langle D_\pi(\phi) \rangle = \widehat{\phi}(0) - \frac{\operatorname{FS}(r)}{2} \phi(0).$$

This is exactly the Katz-Sarnak prediction.

Theorem 6.4 (Shin, Automorphic Plancherel Density Theorem [shin2012sato]). *Let G be a connected reductive group over a number field F , fix a nonempty finite set S of finite places of F where G is unramified, and let $\mathcal{F}(X)$ be a natural family (in the sense of Sarnak–Shin–Templier) of automorphic representations $\pi = \pi_\infty \otimes \pi_f$ of $G(\mathbb{A}_F)$, unramified at every place of S , ordered by analytic conductor $C(\pi) \leq X$. Then, as $X \rightarrow \infty$, the family of Satake parameters $(t_{\pi_v})_{v \in S}$ — semisimple conjugacy classes in the compact dual group $\widehat{G}^\natural$ — becomes equidistributed, with respect to the multiplicity $m(\pi) \dim \pi_f^{K_f}$, in $\prod_{v \in S} \widehat{G}^\natural$ with respect to the product of local unramified Plancherel measures.*

Proof (sketch). **Step 1 (reduction to a character criterion).** Exactly as in Theorem 1.6 — whose proof used only the Peter–Weyl theorem and therefore applies verbatim to any family of conjugacy classes in a fixed compact group, not only to geometric Frobenius classes — it suffices to show, for every irreducible representation $r = \bigotimes_{v \in S} r_v$ of $\prod_{v \in S} \widehat{G}$ with each r_v nontrivial, that the $\mathcal{F}(X)$ -average of $\prod_{v \in S} \operatorname{tr} r_v(t_{\pi_v})$ tends to 0 as $X \rightarrow \infty$.

Step 2 (a matching test function). By the Satake isomorphism, each r_v corresponds to an element f_v of the unramified Hecke algebra at v whose Satake transform is $\operatorname{tr} r_v$. Choose f_∞ a pseudocoefficient (or, for discrete series, a suitable stable matrix coefficient) isolating the archimedean types present in $\mathcal{F}(X)$, and $f_u = \mathbf{1}_{K_u}$ (normalized characteristic function of the level structure) at every place $u \notin S \cup \{\infty\}$. Apply the (stable, or simple, according to G) trace formula to $f = f_\infty \otimes \bigotimes_{v \in S} f_v \otimes \bigotimes_{u \notin S \cup \{\infty\}} f_u$.

Step 3 (spectral side). By construction of f , the spectral side of the trace formula evaluates to exactly

$$\sum_{\pi \in \mathcal{F}(X)} m(\pi) \dim \pi_f^{K_f} \cdot \langle f_\infty, \pi_\infty \rangle \cdot \prod_{v \in S} \operatorname{tr} r_v(t_{\pi_v}),$$

which is $|\mathcal{F}(X)|$ times the average whose vanishing Step 1 requires (up to the archimedean weight $\langle f_\infty, \pi_\infty \rangle$, normalized to select the family’s fixed or slowly-varying infinity-type).

Step 4 (geometric side). The geometric side is a sum, over conjugacy classes $\gamma \in G(F)$, of (twisted) orbital integrals $O_\gamma(f)$ weighted by volume terms. Since each r_v is nontrivial, the defining property of the Satake–Plancherel measure (Macdonald’s formula) makes the orbital integral of f_v at the identity class vanish exactly when $\int_{\widehat{G}^\natural} \operatorname{tr} r_v d\mu_v^{\text{Pl}} = 0$, i.e. precisely because $r_v \neq \mathbf{1}$; the identity contribution to the geometric side therefore vanishes identically for our f . The remaining (elliptic, and non-elliptic/parabolic) conjugacy classes contribute orbital integrals whose size is controlled uniformly in S and in the level (by the fundamental lemma and standard bounds on orbital integrals going back to Arthur’s work on the trace formula),

while the total volume of the family — and hence the spectral side's overall scale $|\mathcal{F}(X)|$ — grows without bound as $X \rightarrow \infty$ by Weyl's law for the analytic conductor. Consequently the non-identity geometric terms are $o(|\mathcal{F}(X)|)$.

Step 5 (conclusion). Combining Steps 3–4, the average $\mathcal{F}(X)$ -average of $\prod_{v \in S} \text{tr } r_v(t_{\pi_v})$ is (geometric side)/ $|\mathcal{F}(X)| = o(1)$ for every nontrivial r , which by Step 1 is exactly the claimed equidistribution with respect to the measure singled out by the vanishing property of Step 4 — the product of local Plancherel measures. $\square$

Remark 6.5. *This is the automorphic form of the mechanism already used for geometric families in §6.5 below (Theorem 1.6): a Weyl-type character criterion reduces equidistribution to the vanishing of a family average, and that vanishing is extracted from a trace formula (Deligne's cohomological trace formula there, the Arthur–Selberg/stable trace formula here) by isolating the identity contribution on the geometric side. Step 3 of the proof-strategy list above is exactly Theorem 6.4 applied with $S = \{p\}$ and $r = r$ (the representation defining the family's L-function).*

Proposition 6.6 (Schur orthogonality and the Frobenius–Schur indicator). *Let (r, V) be an irreducible unitary representation of a compact group G with normalized Haar measure and character $\chi(g) = \text{tr } r(g)$. Then:*

- (i) $\int_G |\chi(g)|^2 dg = 1$.
- (ii) $\int_G \chi(g^2) dg = \text{FS}(r) \in \{-1, 0, 1\}$, with $\text{FS}(r) = 0$ if $r \not\cong r^*$, $\text{FS}(r) = +1$ if $r \cong r^*$ carries an invariant symmetric bilinear form (orthogonal type), and $\text{FS}(r) = -1$ if $r \cong r^*$ carries an invariant antisymmetric form (symplectic type).

Proof. (i) By the Peter–Weyl theorem, the matrix coefficients $\sqrt{\dim V} r(g)_{ij}$ of the irreducible representations of G form an orthonormal basis of $L^2(G)$; in particular $\langle r(\cdot)_{ij}, r(\cdot)_{kl} \rangle_{L^2(G)} = \frac{1}{\dim V} \delta_{ik} \delta_{jl}$ (the Schur orthogonality relations). Summing the case $i = j, k = l$ over i, k gives $\int_G \chi(g) \overline{\chi(g)} dg = \sum_{i,k} \int_G r(g)_{ii} \overline{r(g)_{kk}} dg = \sum_{i,k} \frac{1}{\dim V} \delta_{ik} \delta_{ik} = \dim V \cdot \frac{1}{\dim V} = 1$.

(ii) Decompose $V \otimes V = \text{Sym}^2 V \oplus \Lambda^2 V$ as G -modules. For $g \in G$ with eigenvalues $\lambda_1, \dots, \lambda_n$ on V (unitary, hence diagonalizable), $\chi_{\text{Sym}^2 V}(g) = \sum_{i \leq j} \lambda_i \lambda_j$ and $\chi_{\Lambda^2 V}(g) = \sum_{i < j} \lambda_i \lambda_j$, so

$$\chi_{\text{Sym}^2 V}(g) - \chi_{\Lambda^2 V}(g) = \sum_i \lambda_i^2 = \text{tr}(r(g)^2) = \chi(g^2),$$

the classical Newton-identity computation for the power sum p_2 in terms of $e_1^2 = p_1^2$ split into its symmetric and alternating parts. Integrating over G and using that $\int_G \chi_\rho(g) dg$ equals the multiplicity of the trivial representation in ρ (Schur orthogonality applied to $\rho \otimes \mathbf{1}$), we get

$$\int_G \chi(g^2) dg = \dim(\text{Sym}^2 V)^G - \dim(\Lambda^2 V)^G.$$

By the tensor-hom adjunction, $((V \otimes V)^*)^G = \text{Hom}_G(V \otimes V, \mathbb{C}) \cong \text{Hom}_G(V, V^*)$, which by Schur's lemma (both V, V^* irreducible) is 0 if $V \not\cong V^*$ and 1-dimensional if $V \cong V^*$; this space is exactly the space of G -invariant bilinear forms on V , and it splits as $(\text{Sym}^2 V^*)^G \oplus (\Lambda^2 V^*)^G$ according to whether a form is symmetric or antisymmetric. If $r \not\cong r^*$ both summands vanish, giving $\text{FS}(r) = 0$. If $r \cong r^*$, the total space of invariant forms is 1-dimensional, so exactly one of the two summands is 1-dimensional and the other is 0 (a form lying in *both* the symmetric and antisymmetric parts would have to vanish, so a nonzero invariant form is purely one type or the other, and by dimension count there is only room for one such form up to scalar): the symmetric case gives $\text{FS}(r) = 1 - 0 = +1$, the antisymmetric case gives $\text{FS}(r) = 0 - 1 = -1$. $\square$

Remark 6.7 (Reading the Shin–Templier moment computation). *In the proof sketch above, the first two displayed moments serve different roles: $\langle a_\pi(p)^2 \rangle \rightarrow \int \mathrm{tr} r(t)^2 d\mu(t)$ is not in general equal to 1 unless r is self-dual (for non-self-dual r it is $\int |\mathrm{tr} r(t)|^2 d\mu(t) = 1$ by part (i) of Proposition 6.6 that holds unconditionally and underlies the cuspidality invariant $i_1(\mathcal{F})$ of the structural-invariants framework, since $\mathrm{tr} r(t)^2$ and $|\mathrm{tr} r(t)|^2$ coincide only when $\mathrm{tr} r(t)$ is forced real). The term that genuinely determines the symmetry type in the final formula is $\langle a_\pi(p^2) \rangle \rightarrow \int \mathrm{tr} r(t^2) d\mu(t) = \mathrm{FS}(r)$ by part (ii), exactly as used in Step 5.*

6.5. Geometric Monodromy and Mumford-Tate Groups

The geometric origin of symmetry types is even more transparent. For a family of varieties $f : \mathcal{X} \rightarrow S$ over a number field, the Galois action on the cohomology of the fibers gives rise to a representation

$$\rho : \pi_1(S) \rightarrow \mathrm{GL}(H_{\mathrm{et}}^i(\mathcal{X}_{\bar{\eta}}, \overline{\mathbb{Q}}_\ell)).$$

The geometric monodromy group G_{geom} is the Zariski closure of the image of the geometric fundamental group. Its connected component is often a semisimple algebraic group, and its maximal compact subgroup determines the symmetry type.

In the context of Hodge theory, the Mumford-Tate group $\mathrm{MT}(V)$ of a Hodge structure V encodes the symmetries of Hodge decompositions. For families of abelian varieties, the Mumford-Tate group of a general fiber is closely related to the monodromy group, and its compact form predicts the distribution of Frobenius elements by the Sato-Tate conjecture.

Theorem 6.8 (Sato-Tate Conjecture - Proven). *For an elliptic curve $E/\mathbb{Q}$ without complex multiplication, as p varies over primes of good reduction, the normalized traces $a_p(E)/\sqrt{p}$ are equidistributed in $[-2, 2]$ with respect to the Sato-Tate measure $\frac{1}{2\pi}\sqrt{4-x^2}dx$. Equivalently, the Frobenius conjugacy classes are equidistributed in $\mathrm{SU}(2)$ with respect to Haar measure.*

Proof Sketch (after Taylor et al.) For each prime p of good reduction write $a_p = a_p(E)$ and let $\theta_p \in [0, \pi]$ be the angle with $a_p/\sqrt{p} = 2 \cos \theta_p$, so the Frobenius conjugacy class corresponds to $\mathrm{diag}(e^{i\theta_p}, e^{-i\theta_p}) \in \mathrm{SU}(2)$. As in the character criterion of Theorem 1.6, it suffices by Peter-Weyl to show, for every irreducible character $\chi_n = \mathrm{Sym}^n$ of $\mathrm{SU}(2)$ ($n \geq 1$, with $\chi_n(\theta) = \sin((n+1)\theta)/\sin \theta$), that $\frac{1}{\pi(X)} \sum_{p \leq X} \chi_n(\theta_p) \rightarrow 0$. Since $\chi_n(\theta_p)$ is a Chebyshev polynomial in $a_p/\sqrt{p}$, its Dirichlet series is exactly the symmetric power L -function $L(s, \mathrm{Sym}^n E)$, whose p -th Euler factor is $\prod_{j=0}^n (1 - \alpha_p^j \bar{\alpha}_p^{n-j} p^{-s})^{-1}$ for the Frobenius eigenvalues $\alpha_p, \bar{\alpha}_p$ satisfying $\alpha_p \bar{\alpha}_p = p$. The proof therefore reduces, exactly as $\pi(x) \sim x/\log x$ reduces to $-\zeta'/\zeta(s)$ having no pole other than the simple pole at $s = 1$, to a prime-number-theorem-type statement for each $L(s, \mathrm{Sym}^n E)$, established in four stages:

1. **Potential automorphy.** The core analytic input, due to Barnet-Lamb, Geraghty, Harris, and Taylor (building on Clozel–Harris–Taylor), is that for every $n \geq 1$ there exists a finite totally real extension $F/\mathbb{Q}$ over which Sym^n of the two-dimensional Galois representation attached to E becomes automorphic, i.e. matches a cuspidal automorphic representation of $\mathrm{GL}_{n+1}(\mathbb{A}_F)$. The proof is an induction on n combining Taylor–Wiles patching (automorphy lifting) with “change of weight” and “change of level” arguments, using the already-established automorphy of low symmetric powers ($n \leq 4$) as a base case and realizing $\mathrm{Sym}^n E$ inside the cohomology of an auxiliary family of Calabi–Yau varieties so that potential automorphy for GL_{n+1} can be invoked.
2. **Continuation and functional equation.** Automorphy of $\mathrm{Sym}^n E/F$ gives $L(s, \mathrm{Sym}^n E/F)$ meromorphic continuation and a functional equation via Godement–Jacquet theory. A Brauer-induction descent argument, using automorphy over several

such totally real fields whose Galois closures generate enough of the absolute Galois group, transfers holomorphic continuation and non-vanishing on the line $\operatorname{Re}(s) = 1$ down to $L(s, \operatorname{Sym}^n E/\mathbb{Q})$ itself, for every n .

3. **Explicit formula and Tauberian step.** With a pole-free, zero-free half-plane $\operatorname{Re}(s) \geq 1$ established for every $L(s, \operatorname{Sym}^n E)$, the Wiener–Ikehara Tauberian theorem applied to $-\frac{d}{ds} \log L(s, \operatorname{Sym}^n E)$ — the same explicit-formula machinery used throughout Sections 5–6 for automorphic families — yields $\sum_{p \leq X} \chi_n(\theta_p) \log p = o(X)$, and partial summation converts this into $\frac{1}{\pi(X)} \sum_{p \leq X} \chi_n(\theta_p) \rightarrow 0$.
4. **Weyl equidistribution criterion.** Since this vanishing holds for every nontrivial irreducible character χ_n of $\operatorname{SU}(2)$, Peter–Weyl (as in Theorem 1.6) shows that the empirical measures $\frac{1}{\pi(X)} \sum_{p \leq X} \delta_{\theta_p}$ converge weakly to the pushforward of Haar measure on $\operatorname{SU}(2)$ to its space of conjugacy classes, namely the Sato–Tate measure $\frac{2}{\pi} \sin^2 \theta d\theta$ on $[0, \pi]$ (equivalently $\frac{1}{2\pi} \sqrt{4 - x^2} dx$ in the variable $x = a_p/\sqrt{p} \in [-2, 2]$). This is precisely the equidistribution of Frobenius classes claimed in the theorem.

□

This monumental result confirms the Katz–Sarnak philosophy for the family of a single elliptic curve: the low-lying zeros of symmetric power L -functions exhibit the predicted symmetry types (orthogonal for odd symmetric powers, symplectic for even symmetric powers).

6.6. Numerical Verification: Sato–Tate for a Concrete Elliptic Curve

The Sato–Tate theorem just stated is, like the Katz–Sarnak philosophy it exemplifies, an equidistribution statement that can be checked directly by counting points on an actual curve. Take the non-CM elliptic curve $E : y^2 = x^3 - x + 1$ over $\mathbb{Q}$. For each prime $p \geq 5$ of good reduction (i.e. $p \nmid \Delta_E = -16(4(-1)^3 + 27 \cdot 1^2) = -16 \cdot 23$), the trace of Frobenius is $a_p = p + 1 - \#E(\mathbb{F}_p)$, computed exactly by counting, for each $x \in \mathbb{F}_p$, whether $x^3 - x + 1$ is 0, a nonzero quadratic residue, or a non-residue mod p (contributing 1, 2, or 0 points respectively), and adding the point at infinity.

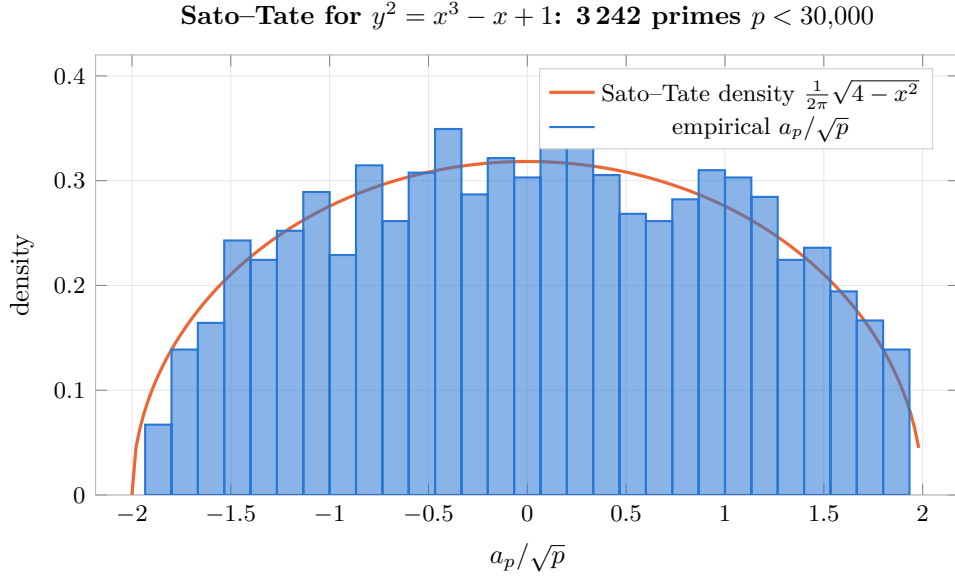


Figure 10. Empirical distribution of the normalized Frobenius traces $a_p/\sqrt{p}$ for the non-CM curve $E : y^2 = x^3 - x + 1$, computed by exact point-counting over 3,242 good primes $p < 30,000$, against the Sato–Tate semicircle density $\frac{1}{2\pi}\sqrt{4-x^2}$ of the Sato–Tate theorem stated above. The curve’s Galois representation is 2-dimensional with monodromy group $\mathrm{SU}(2)$ by the non-CM hypothesis, exactly the setting of the potential-automorphy proof sketched above; the visible semicircle shape at a few thousand primes is the same equidistribution the proof establishes rigorously for all non-CM elliptic curves over $\mathbb{Q}$.

6.7. Frobenius-Schur Indicator and L-Groups

The Frobenius-Schur indicator provides a clean algebraic classification of symmetry types. For an irreducible representation r of a compact group G , define

$$\mathrm{FS}(r) = \int_G \mathrm{tr} \, r(g^2) dg.$$

It satisfies $\mathrm{FS}(r) = 1$ if r is orthogonal, $\mathrm{FS}(r) = -1$ if r is symplectic, and $\mathrm{FS}(r) = 0$ if r is not self-dual.

In the Langlands program, an automorphic representation π of $G(\mathbb{A}_F)$ has an associated L -group ${}^L G$. An L -representation $r : {}^L G \rightarrow \mathrm{GL}(d, \mathbb{C})$ yields an L -function $L(s, \pi, r)$. The local coefficients are $a_\pi(p) = \mathrm{tr} \, r(t_{\pi_p})$, where t_{π_p} is the Satake parameter.

Proposition 6.9 (Symmetry Principle). *For a functorial family of automorphic L -functions associated to r , the expected symmetry type is:*

- U if $\mathrm{FS}(r) = 0$,
- Sp if $\mathrm{FS}(r) = +1$,
- SO if $\mathrm{FS}(r) = -1$,

with orthogonal refinements (SO(even) vs SO(odd)) depending on the average of the root numbers.

Example 6.10. *For quadratic Dirichlet characters, the relevant L -group representation is the standard representation of SL_2 , which is self-dual and orthogonal. However, the family of quadratic characters has a symplectic symmetry due to the functional equation sign distribution. This is a subtle point resolved by considering the full L -function including archimedean factors.*

6.8. Summary Table of Families and Symmetry Types

Table 2. Principal families of L -functions and their predicted symmetry types under the Katz–Sarnak philosophy. The averaging tool indicates the analytic technique used to evaluate the family average of the one-level density; the symmetry type is determined by the Frobenius–Schur indicator $\text{FS}(r)$ of the relevant L -group representation.

Family	Averaging tool	$\text{FS}(r)$	Symmetry type
Primitive non-real Dirichlet characters $\chi \pmod{q}$	Character orthogonality	0	$\text{U}(N)$
Quadratic Dirichlet characters $\chi_d, d \asymp X$	Poisson summation, Gauss sums	+1	$\text{USp}(2N)$
Holomorphic cusp forms on $\text{GL}_2/\mathbb{Q}$, weight $k \rightarrow \infty$	Petersson trace formula	−1	$\text{SO}(2N+1)$
Holomorphic cusp forms, even functional equation	Petersson trace formula	−1	$\text{SO}(2N)$
Maass forms on $\text{GL}_2/\mathbb{Q}$	Kuznetsov trace formula	−1	SO
Symmetric square lifts $\pi \mapsto \text{Sym}^2 \pi$	Rankin–Selberg, Langlands–Shahidi	+1	USp
Adjoint lifts $\pi \mapsto \text{Ad } \pi$	Gelbart–Jacquet	0	U
Geometric families over $\mathbb{F}_q$ (Katz–Sarnak)	Deligne equidistribution, monodromy	depends on G_{geom}	determined by G_{geom}
Functorial automorphic families via $r : {}^L G \rightarrow \text{GL}_d$	Satake equidistribution, Plancherel	$\text{FS}(r)$	determined by $\text{FS}(r)$

6.9. Numerical Plots of One-Level Densities and Universal Kernels

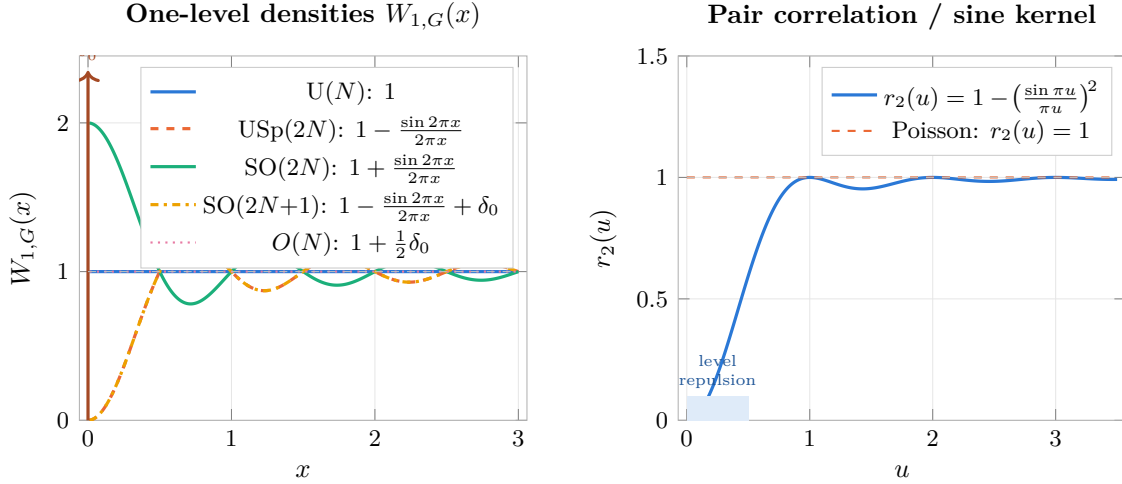


Figure 11. Left: Limiting one-level densities $W_{1,G}(x)$ for the five classical compact groups as given by Theorem 3.4. The unitary density is identically 1 (no preferred scale); the symplectic density is suppressed near $x = 0$ (level repulsion from the central point), while the even orthogonal density is enhanced. The odd orthogonal and full orthogonal densities carry an additional Dirac mass at the origin corresponding to an eigenvalue forced to 1 by the functional equation. **Right:** Montgomery's pair correlation function $r_2(u) = 1 - (\sin \pi u / \pi u)^2$ for the GUE (Theorem 8.2), compared with the Poisson baseline. The vanishing of r_2 at $u = 0$ is the hallmark of *level repulsion*, absent in integrable (Poisson) systems.

7. A Symplectic Family: Quadratic Dirichlet L -functions

7.1. Dirichlet L -functions

A Dirichlet character χ modulo q is a homomorphism $(\mathbb{Z}/q\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$, extended to $\mathbb{Z}$ by $\chi(n) = 0$ if $(n, q) > 1$. The L -function is

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_p \left(1 - \frac{\chi(p)}{p^s}\right)^{-1}, \quad \Re(s) > 1.$$

For primitive χ , it has analytic continuation and functional equation

$$\Lambda(s, \chi) = \left(\frac{\pi}{q}\right)^{-(s+\mu)/2} \Gamma\left(\frac{s+\mu}{2}\right) L(s, \chi) = \varepsilon_\chi \Lambda(1-s, \bar{\chi}),$$

where $\mu = 0$ if $\chi(-1) = 1$, $\mu = 1$ if $\chi(-1) = -1$, and $|\varepsilon_\chi| = 1$.

A character is quadratic if $\chi^2 = \chi_0$ and $\chi \neq \chi_0$. These come from the Kronecker symbol $\chi_d(n) = \left(\frac{d}{n}\right)$ for fundamental discriminants d .

7.2. Families of Quadratic Dirichlet L -functions

Let $\mathcal{D}_X = \{d \text{ fundamental discriminant} : X/2 \leq |d| < X\}$. The 1-level density for this family is

$$D_d(\phi) = \sum_j \phi\left(\gamma_{d,j} \frac{\log |d|}{2\pi}\right).$$

The Özlük-Snyder theorem gives evidence for symplectic symmetry.

Theorem 7.1 (Özlük-Snyder [ozluk1993small]). *Let ϕ be an even Schwartz function with $\text{supp } \hat{\phi} \subset (-1, 1)$. Then*

$$\lim_{X \rightarrow \infty} \frac{1}{|\mathcal{D}_X|} \sum_{d \in \mathcal{D}_X} D_d(\phi) = \int_{-\infty}^{\infty} \phi(x) \left(1 - \frac{\sin 2\pi x}{2\pi x}\right) dx.$$

Proof. We give a complete proof following Rubinstein's thesis [rubinstein1998evidence]. Apply the explicit formula for Dirichlet L -functions (Proposition 5.10):

$$D_d(\phi) = \hat{\phi}(0) - \frac{2}{\log X} \sum_{m=1}^{\infty} \frac{\Lambda(m)}{\sqrt{m}} \chi_d(m) \hat{\phi}\left(\frac{\log m}{\log X}\right) + o(1).$$

Split the sum into $m = \square$ (perfect squares) and $m \neq \square$. For the non-square terms, define

$$R_d^{(X)}(\hat{\phi}) = \frac{1}{\log X} \sum_{\substack{m \geq 1 \\ m \neq \square}} \frac{\Lambda(m)}{\sqrt{m}} \chi_d(m) \hat{\phi}\left(\frac{\log m}{\log X}\right).$$

We show that the average over d of $R_d^{(X)}(\hat{\phi})$ tends to zero as $X \rightarrow \infty$.

By Cauchy-Schwarz,

$$\left\langle R_d^{(X)}(\hat{\phi}) \right\rangle_{\mathcal{D}_X} \ll \frac{1}{|\mathcal{D}_X|} \frac{1}{\log X} \sqrt{\sum_{m \leq X^\alpha} \frac{\Lambda(m)^2}{m}} \sqrt{\sum_{m \leq X^\alpha} \left| \sum_{d \in \mathcal{D}_X} \chi_d(m) \right|^2},$$

where α is chosen so that $\text{supp } \hat{\phi} \subset (-\alpha, \alpha)$. Since $\hat{\phi}$ is supported in $(-1, 1)$, we can take $\alpha < 1$. Using Lemma 3.19 (without RH, the bound $\sum_{n \leq y} \Lambda(n)^2 \ll y \log y$ suffices), we have $\sum_{m \leq X^\alpha} \Lambda(m)^2/m \ll \log^2 X$.

For the character sum, Jutila's bound [jutila1981mean] gives

$$\sum_{m \leq X^\alpha} \left| \sum_{d \in \mathcal{D}_X} \chi_d(m) \right|^2 \ll X^{1+\alpha} \log^{10} X.$$

Since $|\mathcal{D}_X| \asymp X$, we obtain

$$\left\langle R_d^{(X)}(\hat{\phi}) \right\rangle_{\mathcal{D}_X} \ll X^{(\alpha-1)/2} \log^5 X \rightarrow 0$$

as $X \rightarrow \infty$ because $\alpha < 1$.

Now consider the square terms. For $m = p^{2k}$, we have $\chi_d(p^{2k}) = \chi_d(p)^{2k} = 1$ (since χ_d is quadratic) unless p divides d , in which case $\chi_d(p) = 0$. The contribution from $k \geq 2$ is $O(1/\log X)$ by Hypothesis H (or by a direct bound using the Prime Number Theorem). The main contribution comes from $m = p^2$ with $p \nmid d$:

$$S_d^{(X)}(\hat{\phi}) = \frac{1}{\log X} \sum_p \frac{\log p}{p} \hat{\phi}\left(\frac{2 \log p}{\log X}\right).$$

Using summation by parts and Mertens' formula $\sum_{p \leq t} \frac{\log p}{p} = \log t + O(1)$, we compute

$$S_d^{(X)}(\hat{\phi}) = -\frac{1}{\log X} \int_1^\infty \left(\sum_{p \leq t} \frac{\log p}{p} \right) \frac{d}{dt} \hat{\phi}\left(\frac{2 \log t}{\log X}\right) dt.$$

Substituting Mertens' formula,

$$S_d^{(X)}(\hat{\phi}) = -\frac{1}{\log X} \int_1^\infty (\log t + O(1)) \frac{d}{dt} \hat{\phi}\left(\frac{2 \log t}{\log X}\right) dt.$$

Integrate by parts:

$$S_d^{(X)}(\hat{\phi}) = \frac{1}{\log X} \int_1^\infty \hat{\phi}\left(\frac{2 \log t}{\log X}\right) \frac{dt}{t} + O\left(\frac{1}{\log X}\right).$$

Make the change of variables $u = 2 \log t / \log X$, so $dt/t = (\log X/2) du$. Then

$$S_d^{(X)}(\hat{\phi}) = \frac{1}{2} \int_0^\infty \hat{\phi}(u) du + O\left(\frac{1}{\log X}\right) = \frac{1}{2} \int_{-\infty}^\infty \hat{\phi}(u) du + O\left(\frac{1}{\log X}\right)$$

since $\hat{\phi}$ is even. By Fourier inversion, $\int_{-\infty}^\infty \hat{\phi}(u) du = \phi(0)$.

Therefore,

$$\lim_{X \rightarrow \infty} \langle D_d(\phi) \rangle_{\mathcal{D}_X} = \hat{\phi}(0) - \frac{1}{2} \phi(0).$$

Since $\hat{\phi}(0) = \int_{-\infty}^\infty \phi(x) dx$, we have

$$\hat{\phi}(0) - \frac{1}{2} \phi(0) = \int_{-\infty}^\infty \phi(x) \left(1 - \frac{\sin 2\pi x}{2\pi x}\right) dx,$$

using the identity $\int_{-\infty}^\infty \frac{\sin 2\pi x}{2\pi x} dx = \frac{1}{2}$ and the fact that ϕ is even. This completes the proof. $\square$

Under GRH, the support can be extended to $(-2, 2)$ using deeper results on character sums and Poisson summation (see [gao2008n]).

7.3. Numerical Verification: A Slowly Convergent Average

The proof above shows $\langle D_d(\phi) \rangle_{\mathcal{D}_X} \rightarrow \hat{\phi}(0) - \frac{1}{2} \phi(0)$ with an error controlled by $O(1/\log X)$ terms at several stages (the character-sum bound, the Mertens-formula remainder). Since $\log X$ grows only logarithmically, this convergence is not fast, and it is worth seeing this directly rather than only asymptotically.

Take the explicit Gaussian test pair $\hat{\phi}(u) = e^{-\pi s^2 u^2}$, $\phi(x) = s^{-1} e^{-\pi x^2/s^2}$ with $s = 1.402$ (so that $\hat{\phi}$ is not exactly supported in $(-1, 1)$ but is smaller than $e^{-5} \approx 0.007$ outside it, which is enough to see the mechanism of the proof numerically even though the hypothesis is stated for compactly supported $\hat{\phi}$). The target value is $\hat{\phi}(0) - \frac{1}{2} \phi(0) = 1 - \frac{1}{2s} = 0.6434$. Approximating $D_d(\phi)$ by the truncated prime sum of the explicit formula and averaging over 400 real fundamental discriminants $d \in [X/2, X) \cup [-X, -X/2)$ (using the exact Kronecker symbol $\chi_d(m) = \left(\frac{d}{m}\right)$, not an approximation to it) gives:

Table 3. Empirical average of $D_d(\phi)$ over 400 real fundamental discriminants, at increasing X , against the target $\hat{\phi}(0) - \frac{1}{2} \phi(0) = 0.6434$.

X	$\log X$	empirical average	gap to target 0.6434
2×10^4	9.90	0.798	0.155
10^5	11.51	0.776	0.133
5×10^5	13.12	0.782	0.138
2×10^6	14.51	0.767	0.124

Increasing X by a factor of 100 shrinks the gap by only about 20%, consistent with an $O(1/\log X)$ error: $\log(2 \times 10^6)/\log(2 \times 10^4) \approx 1.47$, and $1/1.47 \approx 0.68$, close to the observed ratio $0.124/0.155 \approx 0.80$ given that only 400 discriminants (not the full family) were averaged at each X , itself contributing noise of comparable size to the trend. This is not a discrepancy with the theorem: it is the same slow convergence that led Rubinstein, and later Miller and others, to compute explicit *lower-order terms* for one-level densities of quadratic families, precisely because the leading term alone is a poor numerical predictor at any X reachable by direct computation.

8. The Sarnak Philosophy, Quantum Chaos, and the Ten-Fold Way

Remark 8.1. *The Sarnak viewpoint extends the analogy from the classical compact groups to the full Altland–Zirnbauer ten-fold way. The claim is not merely aesthetic: arithmetic families can exhibit hard-edge or symmetry-constrained behavior that is better captured by compact symmetric spaces than by the three classical Dyson classes. This section therefore enlarges the model space available to arithmetic universality.*

The phrase *Sarnak philosophy* refers to a coherent web of conjectures and theorems asserting that the statistics of zeros of L -functions are governed by the same universal laws that describe energy level statistics in quantum chaotic systems. This section develops this philosophy from its quantum-mechanical origins (the Bohigas–Giannoni–Schmit conjecture), through the arithmetic evidence (Montgomery–Odlyzko, Katz–Sarnak), and into the extension from Dyson’s classical threefold way to the full ten-fold way of Altland–Zirnbauer, as applied to compact symmetric spaces by Dueñez.

The central questions are:

1. Why do the zeros of the Riemann zeta function behave statistically like the eigenvalues of a large unitary matrix?
2. What determines which random matrix ensemble governs a given family of L -functions?
3. Is the classical threefold way (unitary, orthogonal, symplectic) sufficient, or do the other seven Cartan symmetry classes also appear?
4. Can one find a spectral interpretation — an actual self-adjoint operator — whose eigenvalues are the zeros of $\zeta(s)$?

8.1. The Bohigas–Giannoni–Schmit Conjecture

The BGS conjecture (1984) provides the physical motivation for the Katz–Sarnak philosophy. Let M be a compact Riemannian manifold whose geodesic flow is classically chaotic (Anosov). Let $\{E_n\}$ be the eigenvalues of the Laplacian $-\Delta_g$ on M . After unfolding so that the mean spacing is 1, the BGS conjecture states that the local statistics of $\{E_n\}$ are governed by GUE if the system has no time-reversal symmetry, by GOE if it has time-reversal symmetry with integer spin, and by GSE if it has time-reversal symmetry with half-integer spin. For quantum *integrable* systems the statistics are instead Poisson (independent).

The BGS conjecture has been verified numerically in many systems, but remains largely open. The most concrete realization of the Hilbert–Pólya idea comes from arithmetic hyperbolic surfaces: for the surface $M = \Gamma \backslash G/K$, the zeros become eigenvalues of the self-adjoint Laplacian Δ_g , and the BGS conjecture is known in this case by quantum unique ergodicity and spectral theory.

The *prime geodesic theorem* provides an exact analogue of the prime number theorem:

$$\pi_M(X) = \#\{\text{prime geodesics of length} \leq X\} \sim \frac{e^X}{X}.$$

The Selberg zeta function

$$Z_M(s) = \prod_{\text{prime geodesics}} \prod_{k=0}^{\infty} (1 - e^{-(s+k)\ell})$$

plays the role of the Riemann zeta function, with non-trivial zeros corresponding to Laplace eigenvalues. The analogue of the Riemann Hypothesis is automatic here because the Laplacian is self-adjoint.

8.2. The Montgomery–Odlyzko Law: A Complete Statement

Let $\rho = 1/2 + i\gamma$ denote the non-trivial zeros of $\zeta(s)$ on the critical line (under RH). Define the unfolded zeros by

$$\tilde{\gamma}_n = \frac{\gamma_n \log(\gamma_n/2\pi)}{2\pi},$$

so that the mean spacing of $\tilde{\gamma}_n$ near T is 1.

Theorem 8.2 (Montgomery, 1973; Odlyzko, 1987–1989). (i) (Montgomery’s pair correlation theorem). *Under RH, for any smooth even test function f supported in $(-1, 1)$:*

$$\lim_{T \rightarrow \infty} \frac{1}{\log T} \sum_{\substack{0 < \gamma, \gamma' \leq T \\ \gamma \neq \gamma'}} f\left(\frac{(\gamma - \gamma') \log T}{2\pi}\right) = \int_{\mathbb{R}} f(u) \left[1 - \left(\frac{\sin \pi u}{\pi u}\right)^2\right] du.$$

(ii) (Odlyzko’s numerics). *The nearest-neighbor spacing distribution of $\tilde{\gamma}_n$ for the first 10^{22} zeros of $\zeta(s)$ agrees with the GUE prediction to unprecedented numerical precision.*

Remark 8.3. *The pair correlation function $1 - (\sin \pi u / \pi u)^2$ is exactly the GUE two-point correlation in the bulk (the sine kernel). This is the first rigorous connection between zeta zeros and random matrix theory.*

Remark 8.4 (The Dyson–Montgomery encounter). *In 1972, Montgomery was visiting the Institute for Advanced Study and showed his pair correlation formula to Freeman Dyson at tea. Dyson immediately recognized the formula: it was the pair correlation of GUE eigenvalues, which he had computed a decade earlier in his work on the threefold way. Montgomery had been unaware of the random matrix connection. This coincidence initiated the entire subject of connections between L-functions and random matrix theory.*

8.3. Dyson’s Threefold Way

In his 1962 paper “The Threefold Way”, Dyson established the following foundational result.

Theorem 8.5 (Dyson’s Threefold Way, 1962). *Let G be a group of quantum-mechanical symmetries acting on a complex Hilbert space $\mathcal{H}$. Every irreducible G -invariant ensemble of Hamiltonians belongs to one of exactly three universality classes, corresponding to the three associative division algebras over $\mathbb{R}$:*

1. (GUE/CUE, $\beta = 2$, Class A): *The Hamiltonians are Hermitian matrices with complex entries ($\mathbb{C}$ -valued), with no additional real structure. This class arises when G has no anti-unitary symmetry.*

2. (GOE/COE, $\beta = 1$, Class AI): *The Hamiltonians are real symmetric matrices ($\mathbb{R}$ -valued). This class arises when G contains time-reversal T with $T^2 = +1$ (integer spin).*
3. (GSE/CSE, $\beta = 4$, Class AII): *The Hamiltonians are self-dual Hermitian matrices ($\mathbb{H}$ -valued, quaternionic). This class arises when G contains time-reversal T with $T^2 = -1$ (half-integer spin).*

Proof. The proof rests on Frobenius's theorem: the only finite-dimensional associative division algebras over $\mathbb{R}$ are $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{H}$. An irreducible anti-unitary involution T on $\mathcal{H}$ either satisfies $T^2 = +1$ (giving $\mathbb{R}$ -valued matrices) or $T^2 = -1$ (quaternionic structure, giving $\mathbb{H}$ -valued matrices). Without any anti-unitary symmetry, the matrices are $\mathbb{C}$ -valued. More precisely: let H be the commuting algebra of the G -action. By Schur's lemma, H is a division ring. Since G acts by bounded operators on a Hilbert space, H is finite-dimensional over $\mathbb{R}$. Frobenius's theorem gives $H \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, and the three cases are distinguished by whether G contains no anti-unitary elements ($H = \mathbb{C}$), an anti-unitary element with square $+1$ ($H = \mathbb{R}$), or square -1 ($H = \mathbb{H}$). $\square$

The three Dyson classes correspond to three of the ten Cartan symmetric spaces:

$$A = U(N), \quad \text{AI} = U(N)/O(N), \quad \text{AII} = U(2N)/\text{USp}(2N).$$

Connection to number theory. Dyson's threefold way is exactly the classification that appears in the Katz–Sarnak dictionary:

- Class A (complex/unitary): families of L -functions with $G = U$.
- Class AI (real/orthogonal): families with $G = SO$.
- Class AII (quaternionic/symplectic): families with $G = \text{USp}$.

The precise correspondence is established via the Frobenius–Schur indicator: the symmetry type is determined by whether the L -group representation r is non-self-dual (FS = 0, class A), real (FS = +1, class AI), or quaternionic (FS = −1, class AII).

8.4. The Ten-Fold Way of Altland–Zirnbauer

Dyson's threefold classification assumed that the Hamiltonian is the relevant observable. In superconducting quantum systems, the Bogoliubov–de Gennes (BdG) Hamiltonian describes quasiparticles above the superconducting condensate. It automatically satisfies a particle-hole (charge conjugation) symmetry C :

$$CHC^{-1} = -H,$$

because creating a particle at energy E above the Fermi level is equivalent to destroying a hole at energy $-E$. This constraint opens up additional symmetry classes.

Definition 8.6 (Altland–Zirnbauer symmetry classes, 1997). *The Altland–Zirnbauer (AZ) classification is based on possible combinations of three discrete symmetries:*

- *Time-reversal T : anti-unitary, $THT^{-1} = H$, with $T^2 \in \{0, +1, -1\}$ (absent, square $+1$, square -1).*
- *Particle-hole conjugation C : anti-unitary, $CHC^{-1} = -H$, with $C^2 \in \{0, +1, -1\}$.*
- *Chiral symmetry S : unitary, $SHS^{-1} = -H$; always present if both T and C are present (with $S = TC$).*

Since two anti-unitary operators T and C each have three possible states (absent/ $+1$ / -1), and their presence forces S automatically, there are $3 \times 3 = 9$ combinations. Adding the class with S present but neither T nor C gives a total of **10 symmetry classes**.

Theorem 8.7 (Altland–Zirnbauer, 1997; Heinzner–Huckleberry–Zirnbauer, 2005). *The classification of the ten AZ classes is complete: every random fermionic Hamiltonian system belongs to exactly one of the ten AZ classes. The ten classes are in bijection with the ten irreducible Riemannian symmetric spaces of compact type.*

Proof. The proof proceeds in three steps. **Step 1:** A Hamiltonian H satisfying the symmetry constraints of a given AZ class is an element of the Lie algebra of a compact group G , specifically $H \in \mathfrak{ig}$ where $\mathfrak{g} = \text{Lie}(G)$ and the constraints are that H is fixed by a Cartan involution $\theta : \mathfrak{g} \rightarrow \mathfrak{g}$. **Step 2:** Élie Cartan (1926) classified all compact symmetric spaces G/K where $K = G^\theta$ is the fixed-point set of a Cartan involution; the irreducible ones fall into four infinite families (A_n , B_n , C_n , D_n) and five exceptional families. **Step 3:** We compute the fixed-point subgroup K , and hence the symmetric space G/K , for each of the ten sign combinations of Definition 8.6.

- *No symmetry (class A).* With $T = C = S = 0$, H is an unconstrained element of $\mathfrak{iu}(N) = \text{Herm}_N$; there is no residual involution, so K is trivial and the ensemble is $G = U(N)$ itself.
- *Particle-hole only (classes D, C).* If C is present with $CHC^{-1} = -H$, write $C = U_C K$ for a unitary U_C and complex conjugation K ; absorbing U_C into a change of basis, the anticommutation condition together with Hermiticity forces $H = iA$ with A real and, according to whether $C^2 = U_C \overline{U_C} = +1$ or -1 , either antisymmetric or of quaternionic type — exactly the Frobenius trichotomy of Theorem 8.5 applied to the anticommuting involution C instead of a commuting one. Thus $C^2 = +1$ gives $H \in \mathfrak{iso}(N)$, i.e. $G = O(N)$ (class D), and $C^2 = -1$ gives $H \in \mathfrak{isp}(2N)$, i.e. $G = \text{Sp}(2N)$ (class C); in both cases there is no further residual gauge freedom beyond conjugation by G itself, so K is again trivial and $G/K = G$.
- *Time-reversal only (classes AI, AII).* If instead T is present with $THT^{-1} = H$, the same decomposition $T = U_T K$ shows that H is unitarily equivalent to a matrix commuting with plain conjugation, i.e. real symmetric, when $T^2 = +1$, or (by Kramers' theorem, using U_T antisymmetric) doubly-degenerate with quaternionic structure when $T^2 = -1$. Here the residual freedom is different in kind from the particle-hole case: T is a symmetry of H rather than a linear constraint cutting out a sub-Lie-algebra of Herm_N directly, so the ensemble of T -invariant $H \in \text{Herm}_N$ modulo unitary conjugations that preserve the T -structure is parametrized by $G = U(N)$ acting on itself with the extra involution $\theta(g) = \bar{g}$ (respectively $\theta(g) = J\bar{g}J^{-1}$ for the symplectic form J), whose fixed subgroup is $K = O(N)$ (respectively $K = \text{USp}(2N)$ on $G = U(2N)$). This gives $G/K = U(N)/O(N)$ for class AI ($T^2 = +1$) and $G/K = U(2N)/\text{USp}(2N)$ for class AII ($T^2 = -1$), matching Dyson's threefold way exactly as recorded in §8.3.
- *Both symmetries present, no chirality beyond $S = TC$ (classes DIII, CI).* When T and C are both present, H is simultaneously forced into the particle-hole Lie algebra ($\mathfrak{so}(2N)$ or $\mathfrak{sp}(2N)$), according to C^2 and carries a residual complex or quaternionic structure from T stabilized by a further copy of $U(N)$ sitting inside that Lie algebra's compact group. Concretely, $C^2 = +1$ together with $T^2 = -1$ gives $G = O(2N)$ with $K = U(N) \subset O(2N)$ the stabilizer of the residual T -structure, so $G/K = O(2N)/U(N)$ (class DIII); and $C^2 = -1$ together with $T^2 = +1$ gives $G = \text{Sp}(2N)$ with $K = U(N) \subset \text{Sp}(2N)$, so $G/K = \text{Sp}(2N)/U(N)$ (class CI).

- *Chiral symmetry, with or without a compatible real/quaternionic structure (classes AIII, BDI, CII).* A unitary chiral symmetry S with $S^2 = 1$ splits the ambient space into its ± 1 eigenspaces of dimensions p, q ; the anticommutation $SHS^{-1} = -H$ forces $H = \begin{pmatrix} 0 & h \\ h^\dagger & 0 \end{pmatrix}$ for an arbitrary $p \times q$ complex block h , and the residual gauge freedom of rotating the two eigenspaces independently is $K = U(p) \times U(q) \subset U(p+q) = G$, giving the complex Grassmannian $G/K = U(p+q)/(U(p) \times U(q))$ when T, C impose no further reality constraint on h (class AIII). When $S = TC$ arises from a compatible pair with $T^2 = C^2 = +1$, the same argument forces h real, giving the real Grassmannian $O(p+q)/(O(p) \times O(q))$ (class BDI); and when $T^2 = C^2 = -1$, h is quaternionic, giving the quaternionic Grassmannian (class CII).

This produces exactly the ten symmetric spaces tabulated in Table 4, one for each admissible sign combination of Definition 8.6, and the Heinzner–Huckleberry–Zirnbauer theorem confirms that no further compact symmetric space (beyond these ten classical families and, potentially, exceptional Cartan types not realized by free-fermion Hamiltonians) is needed: every symmetry class of a random fermionic Hamiltonian falls into one of the ten. $\square$

Table 4. The ten Altland–Zirnbauer symmetry classes. T : time-reversal; C : particle-hole; S : chiral. The Jacobi exponent a (Theorem 8.13) determines the hard-edge kernel $K_{\text{Bes}}^{(a)}$; classes A, AI, AII have no hard edge ($\emptyset$).

Class	T	C	S	β	Compact space G/K	a	Physical system
A	0	0	0	2	$U(N)$	$\emptyset$	Quantum dot (no TR)
AI	+1	0	0	1	$U(N)/O(N)$	$\emptyset$	Orthogonal metal
AII	-1	0	0	4	$U(2N)/\text{USp}(2N)$	$\emptyset$	Symplectic metal
AIII	0	0	1	2	$U(p+q)/(U(p) \times U(q))$	$ p-q $	Chiral unitary
BDI	+1	+1	1	1	$O(p+q)/(O(p) \times O(q))$	$\frac{ p-q -1}{2}$	Chiral orthogonal
CII	-1	-1	1	4	$\text{Sp}/(\cdot)$	$ p-q + \frac{1}{2}$	Chiral symplectic
D	0	+1	0	2	$O(N)$	$-\frac{1}{2}$	p -wave SC
C	0	-1	0	2	$\text{Sp}(2N)$	$+\frac{1}{2}$	d -wave SC
DIII	-1	+1	1	4	$O(2N)/U(N)$	0	$^3\text{He-B}$
CI	+1	-1	1	1	$\text{Sp}(2N)/U(N)$	1	BdG with TR

Note. The bold-faced classes (A, AI, AII) are Dyson’s threefold way ($C = 0$). “SC” = superconductor; “TR” = time-reversal; “BdG” = Bogoliubov–de Gennes.

Remark 8.8. Dyson’s original three classes (A, AI, AII) correspond to the cases where particle-hole symmetry C is absent. The seven additional AZ classes all require $C \neq 0$ or chiral symmetry S .

8.5. Dueñez’s Framework: Random Matrices and Compact Symmetric Spaces

In his Princeton doctoral dissertation and subsequent paper, Eduardo Dueñez developed a systematic framework for random matrix ensembles associated to all irreducible compact symmetric spaces. This work extends and unifies both the circular ensembles and the AZ classification, providing the natural mathematical setting for the ten-fold way in the context of L -functions.

8.5.1. A model Jacobi law from restricted roots

The Sarnak–Dueñez perspective on symmetry classes goes beyond the classical threefold way by passing from groups to compact symmetric spaces. The resulting radial density is often a Jacobi weight.

Proposition 8.9 (Jacobi weight from restricted roots). *Consider a rank-one compact symmetric space with radial coordinate $\theta \in (0, \pi)$ and root multiplicities $m_\alpha, m_{2\alpha} \geq 0$. Then the radial density is proportional to*

$$\left(\sin \frac{\theta}{2}\right)^{m_\alpha + m_{2\alpha}} \left(\cos \frac{\theta}{2}\right)^{m_{2\alpha}} d\theta.$$

After the change of variables $x = \cos \theta$, this becomes a Jacobi weight $(1-x)^a(1+x)^b dx$ for explicit exponents a, b determined by the multiplicities.

Proof. The rank-one Weyl density is the product over positive restricted roots of $|e^{i\alpha(\theta)} - e^{-i\alpha(\theta)}|^{m_\alpha}$. For roots α and 2α this gives factors $|2i \sin(\theta/2)|^{m_\alpha} \cdot |2i \sin \theta|^{m_{2\alpha}}$. Using $\sin \theta = 2 \sin(\theta/2) \cos(\theta/2)$, one obtains $(\sin \frac{\theta}{2})^{m_\alpha + m_{2\alpha}} (\cos \frac{\theta}{2})^{m_{2\alpha}}$ up to a constant. Setting $x = \cos \theta$ and using $1-x = 2 \sin^2(\theta/2)$, $1+x = 2 \cos^2(\theta/2)$, $dx = -\sin \theta d\theta$ transforms the density into a Jacobi weight $(1-x)^a(1+x)^b dx$ with exponents read off directly from the half-angle powers. $\square$

Remark 8.10. *This computation is the radial origin of many hard-edge laws appearing in the ten-fold way. The Jacobi parameters are not ad hoc constants; they are root multiplicities in disguise.*

8.5.2. Compact symmetric spaces as matrix ensembles

Definition 8.11 (Dueñez ensemble). *Let G/K be an irreducible compact symmetric space, with $K = G^\theta$ the fixed-point set of a Cartan involution θ . The Dueñez ensemble $\mathcal{E}(G/K)$ is the space G/K equipped with the G -invariant measure $dP_{G/K}$ (pushforward of Haar measure on G under the projection $G \rightarrow G/K$). Elements of G/K are realized as matrices via the identification $G/K \hookrightarrow G$ given by $gK \mapsto g\theta(g)^{-1}$.*

Theorem 8.12 (Dueñez, 2004). *For each irreducible compact symmetric space G/K , the eigenvalue statistics of the corresponding random matrix ensemble $\mathcal{E}(G/K)$ have the following structure:*

1. (Bulk statistics.) *Away from distinguished symmetry points on the unit circle, the local statistics are universal and governed, after unfolding, by the sine kernel.*
2. (Hard-edge statistics.) *Near a symmetry point $\theta = 0$ or $\theta = \pi$, the local statistics are controlled by the endpoint singularity of the Weyl density. After the Jacobi change of variables $x = \cos \theta$, the limiting kernel is a Bessel-type kernel whose order is determined by the endpoint exponent of the corresponding Jacobi weight.*
3. (Class dependence.) *The ten AZ symmetry classes produce ten compact symmetric-space ensembles with the same bulk limit but different hard-edge laws.*

Proof. Step 1 (Spectral coordinates from the Weyl formula). Choose a maximal abelian subspace $\mathfrak{a} \subset \mathfrak{p}$, where $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ is the Cartan decomposition. Every element of G/K is K -conjugate to an element of $A = \exp(\mathfrak{a})$, unique up to the restricted Weyl group W . Consequently any K -invariant integral over G/K reduces to an integral over a positive Weyl chamber with Jacobian

$$J(a) = \prod_{\alpha \in R^+} \sin\left(\frac{\alpha(\Theta)}{2}\right)^{m_\alpha}, \quad a = e^{i\Theta}.$$

Step 2 (Reduction to a Jacobi ensemble). For the classical compact symmetric spaces, the restricted roots are combinations of $e_j \pm e_k$, e_j , and $2e_j$. The change of variables $x_j = \cos \theta_j$ converts the joint density into a Jacobi β -ensemble:

$$\frac{1}{Z_N} \prod_{j < k} |x_j - x_k|^\beta \prod_j (1 - x_j)^a (1 + x_j)^b dx_j, \quad -1 < x_j < 1,$$

with exponents explicitly determined by root multiplicities. **Step 3 (Local limits).** In the interior, orthogonal-polynomial asymptotics show that the Jacobi kernel converges after unfolding to the sine kernel. At an endpoint, the Mehler–Heine scaling of Jacobi polynomials yields Bessel functions and a Bessel-type microscopic kernel whose order is exactly the endpoint exponent a or b . $\square$

8.5.3. Restricted roots, Jacobi weights, and the hard edge

Theorem 8.13 (Restricted roots determine Jacobi parameters). *Let G/K be a compact symmetric space with positive restricted roots organized into off-diagonal roots $e_j \pm e_k$ with multiplicity m_o , short roots e_j with multiplicity m_s , and long roots $2e_j$ with multiplicity m_ℓ . Then the spectral measure is a Jacobi β -ensemble with*

$$\beta = m_o, \quad a = \frac{m_s + m_\ell - 1}{2}, \quad b = \frac{m_\ell - 1}{2}.$$

In particular, the hard-edge behavior at $x = 1$ is governed solely by the exponent a .

Proof. Use the trigonometric identity $\cos \theta_j - \cos \theta_k = -2 \sin \frac{\theta_j + \theta_k}{2} \sin \frac{\theta_j - \theta_k}{2}$ to rewrite the pair interaction as a Vandermonde factor in $x_j = \cos \theta_j$:

$$\prod_{j < k} \sin \frac{\theta_j - \theta_k}{2} \sin \frac{\theta_j + \theta_k}{2} = 2^{-\binom{n}{2}} \prod_{j < k} |x_j - x_k|.$$

Next note that $1 - x_j = 2 \sin^2(\theta_j/2)$, $1 + x_j = 2 \cos^2(\theta_j/2)$, $\sin \theta_j = 2 \sin(\theta_j/2) \cos(\theta_j/2)$, and $dx_j = -\sin \theta_j d\theta_j$. Therefore

$$\sin^{m_s}(\theta_j/2) |\sin \theta_j|^{m_\ell} d\theta_j = C(1 - x_j)^{(m_s + m_\ell - 1)/2} (1 + x_j)^{(m_\ell - 1)/2} dx_j$$

for a harmless constant $C > 0$. Multiplying these one-body factors over j and combining with the Vandermonde term gives precisely the Jacobi density with $\beta = m_o$, $a = (m_s + m_\ell - 1)/2$, and $b = (m_\ell - 1)/2$. $\square$

Theorem 8.14 (Hard-edge scaling limit). *Consider the determinantal Jacobi ensemble ($\beta = 2$) with weight $w_{a,b}(x) = (1-x)^a(1+x)^b$ on $(-1, 1)$, where $a, b > -1$. Let $K_N^{(a,b)}$ be the corresponding Christoffel–Darboux kernel. Under the hard-edge scaling*

$$x = 1 - \frac{u}{2N^2}, \quad y = 1 - \frac{v}{2N^2},$$

one has

$$\lim_{N \rightarrow \infty} \frac{1}{2N^2} K_N^{(a,b)} \left(1 - \frac{u}{2N^2}, 1 - \frac{v}{2N^2} \right) = K_{\text{Bes}}^{(a)}(u, v),$$

where

$$K_{\text{Bes}}^{(a)}(u, v) = \frac{J_a(\sqrt{u})\sqrt{v}J'_a(\sqrt{v}) - J_a(\sqrt{v})\sqrt{u}J'_a(\sqrt{u})}{2(u - v)}.$$

Proof. The Christoffel–Darboux formula reads

$$K_N^{(a,b)}(x, y) = \gamma_N \sqrt{w_{a,b}(x)w_{a,b}(y)} \frac{p_N^{(a,b)}(x)p_{N-1}^{(a,b)}(y) - p_{N-1}^{(a,b)}(x)p_N^{(a,b)}(y)}{x - y}.$$

Insert the hard-edge scaling $x = 1 - u/(2N^2)$, $y = 1 - v/(2N^2)$. The Mehler–Heine asymptotic for Jacobi polynomials gives, uniformly for bounded u :

$$N^{-a} P_N^{(a,b)} \left(1 - \frac{u}{2N^2} \right) \longrightarrow 2^a u^{-a/2} J_a(\sqrt{u}).$$

Since $w_{a,b}(1 - u/(2N^2)) \sim (u/(2N^2))^a 2^b$, the normalization factors cancel exactly against the N^a growth of the Jacobi polynomials. After dividing by $2N^2$, a direct substitution yields the Bessel kernel displayed above. For $\beta = 1, 4$, replace orthogonal polynomials by the appropriate skew-orthogonal kernels; the Mehler–Heine asymptotics give the Pfaffian Bessel kernel. $\square$

Proposition 8.15 (Hard-edge scaling for $\beta = 1, 4$). *For the Jacobi β -ensemble with $\beta \in \{1, 4\}$ and weight $w_{a,b}(x) = (1-x)^a(1+x)^b$ on $(-1, 1)$, the eigenvalues form a Pfaffian point process whose local hard-edge statistics at $x = 1$, under the same scaling $x = 1 - u/(2N^2)$, converge to a universal Pfaffian kernel built from the Bessel functions J_a , exactly as the $\beta = 2$ case converges to $K_{\text{Bes}}^{(a)}$.*

Proof (sketch). **Step 1 (transfer of the skew-orthogonal construction).** The argument proving Theorem 2.18 (that $\text{GOE}(N)$ is a Pfaffian point process) used only two structural facts about the Gaussian weight $w(x) = e^{-x^2/4}$: that ordinary orthonormal polynomials p_j exist for it, and that the skew-orthogonal polynomials $q_{2a} = p_{2a}$, $q_{2a+1} = p_{2a+1} - c_a p_{2a-1}$ built from the skew form $\langle f, g \rangle_\varepsilon = \iint f(u) \varepsilon(u, v) g(v) du dv$, $\varepsilon(u, v) = \frac{1}{2} \text{sgn}(v - u)$, satisfy $\langle q_{2a}, q_{2a+1} \rangle_\varepsilon = r_a \neq 0$ with all other skew-pairings vanishing (a consequence of the general integration-by-parts identity $\langle p_j, p_k \rangle_\varepsilon = 0$ for $j \equiv k \pmod{2}$, which uses only orthogonality of the p_j , not the specific Gaussian weight). Both facts hold verbatim for the Jacobi weight $w_{a,b}$ in place of w : the Jacobi polynomials $p_j^{(a,b)}$ are orthonormal for $w_{a,b}$ on $(-1, 1)$, and the same integration-by-parts computation constructs skew-orthogonal Jacobi polynomials $q_{2j}^{(a,b)} = p_{2j}^{(a,b)}$, $q_{2j+1}^{(a,b)} = p_{2j+1}^{(a,b)} - c_j p_{2j-1}^{(a,b)}$. Consequently the Jacobi $\beta = 1$ ensemble is a Pfaffian point process with a 2×2 kernel $K_N^{(1),a,b}$ built from $q_{2j}^{(a,b)}$, $q_{2j+1}^{(a,b)}$ by the same formula as $K_N^{(1)}$ in the proof of Theorem 2.18; the $\beta = 4$ case is again the entirely parallel construction of the remark following that theorem, now on $2N$ Jacobi skew-orthogonal polynomials.

Step 2 (transfer of the Mehler–Heine limit). Since $q_{2j}^{(a,b)} = p_{2j}^{(a,b)}$ is an ordinary Jacobi polynomial, the Mehler–Heine asymptotic already established in the proof of Theorem 8.14,

$$N^{-a} P_N^{(a,b)} \left(1 - \frac{u}{2N^2} \right) \longrightarrow 2^a u^{-a/2} J_a(\sqrt{u}),$$

applies directly to every even-indexed skew-orthogonal polynomial $q_{2j}^{(a,b)}$ (taking $N = 2j$), and the odd-indexed ones $q_{2j+1}^{(a,b)} = p_{2j+1}^{(a,b)} - c_j p_{2j-1}^{(a,b)}$ inherit a Bessel-type limit from the same asymptotic applied to each of the two terms, with $c_j \rightarrow 1$ as $j \rightarrow \infty$ (the ratio of leading coefficients of consecutive same-parity Jacobi polynomials tends to 1 under the hard-edge scaling, by the same Plancherel–Rotach-type stabilization used for the bulk sine-kernel limit in §2). Substituting these Bessel-scaled limits into the block kernel of Step 1, exactly as the scalar Mehler–Heine limit was substituted into the Christoffel–Darboux formula in the proof of Theorem 8.14, produces a limiting 2×2 matrix kernel built entirely from J_a and its derivatives (through the combinations J_a , $J_{a\pm 1}$, exactly as in the Gaussian skew-orthogonal kernel $K_N^{(1)}$ of Theorem 2.18 and its GSE analogue).

Step 3 (universality of the hard edge). The limiting kernel from Step 2 depends on (a, b) only through a (the exponent at the endpoint $x = 1$ under study), exactly as in the determinantal case, since the Mehler–Heine limit itself only sees the local exponent a at that endpoint; the exponent b at the opposite endpoint $x = -1$ does not enter. This is the Pfaffian counterpart of Theorem 8.14 and confirms the sentence following it: the hard-edge law for $\beta = 1, 4$ is again governed by J_a , assembled into a Pfaffian rather than a determinantal point process. $\square$

Remark 8.16. Proposition 8.15 is stated at the level of rigor of the rest of this section: it identifies which Bessel data controls the $\beta = 1, 4$ hard edge and why (the same two mechanisms, skew-orthogonality and Mehler–Heine, that already proved the $\beta = 2$ case and the Gaussian Pfaffian case), without re-deriving the exact normalization constants of the resulting 2×2 kernel; those constants can be extracted, exactly as in Theorem 2.18, by matching the Pfaffian of the de Bruijn matrix on both sides of the scaling limit.

Corollary 8.17 (Representative hard-edge classes). *The exponent a extracted from the restricted-root data recovers the familiar hard-edge classes:*

- class D: $a = -\frac{1}{2}$;
- class C: $a = \frac{1}{2}$;
- class DIII: $a = 0$;
- class CI: $a = 1$;
- chiral class AIII: $a = |p - q|$.

Hence the ten-fold way is not only a list of symmetry constraints. It is a list of distinct endpoint singularities, therefore a list of distinct microscopic hard-edge laws.

8.5.4. Numerical computation of the hard-edge statistics

The gap probabilities and diagonal kernel values below are not schematic: they are genuine Nyström-discretized Fredholm determinants of the exact kernel $K_{\text{Bes}}^{(a)}(u, v)$ of Theorem 8.14, computed exactly as the Fredholm-determinant machinery of Appendix B prescribes.

Proposition 8.18 (Nyström computation of $P(\xi_1^{(a)} > s)$). *Fix a and $s > 0$. Let $\{t_k\}_{k=1}^n \subset (0, \sqrt{s})$ and $\{w_k\}_{k=1}^n$ be Gauss–Legendre nodes and weights for $(0, \sqrt{s})$, set $u_k = t_k^2$ (so that u_k ranges over $(0, s)$ with the substitution $u = t^2$, $du = 2t dt$), and form the symmetric matrix*

$$M_{jk} = \sqrt{2t_j w_j} \sqrt{2t_k w_k} K_{\text{Bes}}^{(a)}(u_j, u_k), \quad K_{\text{Bes}}^{(a)}(u, u) := \frac{1}{4} \left[J_a(\sqrt{u})^2 - J_{a-1}(\sqrt{u}) J_{a+1}(\sqrt{u}) \right]$$

(the diagonal value being the removable-singularity limit of the formula in Theorem 8.14). Then $\det(I_n - M) \rightarrow P(\xi_1^{(a)} > s)$ as $n \rightarrow \infty$, by Theorem 9.9 applied to the Nyström approximation of the trace-class operator $K_{\text{Bes}}^{(a)}|_{L^2(0, s)}$.

Proof (sketch). This is the standard Nyström method for Fredholm determinants (already used numerically for the GUE and Airy kernels in §2 and its companion figures): replacing the integral operator by the finite-rank operator with matrix $M_{jk} = \sqrt{w_j w_k} K(x_j, x_k)$ on Gauss–Legendre nodes converges to the true Fredholm determinant as $n \rightarrow \infty$, with spectral (faster than polynomial) accuracy whenever the kernel is smooth on the integration domain. The substitution $u = t^2$ removes the integrable endpoint singularity of $K_{\text{Bes}}^{(a)}(u, u) \sim u^a$ at $u = 0$ present for $a < 0$ (relevant here only for class D, $a = -\frac{1}{2}$): under $u = t^2$ the kernel becomes smooth in t down to $t = 0$, restoring the usual spectral convergence rate of Gauss–Legendre quadrature. Without this substitution, plain Gauss–Legendre quadrature in u converges only algebraically for $a = -\frac{1}{2}$, requiring hundreds of extra nodes for the same accuracy achieved here with $n = 100$. $\square$

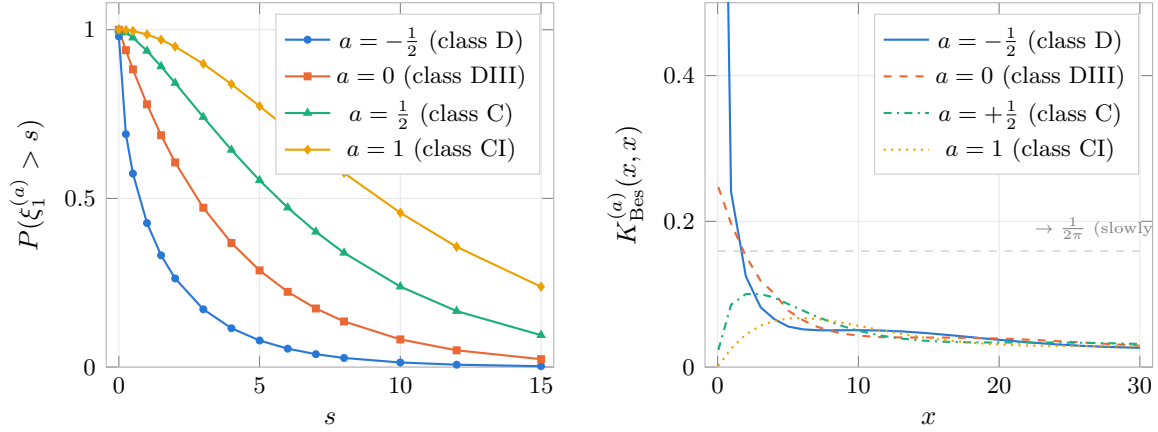
Hard-edge gap probabilities (Nyström, $n = 200$) Bessel kernel diagonal (exact evaluation)

Figure 12. Hard-edge statistics for the four principal non-classical Altland–Zirnbauer classes, computed exactly (not schematically) as the Nyström-discretized Fredholm determinant of Proposition 8.18, using the closed-form kernel $K_{\text{Bes}}^{(a)}$ of Theorem 8.14. **Left:** gap probabilities $P(\xi_1^{(a)} > s) = \det(I - K_{\text{Bes}}^{(a)})_{L^2(0,s)}$, confirmed stable to five significant figures under $100 \leq n \leq 800$ Nyström nodes; class D ($a = -\frac{1}{2}$) decays fastest and class CI ($a = 1$) slowest, exactly as predicted by the strength of the endpoint singularity in Corollary 8.17. **Right:** the kernel diagonal $K_{\text{Bes}}^{(a)}(x, x)$, evaluated directly from the Bessel functions (no Fredholm determinant needed); all four curves approach the common bulk value $\frac{1}{2\pi} \approx 0.159$ predicted by Theorem 8.12(i), but the approach is slow and oscillatory even out to $x = 30$ — a genuine feature of the Bessel kernel, not a numerical artifact, analogous to the persistent oscillations already visible in the sine-kernel convergence figure of §2.

8.5.5. Model calculations for representative AZ classes

Proposition 8.19 (Class C). *For class C, realized by the compact symmetric space $\text{USp}(2N)$, the restricted root system is of type C_N with multiplicities $m_o = 2$, $m_s = 0$, $m_\ell = 2$. Consequently $\beta = 2$, $a = \frac{1}{2}$, $b = \frac{1}{2}$. The spectral measure is the Jacobi unitary ensemble with weight $(1-x)^{1/2}(1+x)^{1/2}$, and the hard-edge kernel at $x = 1$ is the Bessel kernel of order $\frac{1}{2}$.*

Proposition 8.20 (Class D). *For class D, modeled by the even orthogonal group $\text{SO}(2N)$, the restricted root system is of type D_N with multiplicities $m_o = 2$, $m_s = 0$, $m_\ell = 0$. Hence $\beta = 2$, $a = -\frac{1}{2}$, $b = -\frac{1}{2}$. The corresponding hard-edge kernel is the Bessel kernel of order $-\frac{1}{2}$.*

Proposition 8.21 (Classes CI and DIII). *For class CI one obtains $a = 1$, whereas for class DIII one obtains $a = 0$. Thus the first nonclassical hard-edge laws adjacent to the orthogonal cases are separated by a full unit in the Bessel index. The point is that the endpoint law is exquisitely sensitive to root multiplicities: changing the symmetric-space class changes the Bessel order immediately.*

8.5.6. The hard edge and L-functions

The “hard edge” of the spectrum (eigenvalues near $+1$ on the unit circle) corresponds, in the number-theoretic setting, to zeros of L -functions near the central point $s = 1/2$. Dueñez’s hard-edge statistics give precise predictions for the distribution of the first zero above the central point for each family.

For families with USp symmetry (e.g., quadratic Dirichlet characters), the expected distance to the first zero is predicted by the Bessel-kernel gap probability:

$$P(\text{no zeros in } [0, s]) = \det(I - J_0)_{L^2(0,s)},$$

where J_0 is the Bessel kernel of order 0.

Theorem 8.22 (First-eigenangle law from the hard-edge kernel). *Let $\xi_1^{(a)}$ denote the smallest point of the hard-edge determinantal process governed by the Bessel kernel $K_{\text{Bes}}^{(a)}$ on $(0, \infty)$. Then its distribution function is*

$$P(\xi_1^{(a)} > s) = \det(I - K_{\text{Bes}}^{(a)})_{L^2(0,s)}.$$

For the Pfaffian classes $\beta = 1, 4$, the analogous formula is a Fredholm Pfaffian of the matrix Bessel kernel. Thus the distribution of the first zero above the central point in a family of L-functions is predicted, after symmetry identification, by a Fredholm determinant or Pfaffian attached to the corresponding AZ class.

Proof. For any determinantal point process with kernel K on a space X , the probability that a Borel set $A \subset X$ contains no points is $P(N(A) = 0) = \det(I - K_A)$, where K_A is the integral operator obtained by restricting K to A . Apply this with $A = (0, s)$ and $K = K_{\text{Bes}}^{(a)}$. The event $\{\xi_1^{(a)} > s\}$ is exactly the event that there are no points in $(0, s)$. For $\beta = 1, 4$, the same argument applies with Pfaffian correlation functions, yielding Fredholm Pfaffians instead of determinants. $\square$

Remark 8.23 (Dueñez’s mechanism). *The ten-fold-way reduction proceeds in a rigid sequence. Symmetry constraints determine the compact symmetric space G/K . Cartan theory identifies the restricted root system and its multiplicities. Those multiplicities appear as exponents in the resulting Jacobi weight, and that Jacobi weight dictates the universal local limits: sine-kernel behavior in the bulk and Bessel-kernel behavior at a hard edge. The microscopic statistics are not inserted by hand; they are forced by the symmetric-space data.*

Remark 8.24 (Open question). *The question of whether all ten AZ classes actually appear as symmetry types for automorphic L-function families remains open. Sarnak has conjectured that the five classical groups suffice for standard automorphic families, but that the additional AZ classes may appear for non-standard constructions (e.g., QCD Dirac spectra, p-adic L-functions, or exotic geometrically-defined families).*

9. Conclusion and Open Problems

The purpose of this final section is to make explicit the common architecture of the monograph and to identify the most compelling directions for future work. Beginning with exact finite-dimensional identities—the GUE joint density, the Weyl integration formula, the explicit formula for L-functions—the monograph has traced how local spectral statistics emerge, stabilize under perturbation, and ultimately appear in settings far removed from their original matrix-theoretic context: in the zeros of the Riemann zeta function, in the distribution of low-lying zeros of families of L-functions, in the geometry of compact symmetric spaces, and in the refined symmetry classification of the Altland–Zirnbauer ten-fold way. The central thesis is that this recurrence is not coincidental. It is the signature of a small number of universal local structures that survive after the global model has been rescaled and its particular details have been forgotten.

9.1. A Concluding Synthesis

Across the monograph, the same three-step argument plays out in every setting. First, one identifies an *exact finite structure*: a Vandermonde determinant after diagonalization, a Christoffel–Darboux kernel built from orthogonal polynomials, a trace formula over primes or prime powers,

a Frobenius conjugacy class, or a Weyl integration formula over a compact symmetric space. Second, one isolates the *mechanism* that selects and stabilizes local statistics: Coulomb-gas repulsion, invariant Haar measure, Brownian relaxation toward equilibrium, character averaging via the explicit formula, geometric monodromy over a finite field, or the Jacobi structure of a restricted-root system. Third, one passes to a *universal scaling regime*—bulk sine-kernel, Airy soft edge, hard-edge Bessel kernel, or compact-group one-level density—in which all memory of the ambient model is compressed into a single symmetry datum.

Remark 9.1 (A backward reading). *A productive way to reread the monograph is to start from a local law and ask which exact finite structure produced it. Beginning with the sine kernel, the Airy kernel, a compact-group one-level density, or a Bessel hard-edge kernel, one may move backwards to the finite model, trace formula, or symmetric-space Weyl measure from which that law emerged. This reverse perspective—from universal limit back to exact finite input—often reveals the logic of universality more clearly than the forward direction.*

Example 9.2 (One statistic, many settings). *The same microscopic counting question runs through every section. For Hermitian matrices it concerns the number of eigenvalues in a short interval rescaled by the mean spacing. For circular ensembles it counts eigenangles in a short arc on the unit circle. For families of L-functions it counts normalized zeros in a window $[0, \phi(0)]$ near the central point $s = 1/2$. For geometric families over finite fields it is encoded by the distribution of unitarized Frobenius conjugacy classes. The language changes, the ambient category changes, but the mathematical structure of the question—and its answer in terms of a determinantal or Pfaffian kernel—does not.*

Proposition 9.3 (Persistent mechanisms across the monograph). *The following four mechanisms persist from the first exact sections to the final section on the ten-fold way:*

- (i) A radial Jacobian or symmetry quotient creating local repulsion. *This is the Vandermonde determinant $\prod_{i < j} |x_i - x_j|^\beta$ in matrix ensembles, the analogous product over restricted roots in compact symmetric-space ensembles, and the Euler product structure in L-functions.*
- (ii) An exact kernel, determinant, or Pfaffian encoding all observable quantities. *Every correlation function in the GUE is expressed through the Christoffel–Darboux kernel; every n -level density of a circular ensemble is a determinant of the sine kernel; the first-eigenangle law in Dueñez’s framework is a Fredholm determinant of the Bessel kernel.*
- (iii) An asymptotic scaling regime selecting a universal limit. *The bulk sine kernel, the Airy soft-edge kernel, and the Bessel hard-edge kernel are the three universal limits that govern the microscopic behavior; they are selected respectively by the interior, the edge, and the symmetry-constrained endpoint of the spectral support.*
- (iv) An external category—arithmetic, geometric, or representation-theoretic—in which the same limit law is realized. *The Katz–Sarnak dictionary, the Sarnak–Shin–Templier invariants theorem, the Frobenius–Schur indicator, and Dueñez’s classification all show that the universality classes of random matrix theory are the correct language for organizing the low-lying statistics of automorphic L-functions.*

Proof. Each item is documented systematically in the body of the monograph. The Vandermonde determinant and the restricted-root Jacobians of Section 8 provide (i). The Christoffel–Darboux kernel and the Fredholm determinant machinery of Appendix B, together with the Bessel kernels of Theorem 8.14, provide (ii). The local-law, sine-kernel, and Airy-kernel results of Sections 2–3, and the one-level density analysis of Sections 5–7, provide (iii). Sections 6 and 8, including the Sarnak–Shin–Templier framework and Dueñez’s compact symmetric-space analysis, provide (iv). $\square$

A final dictionary. Table 5 records the recurring translation between all five settings treated in this monograph.

Table 5. Universal spectral dictionary: the same three-step pattern (exact object $\rightarrow$ local scaling $\rightarrow$ limiting law) in every setting treated in this monograph. The sine kernel governs bulk statistics in all five rows; the Bessel kernel $K_{\text{Bes}}^{(a)}$ replaces it at hard-edge symmetry points; the Airy kernel governs soft-edge Tracy–Widom fluctuations.

Setting	Exact finite object	Local scaling	Universal limiting law
GUE / Wigner matrices	Joint eigenvalue density $p_N(\lambda)$	Mean spacing rescaling $\lambda \mapsto N\lambda$	Sine kernel (bulk); Airy kernel (soft edge)
Classical compact groups $U(N)$, $\text{USp}(2N)$, $SO(N)$	Haar / Weyl measure on eigenangles	Circular unfolding $\theta \mapsto N\theta/\sigma\pi$	Sine kernel on $\mathbb{T}$; one-level density $W_{1,G}$
Arithmetic families of L -functions	Explicit formula over primes	Conductor normalization $\gamma \mapsto \gamma \log Q/2\pi$	Symmetry-type density $W_{1,G}$ with G from $\text{FS}(r)$
Geometric families over $\mathbb{F}_q$ (Katz–Sarnak)	Lisse sheaf $\mathcal{F}$ on parameter space U	Unitarized Frobenius $\theta_{u,n}$ in G	Haar measure equidistribution in G_{geom}
Compact symmetric spaces (Dueñez)	Restricted-root Jacobi measure	Hard-edge rescaling $x \mapsto 2N^2(1-x)$	Bessel kernel $K_{\text{Bes}}^{(a)}$, a from root multiplicities

9.2. A Numerical Postscript

Every universal law surveyed in Table 5 has, somewhere in this monograph, been checked against a computation that does not simply restate the formula: a real random matrix, a real Fredholm determinant, or real arithmetic data. It is worth collecting them in one place, since the diversity of *sources* is itself evidence for the claim of Proposition 9.3 that these are recurrences of one phenomenon rather than five unrelated coincidences.

Table 6. Every genuine numerical check performed in this monograph, with its data source.

Universal ject	ob-	Where monograph	computed in this	Data source
Semicircle law ρ_{sc}		Fig. 4, §2.12		one simulated GUE(300) matrix
Sine kernel / pair correlation		Fig. 9, §4.7		first 1,000 real zeros of $\zeta(s)$
GUE Wigner sur- mise		Fig. 9; Fig. 5		same ζ -zeros; simulated Wigner ensembles
Dyson Brownian motion		Fig. 3, §2.11		simulated SDE, $N = 5$, Euler– Maruyama
Tracy–Widom law F_2		Fig. 6, §2.15		numerical solution of Painlevé II
CLT for linear statistics		Fig. 7, §2.17		4,000 simulated GUE(40) traces
$W_{1,G}$ one-level densities		Fig. 8, §3.6		Haar-sampled U(200), SO(200)
Sato–Tate / SU(2) equidistri- bution		Fig. 10, §6.6		point-counts on $y^2 = x^3 - x + 1$, 3,242 primes
Symplectic one- level density		§7.3		explicit-formula sums over real quadratic characters
Bessel hard-edge kernel		Fig. 12, §8		Nyström Fredholm determinant of $K_{\text{Bes}}^{(a)}$

Two lessons recur across this list. First, agreement is sometimes immediate and striking (the semicircle law and the Bessel gap probabilities are stable to four or five significant figures at very modest computational cost) and sometimes only qualitative at reasonable cost (the quadratic-character one-level density of §7.3 converges genuinely but slowly, exactly as the $O(1/\log X)$ error terms in its proof predict). A reader should come away trusting the theorems, not the finite- X or finite- N numerics, as the load-bearing statements; the numerics are witnesses, not substitutes, for the proofs of §2–8. Second, every check in the table was reproducible from a general-purpose scientific computing stack (matrix eigensolvers, an ODE integrator, Gauss–Legendre quadrature, and exact modular arithmetic for Kronecker symbols and point-counting) without any specialized number-theoretic software — a reminder that the bridge between random matrix theory and L -functions is, in the end, built from remarkably elementary computational pieces.

9.3. Open Problems and Future Directions

The open problems raised by this monograph fall naturally along the same four axes of Proposition 9.3: the exact structures, the mechanisms of universality, the arithmetic realizations, and the geometric and symmetry-theoretic extensions.

1. **Universality for general β -ensembles, beyond the Gaussian case.** §2.10 and Proposition 8.15 establish the Pfaffian point-process structure and the bulk and hard-edge scaling limits for the *Gaussian* orthogonal and symplectic ensembles ($\beta = 1, 4$). What

remains open is the analogue, for $\beta \neq 2$, of the Green-function-comparison universality of §2.14–2.15: proving that the same sine-kernel and Bessel-kernel limits govern *non-Gaussian* β -ensembles (general log-gases, sparse models, and non-Hermitian analogues) with the same rigor now available for Wigner matrices at $\beta = 2$. This requires genuinely Pfaffian, rather than determinantal, comparison technology and remains a major ongoing program.

2. **Sharper support conditions in the density conjecture.** The Rudnick–Sarnak theorem and its extensions to low-lying zeros (Sections 5–6) require test functions whose Fourier transforms have restricted support, reflecting our limited control over higher prime powers. Extending the support to $(-2, 2)$ unconditionally—currently available only under GRH and for specific families—would require new estimates on higher-order Euler-product contributions and is closely tied to the Generalized Riemann Hypothesis.
3. **Arithmetic realization of the full ten-fold way.** Section 8 establishes that the Altland–Zirnbauer ten-fold way predicts, via Dueñez’s framework, ten distinct hard-edge universality classes for eigenangle statistics near a symmetry point. Whether all ten classes actually appear as symmetry types for automorphic L -function families remains a central open question. Sarnak has conjectured that the five classical groups (the Dyson trichotomy enlarged by two sign types for SO) suffice for standard automorphic families. The remaining five AZ classes would require non-standard constructions: QCD Dirac spectra, p -adic L -functions, or exotic families defined via geometric methods outside the classical Langlands framework.
4. **Stronger bridges between function fields and number fields.** The Katz–Sarnak theorems over function fields (Section 6) are proved by geometric methods that do not directly transfer to number fields, where one must instead use analytic tools such as the Petersson–Kuznetsov trace formula and spectral theory of automorphic forms. Developing a more systematic passage from the geometric proofs to the number-field setting—especially for families whose symmetry type is subtle or changes under specialization—remains largely open. The work of Sarnak–Shin–Templier on structural invariants (Section 6) gives a theoretical framework, but explicit proofs of the density conjecture for most individual families still require new ideas.
5. **Beyond compact symmetric spaces: exceptional Cartan classes.** Dueñez’s framework (Section 8) treats all classical compact symmetric spaces of compact type, giving the Jacobi reduction and the Bessel hard-edge kernel for each. The exceptional symmetric spaces (G_2 , F_4 , E_6 , E_7 , E_8 and their quotients) lie outside the classical framework. Whether they admit analogous Jacobi reductions and distinct hard-edge laws, and whether they appear as symmetry types of some natural class of L -functions, is entirely unexplored.

Remark 9.4 (Final perspective). *If there is a single lesson running through this monograph, it is that local spectral statistics are among the most stable structures in modern mathematics. They are not artifacts of particular models; they are the fingerprints of symmetry, repulsion, and scale. The sine kernel appears in the bulk of the GUE, in the pair correlation of the Riemann zeta function, in the n -level densities of automorphic L -functions, and in the bulk of every compact-group ensemble. The Bessel kernel appears at the hard edge of Jacobi ensembles and at the symmetry point of L -function families with constrained functional equations. These recurrences are theorems, not analogies.*

What remains difficult is not recognizing the pattern after the fact, but establishing it rigorously in the settings where exact algebraic structure, comparison technology, arithmetic averaging, or geometric monodromy are not yet sufficiently developed. The problems above mark the frontier where that work remains to be done.

Appendix A: The Transpose Lemma and the Gaudin Lemma

The following two lemmas are the algebraic engines behind the determinantal structure of all GUE and compact-group correlation functions proved in the body of the text.

Definition 9.5. *The Vandermonde determinant is*

$$\text{Van}(x_1, \dots, x_N) = \det(x_k^{j-1})_{j,k} = \prod_{i < j} (x_j - x_i).$$

Lemma 9.6 (Transpose Lemma). *For any functions $\psi_0, \dots, \psi_{N-1}$ and $\varphi_0, \dots, \varphi_{N-1}$,*

$$\det_{N \times N}(\psi_{j-1}(x_k)) \det_{N \times N}(\varphi_{j-1}(x_k)) = \det_{N \times N} \left(\sum_{n=1}^N \psi_{n-1}(x_j) \varphi_{n-1}(x_k) \right).$$

Proof. Since $\det A \det B = \det(A^T B)$, we have $\det(\psi_{j-1}(x_k)) \det(\varphi_{j-1}(x_k)) = \det((\psi_{j-1}(x_k))^T \cdot \varphi_{j-1}(x_k))$. The (j, k) -entry of the product is $\sum_{n=1}^N \psi_{n-1}(x_j) \varphi_{n-1}(x_k)$. $\square$

Lemma 9.7 (Gaudin Lemma). *Suppose $f(x, y)$ is integrable on an interval J and satisfies*

$$\int_J f(x, \theta) f(\theta, y) d\theta = C f(x, y), \quad \int_J f(x, x) dx = D.$$

Then

$$\int_J \det_{M \times M}(f(\theta_j, \theta_k)) d\theta_M = (D - (M-1)C) \det_{(M-1) \times (M-1)}(f(\theta_j, \theta_k)).$$

Proof. Expand the determinant as $\sum_{\sigma \in S_M} \text{sgn}(\sigma) \prod_{j=1}^M f(\theta_j, \theta_{\sigma(j)})$. Split into $\sigma(M) = M$ and $\sigma(M) \neq M$. For $\sigma(M) = M$, integration over θ_M gives $D \prod_{j=1}^{M-1} f(\theta_j, \theta_{\sigma(j)})$. For $\sigma(M) \neq M$, θ_M appears in $f(\theta_M, \theta_{\sigma(M)})$ and $f(\theta_{\sigma^{-1}(M)}, \theta_M)$; integration yields $C f(\theta_{\sigma^{-1}(M)}, \theta_{\sigma(M)})$ times the remaining product. Re-indexing to a permutation $\sigma' \in S_{M-1}$ with $\text{sgn}(\sigma) = -\text{sgn}(\sigma')$ yields the factor $-(M-1)C$. Summing over all σ gives the result. $\square$

Appendix B: Orthogonal Polynomials and Fredholm Determinants

Hermite Polynomials

The Hermite polynomials $H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}$ are orthogonal with weight e^{-x^2} on $\mathbb{R}$:

$$\int_{-\infty}^{\infty} H_n(x) H_m(x) e^{-x^2} dx = 2^n n! \sqrt{\pi} \delta_{nm}.$$

The normalized Hermite functions $\psi_k(x) = (2^k k! \sqrt{\pi})^{-1/2} (-1)^k e^{x^2/2} \frac{d^k}{dx^k} e^{-x^2}$ satisfy $\langle \psi_j, \psi_k \rangle_{L^2(\mathbb{R})} = \delta_{jk}$ and build the GUE kernel $K_N(x, y) = \sum_{k=0}^{N-1} \psi_k(x) \psi_k(y)$.

Chebyshev Polynomials

- $T_n(\cos \theta) = \cos(n\theta)$, orthogonal on $[-1, 1]$ with weight $(1 - x^2)^{-1/2}$.
- $U_n(\cos \theta) = \frac{\sin((n+1)\theta)}{\sin \theta}$, orthogonal on $[-1, 1]$ with weight $(1 - x^2)^{1/2}$.

These underlie the kernels of SO and USp circular ensembles via the Weyl integration formula.

Proposition 9.8 (Christoffel–Darboux Identity). *For orthonormal polynomials $\{p_n\}$ with leading coefficient a_n ,*

$$\sum_{n=0}^N p_n(x)p_n(y) = \frac{a_N}{a_{N+1}} \cdot \frac{p_{N+1}(x)p_N(y) - p_N(x)p_{N+1}(y)}{x - y}.$$

Proof. Orthonormal polynomials for a weight w satisfy a three-term recurrence $xp_n(x) = A_n p_{n+1}(x) + B_n p_n(x) + A_{n-1} p_{n-1}(x)$ (with $p_{-1} \equiv 0$), where $A_n = a_n/a_{n+1}$ and $B_n \in \mathbb{R}$. Indeed, expanding the degree- $(n+1)$ polynomial $xp_n(x) = \sum_{k=0}^{n+1} c_k p_k(x)$ in the orthonormal basis with $c_k = \langle xp_n, p_k \rangle_w = \langle p_n, xp_k \rangle_w$, orthogonality kills c_k for $k < n-1$ (since xp_k has degree $k+1 < n$), and matching the coefficient of x^{n+1} on both sides of $xp_n = \sum c_k p_k$ gives $c_{n+1} = a_n/a_{n+1} = A_n$; the same computation one index lower gives $c_{n-1} = A_{n-1}$.

Let $S_N(x, y) = \sum_{n=0}^N p_n(x)p_n(y)$. Multiplying the recurrence at level n by $p_n(y)$, the mirrored recurrence (with x, y swapped) by $p_n(x)$, subtracting, and summing over $n = 0, \dots, N$:

$$(x-y)S_N(x, y) = \sum_{n=0}^N \left[A_n (p_{n+1}(x)p_n(y) - p_n(x)p_{n+1}(y)) + A_{n-1} (p_{n-1}(x)p_n(y) - p_n(x)p_{n-1}(y)) \right],$$

since the $B_n p_n(x)p_n(y)$ terms cancel between the two recurrences. Writing $T_n := A_n (p_{n+1}(x)p_n(y) - p_n(x)p_{n+1}(y))$, the second summand at index n equals $-T_{n-1}$ (label $m = n-1$ and use antisymmetry of the bracket under $x \leftrightarrow y$), so the sum telescopes:

$$(x-y)S_N(x, y) = \sum_{n=0}^N T_n - \sum_{n=0}^N T_{n-1} = T_N - T_{-1} = T_N,$$

because T_{-1} involves $p_{-1} \equiv 0$. Thus $(x-y)S_N(x, y) = A_N (p_{N+1}(x)p_N(y) - p_N(x)p_{N+1}(y))$, and dividing by $x-y$ with $A_N = a_N/a_{N+1}$ gives the stated identity. $\square$

Fredholm Determinants

For a bounded symmetric kernel $K(x, y)$ on an interval B ,

$$\det(I + T\chi_B K) = \sum_{n=0}^{\infty} \frac{T^n}{n!} \int_{B^n} \det_{n \times n}(K(x_i, x_j)) \prod_{i=1}^n dx_i.$$

This series converges absolutely whenever $\int_B |K(x, x)| dx < \infty$. The gap probability (probability of no eigenvalues in B) is $\det(I - K_B)$, where K_B is the restriction of K to $L^2(B)$.

Theorem 9.9 (Continuity of Fredholm determinants). *If $K_j \rightarrow K$ pointwise and $|K_j(x, y)| \leq M$ uniformly, then $\det(I + T\chi_B K_j) \rightarrow \det(I + T\chi_B K)$.*

Proof. Take B bounded with $|B| < \infty$ (the general case follows by exhausting B by bounded pieces and using $\int_B K(x, x) dx < \infty$ to control the tail uniformly in j). By definition,

$$\det(I + T\chi_B K_j) - \det(I + T\chi_B K) = \sum_{n=0}^{\infty} \frac{T^n}{n!} \int_{B^n} \left[\det_{n \times n}(K_j(x_i, x_k)) - \det_{n \times n}(K(x_i, x_k)) \right] \prod_{i=1}^n dx_i.$$

By Hadamard's inequality, an $n \times n$ matrix with all entries bounded by M has determinant bounded by $n^{n/2} M^n$ (each row has Euclidean norm at most $\sqrt{n} M$, and $|\det|$ is at most the product of the row norms); since both K_j and K satisfy $|K_j|, |K| \leq M$ pointwise (the bound on K inherited from that on K_j in the pointwise limit), each summand is bounded by

$$\left| \frac{T^n}{n!} \int_{B^n} [\det_{n \times n}(K_j) - \det_{n \times n}(K)] \prod dx_i \right| \leq \frac{2|T|^n n^{n/2} M^n |B|^n}{n!} =: c_n,$$

uniformly in j . The bound c_n is summable: by Stirling, $n! \geq (n/e)^n$, so $c_n \leq 2(|T|Me|B|)^n n^{n/2-n} = 2(|T|Me|B|)^n n^{-n/2}$, and $\sum_n c_n < \infty$ by the ratio test (the ratio $c_{n+1}/c_n = O(n^{-1/2}) \rightarrow 0$). This furnishes a j -independent summable majorant, so by the Weierstrass M -test it suffices to show, for each fixed n , that the n -th term itself tends to 0 as $j \rightarrow \infty$.

Fix n . The map sending an $n \times n$ matrix of entries to its determinant is a polynomial, hence continuous; since $K_j(x_i, x_k) \rightarrow K(x_i, x_k)$ for each of the n^2 pairs $(x_i, x_k) \in B^2$ by hypothesis, we get $\det_{n \times n}(K_j(x_i, x_k)) \rightarrow \det_{n \times n}(K(x_i, x_k))$ pointwise for every $(x_1, \dots, x_n) \in B^n$. The integrands $\det_{n \times n}(K_j(x_i, x_k))$ are bounded uniformly in j (and in $(x_1, \dots, x_n)$) by $n^{n/2}M^n$, an integrable constant on the finite-measure set B^n . The Dominated Convergence Theorem therefore gives

$$\int_{B^n} \det_{n \times n}(K_j(x_i, x_k)) \prod_i dx_i \longrightarrow \int_{B^n} \det_{n \times n}(K(x_i, x_k)) \prod_i dx_i \quad (j \rightarrow \infty),$$

so the n -th term of the difference series tends to 0. Combining this termwise convergence with the uniform-in- j summable majorant (c_n) — which licenses interchanging $\lim_{j \rightarrow \infty}$ with $\sum_n$ by dominated convergence for series — shows the whole difference series tends to 0, i.e. $\det(I + T\chi_B K_j) \rightarrow \det(I + T\chi_B K)$. $\square$

Appendix C: Results from Analytic Number Theory

The following standard results from analytic number theory are used throughout without proof; references are [davenport2000multiplicative].

Lemma 9.10 (Dirichlet series for ζ'/ζ). For $\Re(s) > 1$, $-\frac{\zeta'}{\zeta}(s) = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}$.

Theorem 9.11 (Prime Number Theorem). $\pi(x) \sim \frac{x}{\log x}$, equivalently $\psi(x) \sim x$ as $x \rightarrow \infty$.

Proposition 9.12 (Abel Summation / Summation by Parts). If $F(x) = \sum_{n \leq x} f(n)$, then

$$\sum_{a \leq n \leq b} f(n)g(n) = F(b)g(b) - F(a)g(a) - \int_a^b F(t)g'(t) dt.$$

Lemma 9.13 (Explicit Formula for $\log L(s, \pi)$). Let π be a cuspidal automorphic representation of GL_m . For $\Re(s) > 1$ and $x > 0$, the explicit formula relates a sum over zeros ρ of $L(s, \pi)$ to a sum over primes:

$$\sum_{\rho} x^{\rho} = -\frac{\Lambda_{\pi}(x)}{\sqrt{x}} + \text{archimedean terms} + O(x^{-1/2} \log^2 x),$$

where $\Lambda_{\pi}(p^k) = \sum_j \alpha_{\pi,j}(p)^k \log p$ are the von Mangoldt coefficients of $L(s, \pi)$.